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I INTRODUCTION TO SC300

SC300 is the main controlling program of the **MAN MACHINE INTERFACE 2** (MMI2) system. This manual will describe the functions of SC300 in conjunction with MMI2 and its associated programs, and the various operating modes contained within SC300.

MMI2 is a computer system and software which allow the reading and writing of data from a control system for the purpose of monitoring and controlling a process. MMI2 is not generally an integral subsystem of a process control scheme, but more of an attachment to the system to allow a comprehensive diagnostic window into the process. MMI2 can be thought of as four separate computers contained in one. Only one of these computers may be accessed at any one time. The multiple computers are known as virtual consoles (VCs). Each virtual console executes a foreground program to perform the overall task of monitoring and control. The virtual consoles are named VC1, VC2, VC3, and VC4; and each VC has a particular task to perform. A foreground program is a program with which a user may see and communicate. A background program executes its tasks without directly showing its results to the console monitor or by getting direct commands through the console keyboard or other input devices. Background programs generally get its commands through a foreground program request.

SC300 is the main foreground program and through it, controls several background programs running in MMI2. It resides on virtual console 1 (VC1) on which the user spends most of his time. Communication to SC300 is accomplished through keyboard or touch screen commands, and output can be in the form of CRT displays, printed reports or disk file storage for later retrieval.

A. SC300 OVERVIEW

SC300 is a collection of control subprograms around a kernel of screen displays. The main purpose of SC300 is to show the user variable data in real time and to allow the user to send commands to the process. In addition, another main function is to correlate raw data into intelligible displays for diagnostic purposes.

The main construct of SC300 is an SC300 display screen. An SC300 display consists of a format, active parameters, alarm line, status line, data entry area, and function keys. There are up to 96 screen displays available in each SC300 program with some of those displays being reserved for system use. There are actually up to 90 available screens for custom configurations by the user (in some systems there may be less).

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The screen display is mostly described by the format which is configured separately from the display. The format contains fixed text and graphic characters that do not change in any way to the changing of variable data. Titles, column delimiters, and background colors are typically configured as part of a format. The active data in the form of variables from the MMI2 database is simply placed onto this format to create the active display. The active data is generally numbers or digital states from or to the process being monitored or controlled. These data points can be displayed in many different symbols. For example, a number may be represented in a fixed format such as four digits followed by a decimal point and one digit after the decimal point. This same number may be represented as a graphic symbol showing a variable length line based on the value of the number. A digital state may be represented as a shape which turns different colors based on its state or the words "ON" and "OFF". There are many different symbols available to use.

NOTE: Specific configuration instructions for screen displays are contained in **GEK-85717 SC-300 SCREEN DEVELOPMENT USER'S MANUAL.**

The status line is simply a single line of text which describes the operating system status and shows which VC is active and what program the other VCs are running. In addition, this status line will show which disks are being accessed (actually, which disks have open files) and the system clock. This line will also show any operating system alarms such as a printer or disk drive problem.

The system alarm line will always show the latest event that occurred on the system. This line is in the same format as what is stored in the **ALARMCON** program and will be described later.

NOTE: Operating instructions for **ALARMCON** (VC3) is contained in **GEK-85720 SC-300 ALARM CONSOLE USERS MANUAL.**

The function keys and the data entry area are the main user communication parts to the screen display. Each screen display has at least two levels of function key descriptions. The function key descriptions describe what will happen when the associated function key is pressed. The function key descriptions are broken up into eight separate boxes. Each box has a number in its border which corresponds to the function key on the keyboard. The box labeled "1" is executed when function key 1 is pressed. The eighth function key on a single row of function key descriptor boxes is always the MORE key. The MORE key will toggle the second level of function key boxes onto the display screen. These function keys are operated in the same manner as the first level of boxes.

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The data entry area allows the user to enter a command to the system. Normally, the function keys will have been configured so the user will not have to enter a command directly. This data entry area is normally used as an area for the operator to respond to system prompts after a function key has been pressed. As an example, the user may press a key associated to the descriptor box labeled "CHANGE SETPOINT". The current setpoint will be displayed in the data entry area with a "?" to allow the operator enter the desired new setpoint.

The combination of the screen display and the function keys allow the user to monitor and control the process. Supporting programs in other VCs allow ancillary functions such as analyzing historical trends or reviewing past events.

B. MMI2 DATABASE

The MMI2 database is the central core of the MMI2 system. All data and commands transferred to and from the MMI2 system and the process is contained in this database. The database contains several data types, numbers and digital states. There is a one for one mapping of these values to the operating master control device. If the control device is, for example, a Series Six Programmable Controller (R), for every number in the MMI2 data base, there is an associated register in the Series Six. For every digital state in the MMI2 database, there is an associated input, output, or bit in a register in the Series Six.

A background program provides the communication of data between the master controller and the MMI2 system. This program (DAC) also may scale numbers before they are read into the MMI2 or written out of the MMI2 system. It is therefore possible to have the corresponding MMI2 data point and the master control register to contain different numbers.

A memory map disk file which is provided per system determines the one to one correspondence between MMI2 and the master controller. It is with this file (MMP) that all database configurations are made. These configurations include screen development, recipes, alarms, and historical trending.

C. SC300 PARAMETERS

As described above, SC300 active displays are composed of a format and a list of active data. This list contains the data base point to show on the screen, how to show it (which symbol to use), where it is on the screen, whether or not it is changeable by the user, and whether or not it has a security level associated with it. This collection of information on an active screen data point is called a parameter. Parameter values and therefore, their screen presentation, will be updated as often as

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configured or commanded. The value of the parameter may be changed if the parameter has been configured as changeable.

To change the value of a parameter, the user must first select it by "tabbing" to it. The term "tabbing" is referred as the way a user selects a parameter to change. The user presses the TAB key on the keyboard until the parameter desired is highlighted by a specific color change. Only changeable parameters are capable of being "tabbed". Once the parameter is selected, the user may then activated a change in value by pressing the appropriate function key. In general, the sequence to change the value of a parameter would be:

1. Select the display the parameter is on.
2. Use the TAB or <shift> TAB key to highlight the desired parameter.
3. Press the appropriate CHANGE POINT function key as described by the function key descriptor boxes on the display.
4. Enter the new value by the keyboard.
5. Confirm the change, if required.

If the parameter has been configured with an alarm attribute, the parameter will change colors if the alarm attribute has been satisfied. In many cases, this change in color will be accompanied by an alarm message at the bottom of the display. Similarly, if the parameter alarm condition returns to normal, the color of the parameter will return to normal and a return to normal message will be sent to the bottom of the screen display.

D. SC300 COMMANDS

User commands sent to SC300 are of two forms. The primary way of entering commands into SC300 is by pressing a function key which corresponds to the appropriate descriptor box shown on the top of the screen display. The second way is to type the command directly into the data entry area of the screen display. This second way is not recommended because the command structure is very complicated and often depends on other information such as the attributes of a function key.

NOTE: The commands of SC300 are described in detail in **GEK-85719 SC-300 Command Reference Manual**.

Commands sent to SC300 have enormous impact on the MMI2 system as well as the system it is monitoring or controlling. The range of commands available allow the user to control background as well as other foreground programs and the ability to command the process to perform actions. The command structure allows the user to configure which commands may or may not be available to an operator by security levels or by omitting commands in function keys.

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1. FUNCTION KEYS

Function Keys in SC300 have been formatted at the manufacturer, however; a tool to modify function keys has been included in the MMI2 system. Function keys have different attributes that are definable:

1. Color
2. Description
3. Security
4. Command
5. Key Type (SDCS only)

The foreground and background colors of a function key may be standardized to certain functions. In the Super Drive Control Screen (SDCS) subsystem, the function key colors have been standardized. The description of the function key is simply the text in the descriptor box on the screen display that should explicitly describe the action the function key (upon pressing) would perform. The security level of a function key may be configured such that some function keys may be pressed by certain operators and have no effect when pressed by other operators. This security configuration is dependent on assigning an operator a security login password. This password determines the security level of an operator and therefore; the function keys that will operate if pressed by that operator. This same security level can be applied to individual screen parameters as well as directly entered commands.

The commands "behind" the function key is of the SC300 command set and will be executed when the associated function key on the keyboard is pressed. A function key may have more than one command and when pressed, the commands will be executed in the order they are configured (called gang commands). This implies that a function key may have a "batch" effect on the process. Function keys may be programmed to change the row of function key descriptors to another row of different function key descriptors. This allows a single display to have many function key options. This action of changing function key descriptors is sometimes referred as the changing of a "menu level". A function key may be configured as to what type of button it is (**only in SDCS subsystem**). The key may be programmed as a momentary push button (default), a toggle key (two way selector switch), or a jog type of button (on while down, off while up).

It is not necessary for operators to know the exact format of SC300 commands since function keys make it much simpler to enter commands. There may be times where an operator may desire to enter a command but not have a function key on the current display to do so. At this point he must change displays or call a different set of keys to the screen to find a function key to

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accomplish the desired function. It may be more convenient to type in a command manually in this case. Some of the commonly used commands to enter manually is shown in the next section.

2. TYPED COMMANDS

Commands entered directly into the data entry area of a screen display are known as typed commands. The operator should avoid using typed commands and try to use the function keys if possible. The following list is a description of commonly used typed commands.

DISxx

This command will call a new display screen. The xx is the display number it will show. The display number is shown on the first line of a screen display just to the left of the function key descriptor boxes and to the right of the first line of the data entry area.

UC6REF

This command will change the frequency at which a screen is refreshed. After entering this command, the current refresh rate will be displayed in the data entry area (the number refers to how many seconds between screen refreshes). The next line will have a "?". The operator should then enter the desired refresh rate number. SC300 will ask the operator to confirm the change by asking "OK ?". If he confirms the change by typing "Y", the change will immediately take place. The original refresh rate will be in effect if the operator types "N".

PEN

This command allows the operator to enable or disable a pen on the pen recorder system without showing the pen recorder setup screen. The command will invoke a question and answer session similar to the interactive session when on the pen recorder setup screen. (See the detailed description of the PEN RECORDER later in this document.)

PRT

This command prints the current SC300 display to the printer. This function is different from the Print Scrn key on the keyboard as it will attempt to print the graphic characters on the display as well as the normal text. This is the preferred way of doing a Print Screen Function.

Further explanation of commands will be discussed as the various subfunctions within SC300 are discussed in this manual.

3. SPECIAL KEYBOARD KEYS

In addition to the function keys F1 through F8, there are other

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keys on the keyboard that will by themselves send commands to SC300. These keys are not configurable and send the same commands from system to system. The description of these special keys on the keyboard follows. In addition to those keys discussed below, there are additional special keyboard keys defined for the SDCS (Super Drive Control Screen) system only. These keys will be discussed in detail later in this document.

ESC KEY

This key will abort any data entry except user log in and leave the screen parameter as it was when the data entry sequence was started. The ESC key will also clear error messages that obscure the function key descriptors at the top of the screen.

PGDN KEY

This key will call the next numerical SC300 screen display from the current display to the screen. If the next numerical display does not exist, the function keys will change but the current display will still be displayed. Hitting this key again will then display the TITLE SCREEN which is numerical display number one (1).

PGUP KEY

This key is similar to the PGDN key except it will call the previous numerical display from the current display to the screen.

TAB KEY

Selects the next changeable display parameter on the current SC300 display. If no parameter has been selected, it will select the first parameter.

Shift TAB

Similar to the TAB key except it selects the previous display parameter on the current display. If no parameter is currently selected, it will select the last parameter.

Right Arrow KEY

Same as TAB key except allows the keyboard type ahead buffer to buffer the keystrokes.

Left Arrow KEY

Same as Shift TAB key except allows the keyboard type ahead buffer to operate as normal.

Up Arrow Key

This will toggle the displays shown from the current display to the previously shown display. For example if display 1 was displayed and then display 60 was displayed, pressing the Up Arrow key will toggle between display 1 and display 60.

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F10 KEY

This is a special function key that will always display the TITLE SCREEN (display 1) when pressed.

CTRL KEY

If the MMI2 system has the optional touch screen, the system can be configured so that inputs through the touch screen on the SC300 console can only be done while this key is held down.

Virtual Console switching is accomplished by holding this key down and typing the VC number desired on the numeric keypad (the state of NUM LOCK is not relevant). In addition, holding the CTRL key and pressing the DEL key on the numeric keypad will toggle the size of the active VC from full screen size to a predefined window size.

4. TOUCH SCREEN INPUT

The optional touch screen interface provides an alternative way of user input in addition to the console keyboard. The touch screen is active for the SC300 program and a few other console programs. The console keyboard is always active regardless if there is a touch screen available or not.

The main advantage of touch screen inputs in SC300 is the capability of directly selecting a screen parameter without the the need to "scroll" to it with the TAB key. In addition, pressing the function key descriptor box on the screen emulates the function keypress. Normal data entry is still accomplished by keyboard entry, which include numeric and text entries, even with the touch screen interface active.

Virtual Console switching can be done with the touch screen by pressing the VC descriptor box on the bottom status line of the display. Toggling the screen window size is done by pressing the center of the bottom status line of the display.

On SDCS displays, there are special touch screen keys drawn as part of the SDCS displays which emulate most of the special keys as discussed above. These special touch screen keys emulate keyboard keys which are specially defined for SDCS and will be discussed later.

E. SC300 SECURITY

A security system has been included with the standard SC300 system. The security system is based on the security level of the person currently logged onto the system. Each user of SC300 should be given a user name for the purpose of logging onto the system and establishing a security level for that user. As discussed briefly earlier, function keys may be given a security

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level as well as individual screen parameters. Typed commands all have default security levels defined and are not configurable by the user. Function keys, screen parameters, and typed commands, therefore; can have a security level assigned to them, and for these items to be accessed or used, the user must have an equal or higher level assigned to him upon log in.

There are ten levels of security available. Users can be assigned a user name which establishes a security level from one to nine (1 - 9). Level one is the lowest and nine is the highest of the normal levels. A super level (level 0) can be assigned which will allow access to any function key, typed command or screen parameter regardless of the security level assigned to those items. Similarly, each function key and screen parameter may be given a level from zero to nine where level 0 means that there is no security required and nine requires the highest security.

User names are assigned by the system designer and can be modified by the end user with a special configuration utility which is part of the SC300 configuration package. This utility can only be used by persons who have the Master Password (Super User). A Super User can assign and delete user names as well as change security levels associated with a user name. Once user names have been established, a normal user must log into the run time SC300 system to establish his security level.

Logging into SC300 is done by either pressing the LOG IN function key on an SC300 display or by typing LOG at any SC300 display. A system prompt will over write the function keys of that display, prompting the user to enter his user name. If a valid user name is entered, the security levels associated with that user name will be established. If the user name is invalid, another prompt saying that the log in process has failed and the previous levels are still in force.

Logging off the system is accomplished by pressing the LOG OFF function key on an SC300 display or by typing LOC on any SC300 display. The security levels will be forced to one (1) of the current user. An automatic log off feature can be activated by the system designer or the Super User. This auto log off is assigned an inactivity timer, and if the system keyboard or touch screen has not been used within the assigned time limit, the current user is automatically logged off and the user level returns to level one. This time limit can be assigned by the system designer or the Super User.

Function keys that are assigned a level higher than the level of the current logged in user will show its left border and function key number in red. This means that the function key is not available to the current user. If the key is pressed, a system

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prompt will say that the key is locked. For screen parameters and typed commands, there are no visible screen attribute differences. If the user attempts to change a screen parameter without a high enough security level, a system prompt will state that the user does not have a high enough security level to change the selected screen parameter. If the user enters a typed command without the proper level of security, the command will have no effect on the system and will appear to have been "swallowed" by the system.

The use of the security system is optional and all function keys and screen parameters may be given no security level. This will allow the default security level of SC300 to be established as 0 for all users. This says that any function key and typed command may be used by any user and that any changeable screen parameter may be altered by any user.

F. SC300 SUMMARY

1. SC300 is the main foreground program of MMI2.
2. SC300 is always executing in VC1.
3. SC300 is used to command most other programs in MMI2
4. SC300 consists of
   a) 96 screen displays
   b) Large command set
   c) Direct access to MMI2 data base
5. SC300 screen displays consist of
   a) Background Format
   b) Active parameters
      1) data base point
      2) color selection
      3) symbol selection (how to display)
      4) security selection
      5) changeable attribute selection
5. b) 6) location on background format
   c) Function key descriptors
   d) Latest event
   e) Computer system status
   f) Data entry area
6. SC300 commands are in the form of
   a) Function keys
   b) Special keyboard/touch screen keys
   c) Typed Commands
7. The use of SC300 features may be restricted selectively by assigning security levels to those features and then assigning corresponding security levels only to qualified users of those features.

As discussed above, SC300 is the main controlling program of MMI2. The subprograms or functions of MMI2 that SC300 controls and the SC300 interface to those functions will now be discussed in further detail.

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II PEN RECORDER

A. PEN RECORDER OVERVIEW

The pen recorder subsystem allows the selective historical trending of data in the MMI2 database. This function emulates a sixty four (64) pen chart recorder. Signals must be connected to pens and a chart speed must be selected. Each pen can be assigned a different chart speed.

In MMI2, the signals assigned to pens must come from the MMI2 database and must be configured with special trending attributes before they can be used. This configuration is normally pre assigned before the installation of MMI2, however; the user may add these special attributes to signals not yet configured for trending.

NOTE: Specific information to configure a data base point with trending attributes is contained in **GEK-85721 SC-300 SYSTEM CONFIGURATION MANUAL**. There is a limitation of the amount of signals that can be additionally assigned to be trendable. Consult your General Electric Application Engineer for specific questions on these limitations.

B. PEN RECORDER SIGNALS AND FILES

When a signal is connected to a pen, a disk file is assigned to that signal. The chart speed assigned to that signal determines how often the signal is checked to be recorded. The chart speed can be thought of as the frequency the signal is checked for recording into the disk file. Signals are not automatically recorded each time they are checked. Part of the configuration of the signal specifies a trending "deadband" against which the current value of the signal is checked. If the value of the signal has changed by more than the configured deadband from the last time it was recorded, the new signal value is recorded to the disk file along with the time the signal was recorded. This is a method of data compression which only allows pertinent data to be recorded in the file.

Each disk file is circular in nature, that is, when enough recordings are made to that file so it reaches the end of the file, the next recording will wrap back and overwrite the beginning of the file. The maximum size of a file is configurable, however; it should never exceed 3600 recordings. With this maximum file size, if a signal was assigned a chart speed of one second (the signal will be checked for recording every second - the fastest rate for MMI2) and the signal changed beyond the configured deadband every second, the file will contain one hour of data. Similarly, if this signal changed beyond the deadband limit once an hour, the file will contain

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3600 hours of data, or 150 days of data (nearly five months). This example is extreme and one can expect (depending on the signal assigned) to be between the two examples. The user can control the maximum data storage by assigning variable chart speeds to a pen. The slower the chart speed, the more likely that a pen file will contain data covering a longer period of time. A two second rate means that the the signal will be checked every two seconds which means minimum storage of two hours.

Pen files are automatically named by SC300 so that a signal assigned to pen 1 will create the file PEN.01, and a signal connected to pen 10 will create the file PEN.10. Again up to 64 signals may connected to pens simultaneously which will create 64 pen files (PEN.01 - PEN.64).

NOTE: If more than 12 pens are connected simultaneously, it has been observed that SC300 will not be able to (at times) maintain sample rates assigned below two seconds.

Pen files are viewed with a program running on the Analysis Console (VC2) called SCTREND. This program will give a graphical representation of the values stored in a pen file with respect to time. Up to seven pen files may be viewed simultaneously under a single time scale with SCTREND. Users must specify which pen files to view when running SCTREND by the file names either directly by using the PEN.## files or by specifying a pen file that his been renamed. Pen files may be viewed even while data is still being recorded into the file being viewed.

NOTE: For more information on the operation of SCTREND, refer to **GEK-????? SC-300 SCTREND Users Manual.**

C. CONNECTING/DISCONNECTING PENS

A special SC300 screen display is included to connect and disconnect signals from pens on the pen recorder (PEN SCREEN). This display is called by pressing the appropriate function key command on other displays or by entering the typed command **PEN0** from any SC300 display. This display is a standard display and has function keys defined to accomplish all pen recorder commands. All pen recorder commands can be typed from any SC300 display as well, however; this display is provided as the most convenient method of configuring the pen recorder. Typing **PEN1** from the PEN SCREEN will return the operator to the display from which he entered PEN0.

Configuring the pen recorder from the PEN SCREEN is accomplished by "tabbing" to the desired pen and then pressing the ENABLE/DISABLE function key. If the pen is not currently assigned, SC300 assumes that the operator wishes to assign a

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signal to that pen, so SC300 will ask for a TAG NAME. A tag name is the name of the signal desired to trend. Tag names are defined as part of the control system and is generally common between all devices in the system. As an example, if the master control device is a Series Six Programmable Controller, all variables should have a defined nickname. This nickname will be carried through to the MMI2 system and be now known as a tag name. This tagname is another way of specifying a particular Series Six register or bit and an MMI2 data base point without need for machine dependent addressing schemes.

The operator should enter the tag name of the signal desired to trend. In most systems, help screens with a list of trendable tagnames will be available in the Analysis Console (VC2). If the signal is not legal, SC300 will overwrite the function key area with a message that says "Point not in database". (Remember, pressing the ESC key will clear the error message and restore the function keys.) If the tag name is a legal (configured) trendable signal, SC300 will ask for the period (chart speed). The operator should then enter the desired period. A period of one means that the signal will be checked against the configured deadband once a second, and therefore, has a possibility of recording the signal once per second. A period of five means that the signal will be checked against the deadband once every five seconds and has the possibility of recording this signal once every five seconds. The maximum usable period is probably not more than 60, although longer periods are legal.

Once the period has been entered, SC300 should issue a message to the alarm line that a pen has been connected and is sampling. This same message is sent to ALARMCON so that a record of when pens were connected can be maintained.

If a tagname exists at the pen location on the PEN SCREEN, SC300 assumes that the operator wants to disconnect the signal from that pen. SC300 will ask whether to disconnect the pen. If the operator types "Y" the pen will be disconnected and the pen file will be closed from further recordings until the pen is connected again. If the operator types "N", the pen will remain assigned to the tag name displayed. Each time a pen is connected, the associated pen file will be erased. If the operator wishes to keep the data, he must rename the pen file before connecting a new signal. Renaming a pen file is accomplished using normal DOS commands in a VC not running a foreground program.

A characteristic of the pen recorder is that recordings are first buffered in a RAM area to prevent accessing the disk file too frequently. When the buffer contains twenty (20) records, the buffer is then flushed to the disk file. When a pen is disconnected, any remaining recordings in the partial buffer is flushed to the file. If a pen file is renamed before it is

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disconnected, the remaining records in the RAM buffer will be lost. In addition, when the RAM buffer is flushed and the file is open by another program (such as SCTREND or HISTVIEW), SC300 will not be able to write the new data to the file. If this occurs, SC300 will retry several times to write the new data. If it cannot get control of the file from the other program, the 20 records in the RAM buffer will be erased and lost. If this occurs, an error message will be written to the alarm line of the display as well as ALARMCON to alert the operator that the last 20 samples of a pen were lost due to the open file. This should happen relatively infrequently as long as SCTREND or HISTVIEW is not being aggressively used by the operator.

Pens may be connected and disconnected from displays other than the PEN SCREEN. If function keys to do this are not available on these other displays, the operator may directly type commands to connect and disconnect pens. The typed command to connect or disconnect a pen is **PEN**. This command may be typed at any screen of SC300. SC300 will respond to this command by first asking which pen the operator wants. The operator should enter the pen number to connect or disconnect (1 - 64). If the pen is already connected to a signal, SC300 will ask if the operator wishes to disconnect the pen. If there is no signal connected, SC300 will continue with the same connection procedure as described above.

D. LOAD/SAVE PEN SETUPS

In most cases, an operator will generally have favorite signals to assign to the pen recorder or may have groups of favorite signals. The operator can manually assign signals to pens and then save the pen setup to a disk file. This will allow the operator to quickly assign signals to pens without going through the normal pen connection/disconnection procedure. At the PEN SCREEN, the operator only needs to press the SAVE SETUP function key and name the setup file (do not enter a file name extension as SC300 automatically assigns an extension to the file name the operator enters) to save the current pen recorder configuration. To recall a pen setup, the operator need only to press the LOAD SETUP function key and name the setup file to load. When a setup is loaded, any signals already connected will be disconnected and the associated pen files will be erased before the new pens from the setup file are connected. This means that if the operator wants to save the pen data from the existing setup, he must first rename the existing pen files before loading the new setup.

Again, the saving and loading of pen setups can be done from any SC300 display by using typed commands. The command to save a pen setup is **PEN3**. The command to load a pen setup is **PEN2**. SC300 will prompt the operator for the pen setup filename with both commands; file name to which to save the pen setup for PEN3 and the file name from which to load the pen setup for PEN2.

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PEN RECORDER SUMMARY

1. Pen recorder is a function of SC300 whose purpose is to historically trend both variable analog and digital signals from the MMI2 data base.
2. Only signals configured for trending can be assign to a pen.
3. Signals are named by TAG NAMES.
4. Assigning signals to pens is normally done through the PEN SCREEN and its associated function keys.
5. Summary of typed commands for the Pen Recorder.
   a) PEN - connect/disconnect signals from a pen
   b) PEN0 - display PEN SCREEN
   c) PEN1 - return to previous display from PEN SCREEN
   d) PEN2 - load pen setup file
   e) PEN3 - save pen setup to a file
   f) PEN4 - same as PEN command

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III RECIPE SYSTEM

A. RECIPE SYSTEM OVERVIEW

The RECIPE SYSTEM in MMI2 allows the user to configure setpoints for controlling a process into a list for a certain product. The user may then store different values per this list to store different operating conditions and then reload these values to the operating process to recreate this operating condition. The objective for the recipe system is to facilitate the changing of operating conditions of a control system so that different products made by the control system will be more consistent, and the change of the operating condition itself be made more efficiently.

The recipe system in MMI2 is a very flexible tool to facilitate this objective. The basis of the recipe system is the recipe class. The recipe class is a list of ingredients which define the operating conditions for a line of products. An ingredient in a control system will generally be a setpoint for a process device. This setpoint could be a temperature setpoint to a temperature controller, the speed or current setpoint of a DC or AC drive, or even information meant for a host or inventory computer. An example of classes could be cake and pizza. All cakes have similar ingredients as do pizzas. A recipe uses this common list of ingredients to form variations such as chocolate cake and lemon cake, or pepperoni pizza and sausage pizza. In other words a class is a maximum case of ingredients for a product with each individual recipe under the class specifying the amount of each ingredient including an amount equaling zero.

The recipe system allows 32 classes to be configured with the capability of storing up to 999 recipes per class. The amount of recipes that can actually be stored is more limited by computer hardware than a software limitation. In the SC300 system, each class is named by a three character mnemonic. Each recipe can be named with a 62 character name, however; only the first 18 characters of a recipe name can be seen on an SC300 display.

Recipe classes are defined by an off line configuration tool and become the basis for the recipe system. The running recipe system uses these predefined classes and stores and recalls values per these lists of ingredients.

NOTE: For specific information on creating recipe classes and other recipe support configurations, consult **GEK-85721 SC-300 SYSTEM CONFIGURATION MANUAL** and **GEK-85723 SC-300 RECIPE REPORT FORMAT EDITOR USER MANUAL**.

There are two different types of ingredients: normal ingredients and super ingredients. A normal ingredient is a single setpoint

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which can have only one value per recipe. A super ingredient is a group of other super ingredients and/or normal ingredients. A super ingredient can be thought of as a subdirectory and normal ingredients can be thought of as files in a subdirectory. There can exist up to four levels of super ingredients. In other words, super ingredients may be nested as much as four deep.

SC300 has the capability of moving recipes to three different areas in the system, compare stored and active recipes, edit recipes, and prepare reports for any particular recipe. Several standard SC300 screen displays have been devoted to the recipe system. These displays are all used in conjunction with the above functions.

B. RECIPE STORAGE

Recipes may be moved from and to three different places in the system. The ACTIVE recipe is the current values for the current class executing on the control system. The TEMP recipe is a memory space which can contain one recipe from any one class of recipes. The TEMP recipe is the only place that a recipe may be edited. In addition, the TEMP recipe can be compared to an ACTIVE recipe or a DISK recipe. The third area and main storage area is the DISK recipe. This area holds as many recipes as has been configured. A DISK recipe cannot be compared to an ACTIVE recipe directly nor can a DISK recipe be edited directly.

At least one class of recipes will have been defined prior to the installation of MMI2. In many cases, a control system will only require one class of recipes. There will be no recipes stored for that class in the DISK upon installation. The strategy for recipe storage is to capture known good ingredients from a properly running system and then storing that as recipe.

There are no standard recipe function keys under normal MMI2 systems. The Super Drive Control Screen system devotes an entire function key level to recipe manipulation. Each system may have the recipe functions under a particular SC300 display created for the particular application. There are six recipe action codes available to use when transferring recipes. All or part of these codes may be configured as part of a function key level.

1. Transfer TEMP to ACTIVE
2. Transfer ACTIVE to TEMP
3. Transfer DISK to ACTIVE
4. Transfer ACTIVE to DISK
5. Transfer DISK to TEMP
6. Transfer TEMP to DISK

To capture an ACTIVE recipe, the user must execute action code 2 or action code 4. These can be done through typed commands, however; it is much easier to use the preconfigured function

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keys. When either of the capture ACTIVE action codes are executed, (actually, any of the above recipe action codes) SC300 will first present what it has stored as the class last used. If this is the first time a recipe action code is issued, the class SC300 will assume will be a blank class or " ". SC300 will ask the user if the class displayed is "OK ?". If the class displayed in the data entry area is the class desired, the user simply types "Y". If the user wishes a different class to what is being displayed, he types an "N". SC300 will then ask for the class to be used even if there is only one class defined. The three character mnemonic must be entered at this time. If capturing ACTIVE to TEMP, entering the class is all that needs to be done as the ACTIVE ingredient values will then be copied to the TEMP memory area. If transferring from ACTIVE to DISK, SC300 will ask for the recipe name under which it will be saved after the class has been entered or confirmed.

Recipe naming is accomplished several ways. Names may be entered directly as a response to the "name recipe" prompt, or names may be entered from a standard name screen display. In the above example of capturing an ACTIVE recipe to DISK, or any action code that requires a recipe to be moved to or from the DISK area, a recipe name will be required to identify the recipe being captured. All DISK area recipes are identified by names (SC300 assigns a recipe number to each DISK recipe for its own use and special methods of manipulating recipes by the master control device) so it is important that a naming scheme is developed by the user to allow the unique identification of one recipe to another within a class. Going back to the example of capturing an ACTIVE recipe to DISK; the operator must now identify the name under which the recipe will be stored. The operator can over write an existing recipe or store it as a new recipe.

The data area should now be asking the operator for a name. For the case of over writing an existing recipe, the operator must enter the name of the existing recipe he wishes to over write. This can be accomplished three ways. The operator may enter the first character of an existing name (or the first few characters) and SC300 will search all the existing names for a match. Each time SC300 encounters a match, it will display in the data entry area, the full 18 character recipe name followed by an "OK ?" prompt. If this recipe name is NOT what the operator desires, he enters a "N". This tells SC300 to continue searching the names for another match. SC300 will continue showing the names matching the search criteria until the operator enters a "Y' to the "OK ?" prompt or SC300 cannot find any more names that match. If SC300 cannot find any more matching names, it will issue an error message which overwrites the function key area that informs the operator that a matching name could not be found. If the user had entered a "Y", SC300 will write the ACTIVE ingredient values to those ingredients on the DISK under that name.

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The operator may enter a second type of search criteria and that is to precede the characters with an asterick (*). This will make SC300 search for those characters any where within a recipe name, not just the beginning of a name. The same showing of a matching name and "OK ?" prompts result from this search criteria.

A RECIPE FINDING EXAMPLE :

DISK recipes already stored :

1) 12LB 171 2150FPM 2) 18LB 171 2860FPM 3) 15LB 169 2340FPM
4) 20LB 155 1930FPM 5) 12LB 169 2250FPM 6) 12LB 171 1710FPM
7) 22LB 189 2340FPM 8) 15LB 171 1990FPM

CASE 1 - Operator enters : 12LB

SC300 will find :
12LB 171 2150FPM
12LB 169 2250FPM
12LB 171 2710FPM

CASE 2 - Operator enters : *171 2

SC300 will find :
12LB 171 2150FPM
18LB 171 2860FPM

CASE 3 - Operator enters : *LB 230

SC300 will find :
NONE (SYSTEM ERROR IS DISPLAYED).

A third way of finding a recipe name is by simply entering "?" in response to the "RECIPE NAME ?" prompt. This will automatically switch the current SC300 display to the standard RECIPE NAME DISPLAY (NAME SCREEN). When the NAME SCREEN is displayed, the first sixty (60) recipe names are on this display with the remaining names on a paging system of this display. Each name takes an eighteen character "cell" of the display with three names per line and twenty lines per page. A blank name cell means that the DISK recipe designated for that cell is blank. The display has function keys defined for paging through all the possible name pages for the class defined as well as selecting a recipe and renaming a recipe. Once on this display, the operator uses the paging function keys and the TAB key to point to the desired name. He then must press the SELECT RECIPE function key. SC300 then stores the active values to the ingredients of the recipe under that name. The NAME SCREEN will then automatically switch back to the original display.

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If the user wishes to store the ACTIVE recipe into a blank DISK recipe, he may name this recipe directly on the display he invoked the action code. The operator must simply enter the full 18 character name desired in response to the prompt "RECIPE NAME ?". SC300 first searches the database for a matching name and if it does not find a match, it assumes that the operator is creating a new recipe. SC300 then searches for a blank "cell" and names this cell with the name entered by the operator. The ACTIVE values are then stored into the ingredients under the new name. If the operator wishes to view the name page prior to assigning the ACTIVE recipe to a new (blank) DISK recipe, he may enter "?" to the "RECIPE NAME ?" prompt. This again will automatically switch to the NAME SCREEN and display the first sixty DISK recipe names. The operator must then use the paging function keys and the TAB key to select a blank name "cell" and then press the SELECT RECIPE function key. SC300 will then ask the operator to name the recipe and then store the ACTIVE values to the newly named DISK recipe.

In a similar manner, recipes are moved between storage areas according to the action code issued as described above. The user cannot create a recipe without first capturing a recipe from ACTIVE or modifying an existing recipe on DISK. The movement of recipes between storage areas are an important part of creating, comparing, and modifying recipes.

C. RECIPE NAME SCREEN

As described above, the NAME SCREEN is a standard display of SC300 and is meant to be used as a convenient display for reviewing recipes stored on DISK, selecting recipes and naming recipes. This display has pre configured function keys to accomplish these tasks. The operator may call the recipe NAME SCREEN at any time from SC300 by pressing the appropriate function key sequences as described above. The operator may use a typed command to get to the NAME SCREEN which is "UC70". If SC300 has stored a class name by any previous recipe actions, the NAME SCREEN will display the names for that class. If a different class is desired, the operator must include the class name in the command or "UC7XXX", where XXX is the three character class name. If "UC70" is issued without a class predefined by previous recipe actions, the NAME SCREEN will be displayed and the operator will be prompted to enter a class name. Once a class name is entered, the NAME SCREEN will display the names for that class.

One difficulty that may be encountered by going to the NAME SCREEN via a typed command is that once the SC300 has displayed the NAME SCREEN, it will not automatically switch back to the previous display once the NAME SCREEN actions are completed. The operator must manually return to the desired display by typing

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the "DISxx" command or by pressing the UP ARROW key. The UP ARROW key will return the operator to the previous display. The operator may also press F10 at anytime which will display the title display.

Again, function keys have already been defined for the NAME SCREEN which allow all the NAME SCREEN functions to be conveniently accomplished. The commands the function keys accomplish can be typed as well. Here is a summary of NAME SCREEN typed commands :

1) UC70 - displays the NAME SCREEN and the first 60 names for the defined class. If no class has been defined, the operator will be prompted for one.

2) UC70XXX - Same as UC70 except XXX also defines the recipe class desired.

3) UC71 - displays the next page of names if already on the NAME SCREEN and more than 60 names exist.

4) UC72 - similar to UC71 except displays previous page.

5) UC73 - rename the "tabbed" recipe cell.

6) UC76 - select the "tabbed" recipe in response to an action code.

Recipe naming may be accomplished without the use of the NAME SCREEN. This is not recommended because it is difficult to name or rename recipes without first reviewing what names have already been defined for a class. In general, no pre configured function keys will have been defined for this function prior to MMI2 installation, so this command would only, normally, be accessed through typed commands. The command "UC5" can be typed and SC300 will then ask which recipe number is to be named. (The prompt will also give the legal range of recipe numbers the operator is allowed to enter.) The operator is then allowed to enter a recipe name. The recipe name, in this case, must be wholly entered as this command expects a new name to be given explicitly to the recipe defined. If the class is unknown by SC300, the operator will be prompted for the class name. The user may enter the class directly by issuing a "UC5XXX" command, where XXX is the desired class name. This command should only be used by supervisory personnel because the naming scheme could be harmed irrevocably by using the UC5 command.

D. RECIPE COMPARING

The recipe compare system is another standard screen display. The strategy used for recipe comparisons is based on loading a

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recipe in a temporary memory area and use that recipe (TEMP) as the basis for comparisons and editing. The TEMP recipe may be compared to the ACTIVE recipe, that is the ingredients defined by the class of the TEMP recipe currently executing in the process, or any DISK recipe in the same class as the TEMP recipe. The comparison display (COMPARE SCREEN) is displayed whenever the command to compare DISK to TEMP or ACTIVE to TEMP is invoked by a function key press or by typed command from any display in SC300. The COMPARE SCREEN is organized into four columns; the first containing the ingredient name and/or description, the second containing the value of the ingredient in the TEMP recipe, the third containing the value of the ACTIVE or DISK recipe (determined by the compare command issued), and the fourth column containing the engineering units of the ingredient if any. A fifth column is available showing additional information of the two recipes being compared such as recipe class, recipe names, whether either recipe has been modified, current compare page, and super ingredient level. The first page shown contains as many ingredients that will fit. Additional ingredients are held in subsequent pages and can be viewed by issuing the appropriate paging commands via function keys or typed commands. Again, as in the NAME SCREEN, the COMPARE SCREEN has predefined function keys and three function key levels. These levels are accessed by function keys and the MORE key. The data shown on these screens are the ingredients as defined in the selected class. Only normal ingredients have values associated with them whereas super ingredients will show no values. The color of the first column ingredient descriptions will distinguish between normal and super ingredients. Normal ingredients are shown with green text and super ingredients are shown with cyan (sky blue) text. If the value of a normal ingredient in TEMP is different than the value of the ACTIVE or DISK ingredient being compared,the description of that ingredient is shown in reverse video (black text on a green background). This reverse video effect is to allow the operator to scan ingredients and tell immediately if there are differences. Similarly, super ingredients will be shown in reverse video (black text on a cyan background) if any ingredient contained within the super ingredient has a miscomparison.

1. NORMAL INGREDIENTS

As mentioned above, normal ingredients are the only ingredients in a recipe which can have a value. This value can be an analog value as well as a digital state. The value of the ingredient is transferred when the recipe is moved from one storage area to another. Although the ingredient description is a fixed color and character length, the ingredient value itself is configurable. The presentation of the normal ingredient value and its color is defined by a system configuration tool in much the same manner as an SC300 screen parameter.

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NOTE: The detailed information on Recipe ingredient configuration is contained in **GEK-85721 SC-300 SYSTEM CONFIGURATION MANUAL.**

2. SUPER INGREDIENTS

Super ingredients can be expanded to show their contents by issuing the proper command, either via a function key or a typed command. Because, the expansion of a super ingredient can only be viewed at the COMPARE SCREEN, the pre configured function keys will accomplish this. Recipe class definitions can be configured to allow four levels of super ingredients. In other words, a super ingredient can contain another super ingredient which can contain another super ingredient which can contain another super ingredient. This fourth level super ingredient can only contain normal ingredients. Super ingredients in the first through third levels may contain can contain up to 20 of any mix of super and normal ingredients. The status column of the COMPARE SCREEN will show the super ingredient level on which the operator is displaying.

3. COMPARING TO ACTIVE RECIPES

When comparing TEMP to ACTIVE, the values shown on the COMPARE screen for the ACTIVE ingredients are captured at the time the compare command was issued. These ACTIVE values do not automatically refresh at the refresh rate assigned as on normal SC300 displays. The operator must issue a command to manually update the ACTIVE recipe values on the COMPARE SCREEN. This command is a pre configured function key of the COMPARE SCREEN and when pressed, the ACTIVE ingredient values are read from the data base and the new values are shown

E. RECIPE EDITING

The editing of a recipe can only normally be done from the COMPARE SCREEN. If MMI2 has the SDCS (SUPER SCREEN) system included, drive setpoints in recipes should be only edited with the provided SDCS CHANGE SETPOINT TEMP level of function keys which is described later in this document. On the COMPARE SCREEN, only normal ingredient values in the TEMP recipe can be edited. The operator must load the TEMP recipe from ACTIVE or DISK with the recipe movement action codes described above and use either the COMPARE ACTIVE TO TEMP or COMPARE ACTIVE TO DISK command to call the COMPARE SCREEN. Since only normal ingredients can be edited, the operator must expand super ingredients to show the actual normal ingredient before he can edit it. The operator must "tab" to the normal ingredient he wishes to edit and then issue the EDIT TEMP command through the pre configured function key on the COMPARE SCREEN or through the typed command "UC17". SC300 will display the current value in the data entry area followed by a "?". The user enters the new value (1 or 0 for a

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digital state), and then confirms the change by responding to the "OK ?" prompt. When the TEMP recipe is edited, the fifth status column updates to show that the recipe has been edited.

The operator continues this process until all desired ingredient values are changed. He then uses the recipe movement action codes to move the modified TEMP recipe to ACTIVE or to DISK. If he moves it to disk, he may overwrite an existing recipe or store it as a new recipe using the methods described above.

F. RECIPE REPORTS

The user has the opportunity to configure a customized recipe report for every defined class. This report can be printed utilizing one of the movement commands: REPORT ACTIVE or REPORT DISK. The report can be configured to have descriptive notation of the product or machine configuration for which the recipe values are set. The report of the ACTIVE recipe will pull the ingredient values from the ACTIVE recipe currently executing on the process. The report of a DISK recipe will report the ingredient values from a DISK recipe that the operator must also choose. Choosing a DISK recipe for a report is the same procedure used for choosing a DISK recipe to move to ACTIVE or TEMP.

NOTE: Specific information on how to create a recipe report format and the options available in creating the report is described in **GEK-85723 SC-300 RECIPE REPORT FORMAT EDITOR USER MANUAL**.

G. RECIPES IN SDCS

Recipe handling in the Super Drive Control Screen system is handled the same as recipes in a normal SC300 system. SDCS allows the comparing and editing of drive related recipes (TEMP) on the SUPER SCREEN in addition to the normal COMPARE SCREEN functions. These features will be discussed further in the SDCS section of this manual.

H. RECIPE SYSTEM SUMMARY

MMI2 systems are generally used in conjunction with a manufacturing process. These manufacturing processes are creating products or grades of products based upon the operating condition of their system. The operating condition of the system can be described by the various setpoints of local controlling mechanisms that make up the system. For example, a certain grade of extruded plastic could be described by the type of raw material being used, the temperatures necessary to melt the raw material, the pressure required for extrusion, and the cooling and curing time of the finished product. The operating condition

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of the manufacturing control system could now be described by the temperature setpoints of the melting devices, the pressure setpoints of the extrusion devices, and the temperature and time setpoints of the cooling devices. These setpoints, if recorded and reused for that grade should be able to recreate the product fairly consistently.

The objective of the recipe system is to facilitate the recording and re-employing of process setpoints for the creation of different products in a manufacturing control system. Recipes, if easily manipulated, can offer a very efficient way of changing the operating condition of the control system to make consistent grades of products.

The recipe system controlled by SC300 has the following features to successfully achieve this objective:

1. Capture known good operating recipes and storing them.
   a. Utilize English naming convention to functionally describe the recipe.
   b. Utilize ingredient partitioning to allow large recipes to be more easily understood and handled.
2. Reload stored recipes to the control system
3. Allow comparing of recipes
4. Allow manual editing of recipes
5. Allow custom reports of ingredient values to be created for hardcopy records.
6. Allow variations in the ingredient definition to group recipes into major sub product definitions
7. Allow large recipe storage capabilities.

H. SUMMARY OF RECIPE TYPED COMMANDS

UC7 commands

UC70	Display the NAME SCREEN and show the first 60 recipe names.
UC71	Display the next 60 recipe names on the NAME SCREEN.
UC72	Display the previous 60 recipe names on the NAME SCREEN.
UC73	Rename the "tabbed" recipe on the NAME SCREEN
UC75#	Perform recipe action code (# = action code)
# = 1	Move the TEMP recipe to ACTIVE
# = 2	Move the ACTIVE recipe to TEMP
# = 3	Move a DISK recipe to ACTIVE
# = 4	Move the ACTIVE recipe to DISK
# = 5	Print a custom report (configuration of a report format prior to the execution of this command is required) for a DISK recipe
# = 6	Print a custom report (report format configuration again is necessary) for the ACTIVE recipe

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UC75#	Continued
# = 7	Move a DISK recipe to TEMP
# = 8	Move a TEMP recipe to DISK
UC76	Select a recipe from the NAME SCREEN

UC5 commands

UC5	Name a recipe from the known recipe class
UC5XXX	Name a recipe from the XXX recipe class

UC1 commands

UC10	Display the COMPARE SCREEN and compare the ACTIVE recipe to the TEMP recipe.
UC11	Display the COMPARE SCREEN and compare a DISK recipe to the TEMP recipe.
UC12	Go to the next page of the COMPARE SCREEN.
UC13	Go to the previous page of the COMPARE SCREEN.
UC14	Expand a super ingredient on the COMPARE SCREEN.
UC15	Contract a super ingredient on the COMPARE SCREEN.
UC16	Update the values of the ACTIVE recipe on the COMPARE SCREEN.
UC17	Edit the "tabbed" TEMP recipe value on the COMPARE SCREEN.

This summary shows the commands capable of being entered directly by the user. It is recommended that function keys be used instead of typed commands whenever possible to avoid confusion of commands and their validity depending on which SC300 display the command is being executed.

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IV ALARM/EVENT SYSTEM

A. ALARM/EVENT SYSTEM OVERVIEW

The annunciation of abnormal events in a control system is a primary diagnostic tool of MMI2. The alarm/event system of MMI2 is contained in a background program and reports to the ALARMCON program running in the Alarm Console (VC3). This system depends on the user for the configuration of signals it should scan for reporting alarms and events. Alarm attributes may be configured with any signal, either analog or digital, to allow the annunciation of high and low limits, events or alarms based on the state of a digital signal, and the triggering of an alarm action based on the alarm condition of a signal.

The alarm/event system scans all signals defined with an alarm attribute at the defined scan rate. The scan rate at which a signal is monitored can be as fast as once per second. When a signal is sensed to be in an alarm state, a status is tagged with the actual value in the data base for SC300 to handle appropriately on screen displays, as well as the alarm system sending an alarm message to the Alarm Console, to the alarm line on the current SC300 display, and to the console printer.

The configuration of a signal's alarm attribute is critical to insure the alarm or event reporting properly. Alarms may have four priorities. Priority of alarm is the main attribute that will determine the color of an alarm message, the routing of the alarm message, and the color of the signal's representation on an SC300 display should it go into an alarm state. URGENT and CAUTION alarms are always routed to the Alarm Console Active Alarm Display and will remain on that display until the alarm condition has been reset. ATTENTION and EVENT alarms are not routed to the the active alarm display of Alarm console but are routed to the Alarm Console Alarm History Display as are URGENT and CAUTION alarms. When the alarm condition of any of the four priorities are reset, a return to normal message is issued to the Alarm Console Alarm History display, the alarm line of the current SC300 display and the console printer.

NOTE: The configuration of alarm attributes are further described in **GEK-85721 SC-300 System Configuration Manual.**

NOTE: For further information on the Alarm Console refer to **GEK-85720 SC-300 Alarm Console User Manual.**

In general, alarm priorities URGENT, CAUTION, and ATTENTION are considered alarms and EVENT alarms are considered events. Alarms should describe abnormal conditions of a process with the severity of the abnormality described by the above three alarm categories. Event alarms should not necessarily describe an

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abnormal condition, however; EVENTS should convey an important occurrence in the process to demark a change in the way the process is functioning. These four alarm categories may be used to configure automatic color or display attribute changes of the signal representation on an SC300 display so an operator may quickly determine by scanning the normal SC300 display that an abnormal condition exists and that further investigation which may include looking at the Alarm Console or other SC300 displays may be necessary.

B. ALARM MESSAGES

Alarm messages generated by the alarm/event system is the first item that should be consulted when investigating a process problem. Alarm messages are formatted to give information to the user, the time, item, type and description of the alarm condition. From this information, the user can make a determination of what other MMI2 tools will be required to fully analyze the problem, or by consulting events and/or alarms leading up to the problem, the user may make further intelligent decision as to what caused the problem.

The format of an alarm or event message changes from type of alarm and whether or not the alarm has been produced by a configured alarm action. The color of the alarm message is also variable according to the alarm priority but the colors can be set so that alarm priorities are not distinguishable by color alone.

NOTE: For more details on color configurations for alarm messages and alarm color attributes of screen parameters, consult **GEK-85722 SC-300 Alarm/Event Color Configuration Manual.**

1. ALARM MESSAGE DATE AND TIMES

All alarm/event messages begin with the time and date when the alarm system has sensed the signal has gone into an alarm state. The date and time is in the format:

HH:MM:SS DDMMMYY

where HH is the hour of the day 00 = midnight, 11 = 11:00 am, and 23 = 11:00 pm, MM is the minutes of the hour, 00 to 59, SS is the seconds of the reported minute, 00 to 59, DD is the day of the month, MMM is a three character abbreviation of the month (i.e Jan = January, Apr = April, Dec = December, etc.), and YY is the last two digits of the year. The time and date reported in an alarm message is the time the alarm scanner has sensed a signal has transitioned into an alarm state. The actual occurrence of the alarm may actually have occurred within two times of the configured alarm scan rate. For example, if a signal was

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configured with an alarm scan rate of two seconds, the alarm system would check the value of the signal once every two seconds to determine if the signal has met its alarm condition. If the alarm has satisfied the configured alarm condition(s), the alarm system will build the alarm message beginning with the time and date of the alarm. This time and date will show the time at which the alarm system began to build the message, therefore; the actual time the signal went into its alarm state could have been just after an alarm scan and the time stated in the message could be an additional scan rate depending on the scan load of the system. In addition, because data is being updated in the system at least every second (more typically, two or three times a second), a quickly changing signal could go into and out of its alarm state before the alarm system has had an opportunity to check it. The alarm would have, in this case, been missed.

2. ALARM TYPES AND CODES

The next two to four characters in a message will show the general type of alarm that is being shown. The possible codes are listed below:

Type	Code	Description
1	**OK**	Return to normal message
2	**DA**	Digital Alarm
3	**DK**	Digital Acknowledge (reserved for future use)
4	**HI**	High Alarm
5	**HK**	High Alarm Acknowledge (reserved for future use)
6	**LO**	Low Alarm
7	**LK**	Low Alarm Acknowledge (reserved for future use)
8	**MDCH**	Drive Mode Change Alarm
9	**SPCH**	Setpoint change message
10	**GENI**	Genius Alarm
11	**##**	Drive fault or SC-300 Program alarm

Type 11 alarms use a two digit number in conjunction with the text string following it to indicate what drive or SC-300 program has generated the fault message.

Alarm types 1, 8, 9, and 10 are always considered EVENT type alarms and cannot be configured to have any other alarm priority. This is due to the fact that no return to normal condition for these alarms is available.

The following sub-sections describe each alarm category.

a. Digital Alarms

```
15:20:38 03Feb88 DA U 272 COMFAIL   COMMUNICATIONS              STALLD
```

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The above alarm format is used for digital alarms. Digital alarms are used for indicating faults when one state indicates normal status and the other is a fault condition.

The alarm priority is indicated by the 'U' character. In this example, the priority is 'Urgent'. The number of the alarm is 272. This number can be used to disable the alarm from the SC-300 console, if desired. The 'COMFAIL' text is a 1 to 9 character nickname (tagname) that is unique for every alarm point in the system. It is followed by an english description of the alarm point and its new state. In this case, the state is 'STALLD'. When the alarm returns to normal, the state would probably indicate 'HEALTY'.

b. Analog Alarms

Analog alarms are similar to digital alarms except that they provide the current value of the point when the alarm text message was created. In addition, the engineering units for the point are shown immediately following the current value.

```
15:27:00 03Feb88 LO C E1B1TFB  EXTRUDER 1 BARREL 1 TEMP FDBK 100.00  DEG
```

The LO in the message defines that the signal whose name is E1B1TFB has gone below the configured low limit of the signal and thus is a LOW ALARM. If the characters were HI, the alarm would be a HIGH ALARM. The C following the LO is the alarm priority of Caution.

c. Series Six Genius Alarms

Genius alarms are always indicated by the word 'GENI' in the alarm code field. Genius alarms can indicate faults for bus controller(s), blocks, and points. Genius alarms must be supported in the Series Six programmable controller ladder logic program in order for these messages to be generated.

d. Drive Alarms

```
15:27:00 03Feb88  4 U FIRST PRESS   LOSS OF AN AC PHASE                F22
```

The 4 in the above message indicates the drive number with this message is associated. The description of the alarm includes an english description of the drive along with an expanded fault description. The alarm priority is U or Urgent and should distinguish between a trip or annunciated type fault. A quirk of the alarm system only allows the first alarm sensed to be given a message. If an annunciate (C or Caution) type alarm was displayed, and the alarm progressed into a trip type alarm, a message would not be built for the trip type alarm. The fault number shown at the end of the message is the same number one

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would find at the drive itself and all reference material to further describe and give possible solutions to the fault, would be applicable.

e. Drive Mode Change Events

If this function is supported in your configuration, the system will generate an event whenever a command sent to one or more drives is sensed. The message will indicate the source of the command, the destination(s) of the command and the command that has been requested.

```
15:28:00 03Feb88 MDCH MSTR CTL P to 1st PRESS                    RESET
15:27:12 03Feb88 MDCH FELT OS    to FELT SECT                    RUN FWD
15:27:47 03Feb88 MDCH MSTR CTL P 1st Dryer *REEL       (BCAST)   SLOW FWD

15:47:47 03Feb88 SPCH MASTER OS  Wire Turn *REEL       MMS       2500.0 FPM
```

The first message indicates that a request to reset the First Press had been issued by the SERIES SIX at the time indicated. The second message shows that a request to Run Forward the First Press, the Paper Roll #1 and all drives between those two drives from the FELT intelligent operator station had been issued. Because the message did not indicate a range of drives, the request was sent to an entire preconfigured zone or section of drives. The third message indicates that a request to put the First Dryer, the Reel, and all drives between those two, into Slow mode. The (BCAST) and * indicate that the request transcended zone or section boundaries and was issued to the entire line. The start of progression is indicated as the First Dryer and the end of progression is indicated as the Reel and the source of the command was the Series Six. The "P" in the source indicates the command came from LAN channel P.

The fourth message is a setpoint change message denoted by the SPCH. These are a class of Drive Mode Change Events that have a slightly different format than the Mode Change messages. Again, the source of the new setpoint is shown first, in this example, the MASTER intelligent operator station. The destination is next and in the message above includes the Wire Turning roll, the Reel, and all the drives between. The * indicates a broadcasted setpoint change. The setpoint being changed is the Master Machine Setpoint (MMS) and the new setpoint is 2500.0 FPM.

f. Internal Setpoint Change Messages

```
15:25:06 03Feb88 SPCH     PA01SET   PARISON POINT #01  0.0000 to  56.00 %
15:26:01 03Feb88 SPCH     E1MXTSP   EXT 1 M-PIPE MELT  0.0000 to  230.0 DEG
```

Whenever a point is changed by the operator using the CHANGE POINT function key, or a SETPT INC or a SETPT DEC function key, an english message alarm is generated. This message records the

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setpoint change showing the nickname (tagname), its description, the old value, and the new value with its engineering units (if applicable).

g. SC300 Program Alarms

```
15:23:49 03Feb88 72 E SC-SETUP      Execution from DISK complete        F93
15:27:00 03Feb88 75 E RECORDER      Pen channel started and sampling    F203
```

Several of the programs in the SC-300 system generate alarm or event messages to indicate system conditions. For example, when

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a new pen is connected in the recorder system, an event message is generated indicating that a pen channel has started. When recipe operations complete, an event message is generated to indicate the request has been processed.

C. OTHER ALARM SYSTEM FUNCTIONS

In addition to the normal creation of alarm messages to annunciate an alarm or event, the display of the signal in an alarm state can be configured, as part of an SC300 display parameter, to change its display attributes to show when the signal is in an alarm state. This display attribute is usually a color change, where the color can be configured differently by alarm priority.

The configuration of an alarm on a signal can contain other information to further help the user analyze the cause of an alarm or system problem. The user can configure an alarm to trigger a digital state in the database which could then be sent to the master control device to further signal an operator that an alarm has occurred. This associated digital output may be used to invoke a fault recovery procedure in the control system.

An alarm may be configured suppress the scanning of other alarms to prevent conflicting or ambiguous or nuisance messages to be generated as a result of this alarm from occurring. For example, if it is known that if a drive has a trip fault, the mechanical system will automatically reconfigure itself to a different state. The trip fault may be used to disable the alarm message that the mechanical system has been reconfigured thereby eliminating any confusion that the reconfiguration was the cause of the fault rather than an effect of the fault.

In addition to the above, alarm action procedures may be invoked to automatically switch the display screen of SC300 to a particular SC300 display or a particular virtual console to be displayed.

D. ALARM SYSTEM SUMMARY

Alarms and events, when used in conjunction with pertinent system occurrences can be a useful tool in allowing operating and maintenance personnel to diagnose system problems quickly and efficiently. In general, the alarm system should annunciate the problem itself and with preceding alarms and events, the cause of the problem should be easily narrowed to a small amount of possible causes. Utilizing the pen recorder and other maintenance tools of MMI2 and other diagnostic tools, the cause should be quickly analyzed. It is important then to configure those alarms and events to allow this kind of analysis. The alarm system has the following features to aid in this objective.

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1. Provide descriptive messages
   a. Time/Date Stamps
   b. Signal Identification
   c. Diverse amount of alarm types
   d. Four different alarm priorities
2. Provide SC300 display alteration for signal in an alarm state
3. Allow an alarm state to disable other alarms
4. Allow special alarm action procedures to be invoked by an alarm or event.
5. Allow a digital state to change in association with an alarm to activate master control fault recovery or external alarm annunciation.

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V SUPER DRIVE CONTROL SCREEN (SDCS)

A. SDCS OVERVIEW

The Super Drive Control Screen (SDCS) is a subsystem of SC300 and not simply an SC300 screen display. SDCS was created especially for the monitoring, control and problem diagnosis of the General Electric DC-300 family of drives. The programs and operating procedures contained within SDCS is very specialized for this purpose and cannot be transferred to the control of other drives or control systems. SDCS depends on the drive system architecture of the DC-300 control system including peripheral control devices such as the master control system and Drop In Operator Stations.

SDCS gets its name from its central construct, again an SC300 display screen (the Super Screen). The Super Screen is a standard SC300 display screen capable of all functions of a normal SC300 display and more. The Super Screen can show the data for up to eighteen drives per screen, and therefore; drive systems having more than 18 drives will include more than one super screen. The display parameters on the Super Screen is limited to one type of symbol which overrides SC300's normal display techniques. This symbol actually removes control from SC300 and allows the display symbol programming to control the screen display.

The operational intent for the Super Screen is to constantly display the operating data of the drive system on one screen while showing operating and monitoring options on several different function key levels interchangeable on the same display. The function keys for SDCS have been expanded to show two rows of function keys instead of the normal one row of keys. This special two row function key system is unique to SCDS and will not be used in any other part of SC300. This two row function key system gives the user a tremendous amount of operating and monitoring functions. These functions may be enabled or disabled depending on the amount of control desired at the MMI2 console over the process control system.

SDCS, however, is not just the Super Screen, but is a system of programs running concurrently with SC300 to provide many functions in addition to normal SC300 sub functions already described.

1. Special Data Display
2. Drive Command Control
3. Special Recipe Features
4. Added Alarm Capability
5. Drive Historical Trending Retrieval
6. Special System displays

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The combination of these features allow the monitoring and control of the complete drive system from SC300 and the SDCS screens if desired. In addition, the features of SDCS allows comprehensive diagnostic capabilities for each individual drive as well as the entire drive system.

B. SDCS DATA COMMUNICATION

Data for SDCS is communicated to and from MMI2 and the control system as described earlier for SC300 data communication, however; data from the drives to the master control device is a special communication protocol meant to allow a highly efficient way of communicating data from MMI2 to the drives. The heart of data communication is the Drive Local Area Network (LAN). There is a control board in the master control device that allows the access of data from the LAN. All drives in the system, as well as Drop In Operator Stations, will be connected to this LAN to send feedbacks and status to the master control device and each other as well as to receive commands and setpoints from the master control device or Drop In Operator Stations. The control card in the master control (SuP) allows programs in the master control (and MMI2) to send commands and setpoints to the drives as well as receive and decode status and feedbacks from the drives. SuP places the decoded data in a central memory area of the master control to allow this data to be read by MMI2 and other devices which care about this data, such as a process control host computer. Once this data is in this memory area of the master control device, MMI2 will read this data through its normal means along with other data for the normal SC300 functions. Sending commands and setpoints to the drives from MMI2 is accomplished in the opposite manner where SC300 will send data to the master controller by its normal means. SuP reads the memory area and picks up those pieces of data it was taught to care about. SuP then converts this data into commands and setpoints and sends them to the appropriate drives or Drop In Operator Station.

The function of SuP is simply a data interpreter. SuP understands two languages, the drive language and the master control language, and converts these languages into simple pieces of data. Because of this, the control of drives in this system becomes totally data driven and only requires simple data protocols to accomplish very complicated control functions.

Data can now be classified into four different categories :

1. Commands to the Drives
   RUN, STOP, JOG, etc.
2. Setpoints to the Drives
   Master Machine Speed, Tension Reference, Etc.

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3. Status from the Drives
   RUNNING, FAULTED, IN CURRENT LIMIT, etc.
4. Feedbacks from the Drives
   Speed, Load, Tension, Motor Volts, etc.

All data can now be "mapped", that is, the data for all four categories of data may be placed into a unique and fixed memory area of the master control. For status and feedbacks, there will be a memory area mapped for each drive since each drive can have unique status and feedbacks. For commands and setpoints, it is only necessary to have a common area for all the drives. The command is established and another qualifier is created to tell SuP which drives should receive the command or setpoint.

For example, if the operator wants to tell the last three drives in the line to STOP, he must simply set the STOP bit in the memory map and then establish that that bit should go only to the last three drives in the line. This amounts to setting one bit and sending one word of numeric data from SDCS to the master control. In the master control, Sup watches for changes in the memory map that SuP was pre-taught to watch. Upon seeing a change, SuP will pick up the command and the word which specifies the drives to receive the command. SuP then formats a data packet to send on the LAN and then sends it when it has the token. All drives will receive the packet and then decode it. When a drive decodes this packet, the drive will first determine whether or not the data in the packet is meant for "him". If the packet is meant for that drive he implements the command, in this case, STOP. If the drive "sees" that this command is not meant for "him", the drive simply ignores it.

SDCS and the function key levels allow the operator to send these commands and setpoints without requiring knowledge of this memory map or communication protocol. To receive data, SDCS utilizes the normal communication scheme used by SC300. (There are special communication methods for SDCS in addition to what is described here and will be described in detail later.) The drives send status and feedbacks only if the status and feedbacks change. This allows only pertinent and new data to be transmitted on the LAN. This data is then transferred to the master control memory map by SuP. MMI2 using its normal communication scheme, reads this data from the map and transfers the data into the SC300 data base. SDCS reads this data base and updates the Super Screen and other functions requiring drive data.

C. SUPER SCREEN DATA DISPLAY

The data on the Super Screen is formatted in a fixed manner for each drive. Each drive will have one line of the Super Screen dedicated to displaying its data. The data is arranged in

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seven separate columns with each column displaying a fixed category of drive data. The lines of data have an alternating background color of dark blue and light blue so it easier to follow a line of data for a single drive from the first column of data to the seventh column of data.

1. DRIVE NAME COLUMN

The first column of data shows a fifteen character drive name. The name will not change based on any status of the drive, however; the name column is important because it will display pertinent information based on the colors displayed. When an operator sends a command to a drive, he must select the drive that will receive this command. This is done by using the TAB or Shift TAB keys. When a drive has been selected, the name of the drive will change from its normal color to a background color of brown and use a bright white for the color of the characters. In addition, if the "tabbed" drive is part of a group of drives making up a mechanical section, the characters of the other drives' names will be highlighted with a bright white color.

For example, a paper machine press section may consist of four driven rolls. These rolls drive a felt which supports the paper during the pressing action of this section. The felt mechanically ties these four driven rolls together such that an increase or decrease in power input from one roll will affect the operation of the other three rolls. Although each of the four drives' data will be displayed on separate lines of the Super Screen, if the operator "tabs" to any one of the four drive names, the other three names will be displayed with bright white characters instead of their normal colors. The background colors of these other drives in the same press section will not be altered. This highlighting of drive names is meant to alert the operator of the mechanical sections, or zones, in the drive line. It will become apparent later in this document that the operator awareness in zone configuration is important when sending commands to multiple drives.

2. MODE COLUMN

The second column shows the drive's mode of operation. The mode column is very important and shows the drive's mode and status with a combination of mnemonics and colors. This column display is actually a compression of a multitude of status bits available from the drive. SuP actually does the compression and encodes a sixteen bit word with the selected status bits chosen for this mode word. This mode word is then converted into a symbol for the SDCS display screen and is shown as a seven character block with the characters in a foreground color and the block with a particular background color.

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There are four major modes available :

1. Drive is not communicating on the LAN
2. Drive has a fault (distinguishes between a trip fault and annunciated faults)
3. Drive is in an independent mode
4. Drive is in a coordinated mode.

The mode column will show when the drive is not communicating on the LAN with the characters, DRV OFF, in white characters with the background in black. In addition, the rest of the line of data for that drive will be blanked (no characters on a black background in the subsequent five columns) to reinforce that the drive is not communicating on the LAN.

When the drive declares a trip fault (a fault condition that removes power from the motor), the background color of the mode block will turn red with a white three to five character mnemonic that describes the fault that occurred. If the drive declares an annunciated fault (a cautionary fault condition that allows the drive to keep running), the mode block will turn brown with a yellow three to five character mnemonic that describes the particular fault that occurred. For example, if a drive is accelerating and the load is above its rated current capability, after a preset time (depending on how much the current is over rated load), the drive will declare an annunciated fault, TOCAN. TOCAN simply means that the drive is over loaded and will trip if the load is not brought below rated current. If the load is not reduced, the TOCAN indication (which is yellow characters on a brown background in the mode column) will proceed to a trip fault indication of TOC (which is white characters on a red background). At this point, power would be removed from the motor and the fault indication would remain in the mode block even though the operating mode of the drive has changed (from running to a stop). This is because the fault indication takes priority on normal mode descriptions.

Normal mode indications will be displayed if the drive is communicating on the LAN and does not have any faults. The drive could now be in an independent mode or a coordinated mode. Independent mode is annunciated by a light gray background in the mode block and coordinated mode is annunciated by a green background in the mode block. If a drive is in independent mode, it can only respond to a command directed to that particular drive. Coordinated mode allows the drive to respond to directed commands as well as group or broadcasted commands. For example, if there are three motors mechanically coupled together at a gear box, and these three motors are configured to be in a common zone of drives, the drives must respond to commands as a group. A zone run command sent to this zone will make all three drives run simultaneously if all three drives were in the coordinated mode.

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If one of these drives were in independent mode and the other two were in coordinated mode, only the two in coordinated mode would respond to the zone run command. The drive in the independent mode would throw out the RUN command message. The intent of independent mode is to allow drives configured in a zone to be broken up to individual drives for maintenance or an abnormal running condition.

In either independent or coordinated modes, the characters in the mode block are broken into two separate three character mnemonics. The first mnemonic in the block will show the operating condition and the second mnemonic will show the active regulator in control of the drive. This second mnemonic is important because it determines the type of data shown in the next two data columns on the data line for that drive.

First three character normal mode mnemonic can be:

STP - Drive is stopped (no power to the motor) and is NOT ready to RUN. (run permissives are not satisfied although the drive is not faulted)

RDY - Drive is stopped (no power to the motor) and is ready to run. If a run command was invoked to this drive, it would run.

JFD - Drive is jogging in the forward direction.

JRV - Drive is jogging in the reverse direction.

SLO - Drive is running at the internal SLOW reference. (This indication would be showing regardless of direction)

RUN - Drive is running in the forward direction at its RUN reference.

REV - Drive is running in the reverse direction at its RUN reference.

STU - Drive is running and has the slack take up reference applied. (additional internally generated forward reference)

STD - Drive is running and has the slack take down reference applied. (additional internally generated reverse reference)

Second three character normal mode mnemonic can be:

SPD - Drive is speed regulated with the draw ratio reference frozen at unity (1.0).

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DRW - Drive is speed regulated and will use the draw ratio reference to modify the base speed reference.

CUR - Drive is current regulated. This means that the drive can be a current follower or a current tension drive. A current follower will regulate a proportion of another drive's current and a current tension drive will regulate a proportion of a fixed commanded current reference.

XXX - Drive is process regulated. The process regulator is a multifunction regulator which can have its output sent to the speed regulator or the current regulator. An example of a process regulator is a tension regulator using an external load cell or strain gauge as its feedback. The actual mnemonic is configurable by the user and probably would be TEN for the above load cell regulator example.

BDP - The drive is running as a biased droop regulator. This is a special regulator that allows the drive to run as a speed regulator normally and apply a droop reference only when its current is above a configurable threshold. When this regulator is in control, a load adjust setpoint is used as the setpoint for this regulator.

ORW - The drive is running as a current regulator and the speed over ride has taken over the operation of the drive. If this is showing in the mode block, it means that the drive is in a CEMF or speed limit and the speed or CEMF regulator is actually in control regulating the limit setpoint.

Normal mode blocks would display a combination of the two mnemonics. For example, if a mode block was displaying RUN DRW on a green background, this means that the drive is coordinated, running forward, speed regulated, and with the draw ratio reference modifying the base speed reference. There are two modes where the color of the mode block will change to a brown background and yellow characters besides an annunciated fault. These modes are when ORW and/or STP is being shown at any time. The brown background color and yellow characters simply imply that the drive's operating mode is not normal, however; it is also not necessarily in a state which requires immediate attention. The STP says that the drive is not ready to run but the drive is not faulted. In most cases, this means that a system permissive, such as a limit switch, or thermal device is not satisfied to allow the drive to be put into a run mode. ORW simply means that the drive's speed regulator has taken over for the current regulator because it has sensed the speed or voltage of the motor has gone over the preset limits. The color of the mode block is used to alert the operator of normal and abnormal conditions at a glance of the Super Screen.

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3. ACTIVE REGULATOR SETPOINT COLUMN

The third column on the SDCS display screen will display the setpoint in engineering units of the active regulator. The active regulator is designated by the second mnemonic shown in the mode column.

ACTIVE REGULATOR -

SPD - The setpoint is the base speed setpoint for the drive. In a coordinated line of drives, only one drive will be a SPD regulator and will generally be the "PEG" section for the entire line. A "PEG" section means that the drive will run at the fixed speed specified by the setpoint in this column. All other drives will "follow" this speed and have the capability of a speed modifier. This modifier will allow the drives to follow at a proportion of the "PEG" speed reference.

DRW - This setpoint is the ratio modifier of the base speed specified at the "PEG" section. The number here is based on 10000 equaling a unity ratio. For example, if the "PEG" section has a reference of 900 feet per minute (FPM), and this drive has DRW as the active regulator with a setpoint of 10000, the drive will run at the same speed, 900 FPM, as the "PEG" section. If the setpoint was 11000, this drive would run at 990 FPM. If the setpoint was 9000, this drive would run at 810 FPM. In other words, the speed of a drive with its active regulator at DRW will run at the speed specified by:

$$\text{SPEED} = \frac{\text{"PEG" SPEED X DRW REFERENCE}}{10000}$$

CUR - This setpoint specifies a current reference for the drive. In actuality, the reference specified is a ratio of an external or internal fixed reference. In a current follower mode, the actual current reference is generated by another device, normally another drive. The CUR setpoint then specifies a ratio of this reference for the current regulator to use. If the lead drive sending the follower reference increases or decreases its current, the current follower drive will increase or decrease its current in the proportion specified by this ratio. A current tension drive will use a ratio of a fixed current reference, normally 100% of rated load. This allows the ratio to be a direct current reference for the drive, in percent of rated motor load current.

XXX - The process regulator setpoint depends on the configuration of the regulator itself. For SDCS, the setpoint will be

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displayed in a format of up to 5 significant digits with four digits to the left of the decimal point and one digit to the right of the decimal point. The largest number can be 99999 and the smallest non zero number can be 0.1.

BDP - The bias droop reference is the load adjust reference number when droop is active in the drive. The number has two meanings. When the droop is in effect, that is, when the load of the drive has exceeded the bias reference, the number specifies a percent of rated motor load that will be regulated by the drive. When droop is not in effect, the number, although still in percent load engineering units, will specify a lead speed for the drive above the commanded base speed (with the DRW reference modifier affecting or not affecting this speed reference) that the drive should run. The bias setpoint and droop setpoint are contained in the drive and are not modifiable from the SDCS screen display.

ORW - When the speed or CEMF override takes control of the regulation of the drive, the CUR setpoint will still be displayed in the setpoint column. This allows the operator to increase or decrease the CUR setpoint while the CUR regulator is overridden.

The setpoint column will only display the setpoint number for the active regulator. A three character mnemonic which describes the engineering units of that number (such as FPM and %) is displayed with the fourth column of data, the active regulator feedback. It is intended that the engineering units for the numbers in both the Setpoint and Feedback columns be the same, and therefore; when the regulator is functioning properly, the numbers in both columns should be the same or very close to being the same (within the precision or resolution limits of the active regulator).

On the Super Screen, the setpoint column is the only column which an operator may enter data. Although commands may be directed to a drive or a group of drives, these commands are done through function key presses rather than modifying anything on the SDCS data line. Setpoints are the only number on the data line that may be modified. When an operator enters a setpoint, the number entered is checked against an entry limit configured as part of the MMI2 system. If a number entered exceeds a high or low limit, a system error message is displayed over the function keys which will display the limits for that setpoint. When this happens, the number entered by the operator is discarded and the previous setpoint will still be in force.

4. ACTIVE REGULATOR FEEDBACK COLUMN

As described above, the number for the active regulator feedback column should match the setpoint when the active regulator is

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functioning properly. The active regulator will determine which setpoint and feedback will be displayed, and depending on the configuration of the drive system, the actual numbers may be a result from an internal algorithm rather than the direct feedback from a drive. The following description further details the methods used for showing feedbacks and setpoints.

SPD - A drive whose active regulator is SPD, will have the feedback for that regulator as simply the speed feedback from the drive. There are some considerations on the numbers shown for this regulator. The drive will utilize a setpoint in its own units and not any particular engineering unit. This unit of speed setpoint as well as most other numbers used in the drive are known as counts, where counts, as related to speed, may be related on a one to one basis with a number in engineering units. For example, 6000 counts at the drive may be equated to 2000 feet per minute (FPM). This implies that if an operator enters a number such as 1000 FPM at SDCS, the drive will regulate 3000 counts. Similarly, for feedback, 3000 counts will also be related to 1000 FPM. The conversion from engineering units from SDCS to the drive is accomplished by the data communication program which transfers the data between MMI2 and the master control device. SuP will actually read and write numbers already converted into drive "count" units. If the drive is utilizing a tachometer to sense speed, the drive will sample the data coming from the tachometer and apply scaling constants stored in the drive to convert the raw tachometer data to counts relating to the setpoint applied. For example, if the drive relates 6000 counts to 2000 FPM, the tachometer will return a number which corresponds to the rotational speed or RPM that relates to this linear speed. This relationship between RPM and FPM is kept in scaling constants in the drive so that the RPM can be converted to counts which relate to the desired FPM units. This resolved number in counts is sent via SuP to the master control and then to MMI2 through the communication program in MMI2. This program then converts the number in counts to a number in FPM and puts it in the MMI2 database. SDCS fetches this number and without further ado, displays it on the Super Screen. Therefore, regardless of the device which senses speed, the corrected value for the engineering units desired to be displayed will be shown in the feedback column.

DRW - If the active regulator is DRW, one could immediately dispute that this is a regulator at all, and would be correct. The draw ratio that is shown in the setpoint column for this active "regulator" is simply the ratio modifier to the base speed reference. All drives, regardless of active regulator, will be cognizant of what the base speed reference for the line is. This base speed reference or Master Machine Speed reference is the reference for the "peg" section. Every other drive coordinated with this "peg" section will have a copy of this number which

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should be the same in all drives. If the draw ratio was equal to 0000 in all other drives, the drives would run at the same speed s the "peg". The feedback column for a DRW regulator is not a egulator feedback but the actual value of the DRW reference eing used by the drive. In a steady state mode of operation, he setpoint and the feedback will always be the same. If the perator changes the setpoint, at the moment of the change, the etpoint and feedback will not match. A draw reference ramp will e employed to the setpoint and the results of that ramp will be isplayed in the feedback column. If the draw number should each a low or high limit in the drive, the feedback column will isplay the draw at that limit. If the setpoint and feedback umbers do not match, the most likely cause is that the setpoint as been changed and the actual draw number is ramping to the etpoint value. The second most likely cause is the draw number as hit a limit while ramping to the setpoint.

CUR - The system designer will have specified whether this regulator is configured as a current tension drive or a current follower drive. If the regulator was configured as a current follower, the lead drive sending this drive its current reference would also be specified. The number in the feedback column will e represented as the percentage of the follower drives current o that of the lead drive's current or :

$$\text{CUR Feedback} = \frac{\text{\% of follow drive's rated current}}{\text{\% of lead drive's rated current}} \times 100\ \%$$

his number should now directly relate to the setpoint which is he ratio of current reference sent by the lead drive. Here is n example :

RIVE A (lead drive) rated current = 500 A
RIVE B (follow drive) rated current = 200 A

unning conditions:
RIVE A load = 60% of rated (300A)
RIVE B CUR setpoint = 80%

he base current reference being sent by DRIVE A is 60% of full eference. DRIVE B employs a ratio to this reference of 80%. RIVE B now runs at 80% of 60% of its rated load or 48% rated oad (96A). The DRIVE B CUR regulator feedback is:

$$\text{CUR Feedback} = \frac{48\%}{60\%} \times 100\% = 80\%$$

his feedback now matches the setpoint and shows that DRIVE B is egulating current properly. The setpoint for a current follower

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drive simply asks the regulator to regulate a proportion of current dictated by the lead drive. For the above example, the setpoint is stating "Regulate current to 80% of whatever the lead drive's current reference is". The actual load of DRIVE B is 48% of rated current. This number will be displayed in the next column of the data line labeled LOAD. For current follower drives, CUR feedback is NOT the same as LOAD, and will be properly showing the actual feedback number it should be regulating in the active regulator feedback column.

For current tension drives, the setpoint is a proportion of rated current, and therefore; the CUR feedback number and the LOAD number will be the same. A setpoint of 30% is asking the drive to regulate 30% of rated load. The actual load for that motor will then be displayed in the CUR feedback column.

XXX - The process regulator is configurable for the regulation desired. The actual regulator is being implemented in the drive itself, so the feedback to the regulator must be connected to the drive. The feedback may be of several forms and from a multitude of sensors. The actual feedback signal will be converted to drive counts similar to the example given for a tachometer (analog or digital) by hardware means and then employing scale constants to relate the feedback counts to the setpoint counts. For example, the drive has its process regulator configured to regulate tension as sensed from a load cell. The load cell sends to the drive an analog signal of zero to ten volts DC to show a tension of zero to five pounds per linear inch (PLI). The setpoint is scaled such that 4096 counts equals 5 PLI. The setpoint on SDCS is shown as a number from 0.0 to 5.0 PLI. The communication program sending this number to the drive converts the SDCS number to drive counts by multiplying the SDCS setpoint by 819.2. So an SDCS setpoint of 2 PLI will be converted to 1638 counts in the drive (notice that the scaled number has been rounded to the nearest integer). Because of rounding error, the setpoint actually specifies 1.9995 PLI. The regulator will not be satisfied until the feedback reaches 1638 counts which directly cancels the setpoint. Since the internal drive scale factors will scale 5 PLI = 10 VDC = 4096 counts, 1638 counts of feedback requires that the load cell output 3.999 VDC. Although SDCS has specified 2.0 PLI as the setpoint, the drive will actually regulate 1.9995 PLI, but the feedback will be rounded to the nearest tenth of a PLI and SDCS will display a feedback of 2.0 PLI.

BDP - Currently, the feedback column only mimics the setpoint column. This is due to the multiple modes in which a drive in bias droop may be. The determination of a properly regulating bias droop drive cannot be determined simply by looking at the data on the SDCS data line. Operating conditions, load and speed feedbacks must all be utilized to make this determination.

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ORW - As described earlier, the feedback column will display the CUR regulator feedback even if the override takes over. This mode shows an abnormal condition which says that the current reference specified could not be regulated due to the speed of the motor (or CEMF) reaching a high or low limit. Although this is an abnormal condition, it may not be unexpected for the operating conditions existing for that drive. See the description for CUR regulator feedback for more information on the feedback displayed for this mode.

5. LOAD COLUMN

The fifth column on the SDCS data line for a drive is the percent of rated motor current for that drive. It is a direct feedback from the drive. The numbers represented in this column are normally configured with an alarm attribute which is a high or low limit alarm for this number. If this has been done, the column containing this number will turn red with the number in bright white characters if the load number satisfies the alarm attributes configured.

6. DRAW COLUMN

The draw column is specifically for coordinated drive systems and has a high amount of configurability. Draw is defined as the speed difference between two drives and may be displayed as a percentage difference (percent draw) or just the difference in speed in the speed engineering units (linear draw). The two drives are identified as the lead drive and the follow drive. The lead drive is the drive whose line of data that is currently being observed. The follow drive is normally the "Peg" section but can be configured to be any drive displayed on any SDCS display. The data in the draw column will change depending on whether the operator has "tabbed" to a drive on an SDCS display or not. If the data line has not been "tabbed", the draw column in that line will be showing the speed difference of that drive to the "peg" drive. If the data line is "tabbed", the draw column will show the speed difference of that drive to the configured follow drive. When a data line is "tabbed", a white arrow will be displayed to the right of the draw column. This arrow will be pointing up or down which indicates the screen position of the follow drive. The follow drive will have a yellow arrow to the right of its draw column. If no yellow arrow is found on the display with the white arrow, the follow drive is on another display. If the white arrow is pointing up, the follow drive is on a previous display. If the white arrow is pointing down, the follow drive is on a following display. The white and yellow arrows only indicate how the draw has been configured and do not have to be on the same display.

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Draw can be displayed three ways, linear draw, percent draw, or blanked.

$$\text{Linear Draw} = \text{Lead Drive Speed} - \text{Follow Drive Speed}$$

$$\text{Percent Draw} = \frac{\text{Lead Drive Speed} - \text{Follow Drive Speed}}{\text{Lead Drive Speed}} \times 100\ \%$$

There are a few reasons that a draw column display may be blanked. The drive may be the "peg" drive which has no draw. The drive may not be a part of the coordinated line (an auxiliary drive). The drive may not have a tachometer so its speed feedback is not accurate. The user may not desire to see draw data for a particular drive.

There is a function key menu level as part of SDCS that allows the configuration of the draw column. This function key level and the configuration procedures will be explained later in this document.

7. SPEED COLUMN

The last column of the SDCS display shows the actual speed of the drive. This number is a direct feedback from the drive and may be the results from any tachometer or encoder, or may be an approximation of speed based on the voltage or CEMF feedback from the motor. In general, the number in the speed column will be blanked if the drive is not sensing speed from a tachometer. Approximated speed, if displayed, may give inconsistent values depending on motor load and motor type. This may lead to conclusions of having problems when no problems exist. The engineering units of this column will be assumed for the draw column when numbers there are represented as linear draw. It is also assumed that all drives in a coordinated line have the same engineering units for speed for the purposes of calculating draw for the draw column.

D. SDCS FUNCTION KEYS

The function keys for SDCS displays are specialized for the SDCS system. These function keys contain the commands for all of the SDCS functions which include drive control as well as diagnostic program control. These function keys are distinguished from normal SC300 function keys by the fact that SDCS function keys have two rows of descriptor boxes and normal SC300 function keys have a single row of descriptor boxes. SDCS displays are normal SC300 displays so each SDCS display have a single row function key set. This normal function key set is called the END LEVEL of function keys. The first two row level of function keys is called the HOME LEVEL of function keys and this level allows the

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operator to access the other 16 levels available in the SDCS function key system. Some of these 16 levels are not yet defined and are reserved for future use. Each of the 16 levels of function keys have a specific set of tasks associated with them. Each level may have up to 16 function keys programmed to use with the SDCS system. When an SDCS screen is called by SC300, the END LEVEL of keys will be displayed. One of the keys on the END LEVEL will be labeled OPERATE KEYS. This function key will call the HOME level of keys, some times called the dispatch menu. The HOME LEVEL of keys now have labels to allow the operator to select new levels to be displayed on the SDCS screen. There are predefined keyboard keys that will be in effect only when an SDCS display is being shown on SC300 :

END key - Pressing this key will bring the END LEVEL of keys to the screen.

HOME key - Pressing this key will bring the HOME LEVEL of keys to the screen.

PLUS key - Pressing a key labeled "+" will switch the display to the next SDCS display if one exists. If the drive system being controlled has more than 18 drives, there will be more than one SDCS display. If the last SDCS display is displayed, pressing the + key will display the first SDCS display.

MINUS key - Pressing a key labeled "-" will switch the display to the previous SDCS display if one exists. If the first SDCS display is being displayed, the last SDCS display will be shown.

These special keys are in addition to the special keys already defined for SC300, but again, are only in effect when an SDCS display is being shown on SC300.

For Touch Screen users, the HOME, END, PLUS and MINUS keys will be duplicated on each SDCS display. The PLUS key will be labeled NEXT and the MINUS key will be labeled PREV on Touch Screen Displays. Touch Screen input from these screen keys will duplicate the actions described above.

Function keys in the SDCS system have been color coded to simplify and further describe the intended function of each key. Brown keys will select new menu levels or call new displays. Green keys will send a command or commands to the operating drive system and will have a direct effect on the process. Light gray keys will manipulate data but will not send data outside of the physical MMI2 system. Light blue keys are mode selection keys. These keys will change the meaning of other keys on the same level or may redefine the range of devices to which a command may be sent. Black keys are function keys that are currently not available for use. These keys may have permissives which when

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satisfied will allow the keys to return to a non Black color. Dark blue keys show that a function key has been pressed or the function described by the function key is active. Red keys are function keys that were black and pressed anyway, or functions that are active but illegal. This function key color coding will have been maintained throughout all SC300 function keys if the system has the SDCS system as part of it.

The following descriptions will describe the levels that have been defined to date for SDCS. These descriptions will be for a maximum case configuration. Most systems will have function levels that do not contain some of the keys that will be described here. If this is the case, those keys that were eliminated from a particular system was done for particular application or functional requirements. SDCS keys, like normal keys can be altered by the Function Key Editor tool included with MMI2 systems. Users are cautioned to take extreme care when doing any alteration to this set of function keys as the configuration requires good knowledge of MMI2 and the particular drive system it controls.

1. HOME LEVEL

This is the first level of the SDCS function keys. Because SDCS function keys have two rows, an alternate way of pressing function keys were developed. The top row of keys are accessed by pressing the associated function key number. The bottom row keys are accessed by pressing and holding the ALT key and then pressing the associated function key number. For Touch screen systems, touching the appropriate function key descriptor box is all that is necessary. The HOME LEVEL keys are used for choosing the SDCS function key level desired for control and maintenance of the drive system.

1OPERATE	2REFRNCE	3 DRIVE	4 SETPT	5CHG REG	6CONFIG	7 DRIVE	8EXTD
.LINE	LINE	OPERATE	ACTIVE	ACTIVE	RECIPES	HISTORY	FUNC
.ZONE	ZONE	REFRNCE	TEMP		SPD&DRW	PERMS	

The two row function keys are mainly described by the top line. In most cases the bottom key is connected to the top key in the description. For example, in the keys shown above, the top function key number 1 is described OPERATE LINE. The bottom key actual takes the first line description and replaces the second line with what is in the bottom descriptor box. The bottom function key number one should be described then as OPERATE ZONE. Using this convention, the bottom function key number six should be described as CONFIG SPD&DRW which means that the function level that is displayed after pressing ALT-F6, will present function keys that will configure the speed and draw columns on SDCS displays.

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The following instructions will describe the levels which can be called by the HOME LEVEL keys in detail. Again, the descriptions below is based on a maximum case description that may not entirely be pertinent to a specific application of SDCS. The menu levels are always customized by deleting extraneous functions for each MMI2 application.

2. OPERATE FUNCTION KEY LEVELS

These levels are for sending commands to the drive system primarily for movement control, although some referencing commands are included on these levels for convenience. Movement commands are defined to be RUN, SLOW, JOG, and STOP type of commands. There are three levels of function keys devoted to OPERATE COMMANDS. Most keys are duplicated on each level, however; the level of keys used determines the range of drives to which the commands may be sent.

a. Line Operate Level

1 LINE	2REFRNCE	3 RUN	4 JOG	5 SLOW	6TENSION	7 SPEED	8CSTP
DN-STRM	LINE	FWD	FWD	FWD	ON	INC	
UP-STRM	ZONE	REV	REV	REV	OFF	DEC	STOP

The function key level depicted above shows the maximum case OPERATE LINE level. Function key one are light blue keys, function key two are brown keys and function keys three to eight are green keys. The OPERATE LINE level of keys will allow the operator to send the commands as described by the green keys on this menu to, at the minimum, one zone of drives. If the LINE DN-STREAM key is selected (dark blue), the command will be sent from the drive that is "tabbed" to the last drive in the line, which is the last drive on the last SDCS display. If the LINE UP-STREAM key is selected, the command will be sent from the "tabbed" drive to the beginning of the line, which is the first drive listed on the first SDCS display. If neither the LINE UP-STREAM or LINE DN-STREAM keys are selected, the command will be sent to the "tabbed" drive and any other drive defined to be in the same zone. (Remember, drives in the same mechanical zone will have their names highlighted on the SDCS display with bright white characters.) The beginning of the line and the end of the line does not necessarily have to be the first and last drives listed on the SDCS displays. A special register that defines the beginning and the last drive can be configured, and those drives as defined by this register will be used by this level of keys.

NOTE: The from/to register is only read during power up initialization of MMI2 and therefore, the beginning and ending drive configuration is always the same while MMI2 is running. In

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other words, the from/to register cannot be changed to allow different beginning and ending drive configurations for this level while MMI2 is running.

The second function key allows the user to select the LINE REFERENCING or the ZONE REFERENCING LEVELS without the need to return to the HOME level to activate them. If the operator requires any of the other SDCS levels, he must return to the HOME level by pressing the HOME key on the keyboard (or the HOME screen key for Touch Screen systems) and select those levels from there.

The next three keys are movement commands where the top keys describe which forward movement and the bottom three keys describe which reverse movement will be commanded. In all three cases, only the command as described by the function key will be sent and therefore; the drives will execute the command with the active regulator that is currently in force.

The next function key, TENSION ON and TENSION OFF, is a special command that requires that the drives be specially configured to understand and execute these commands. The DC-300 drive may be configured to have a TENSION mode available. If the TENSION mode is available, the drive must be configured such that when a TENSION ON command is received, the drive will switch to the CURRENT regulator for current tension or to the PROCESS regulator for a process tension mode that for example uses load cell feedback. Only those drives configured with TENSION available will respond to this command. TENSION OFF will turn the active tension regulator, either CURRENT or PROCESS, off and will make the speed regulator active.

The next function key is actually a referencing key and is configured on this level for convenience. This key will tell all the drives as part of the entire coordinated line to increase or decrease their master machine speed setpoint. This is the same as telling the drives to increase or decrease master speed like systems with a master speed motorized potentiometer. All drives must receive identical master speed setpoints to remain coordinated.

The last function key is the STOP function. CSTP will cause the drives commanded to COAST STOP, that is regardless of regenerative capabilities, the drives will simply remove power from their motors and the motors will coast to a stop. CSTP may be replaced by QSTP which can be configured as a coast stop for non regenerative drives and current limit regenerative stops for those drives with regeneration capability. STOP will cause the drives to execute a normal stop which will insert a timed deceleration rate to the drives. The drives will follow this rate to a stop condition.

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This level, when it is used to send commands to multiple drives, requires that all drives expected to execute these commands must be in the coordinated mode (mode columns must all be displaying green background colors). Drives that have their mode column background color as light gray will ignore LINE OPERATE COMMANDS.

WARNING

The OPERATE levels will allow the operator to start and stop multiple drives, up to and including all the drives in a coordinated line. In many cases, the physical machine that will begin to operate will be out of visual contact by the operator initiating these commands on MMI2. **Extreme caution** should be exercised when starting drives in sections that cannot be seen to prevent personnel injury.

b. Zone Operate Level

1 ZONE	2REFRNCE	3 RUN	4 JOG	5 SLOW	6TENSION	7 SPEED	8CSTP
DN-STRM	ZONE	FWD	FWD	FWD	ON	INC	
UP-STRM	LINE	REV	REV	REV	OFF	DEC	STOP

The zone operate level has identical keys as the LINE OPERATE LEVEL keys. This level of keys can only control drives within a single zone of drives. For example, the ZONE DN-STRM and the ZONE UP-STRM keys are not selected. The user has "tabbed" to the third drive of a five drive zone, and pushed the RUN FWD function key. The command to RUN FWD would be sent to all five drives in the zone. If the ZONE DN-STRM key was selected, the third through fifth drive as listed on the SDCS display in the zone would be sent the command to RUN FWD. Similarly, if the ZONE UP-STRM key was selected, the first through third drives as listed on the SDCS display in the zone would be sent the RUN FWD command. The ZONE OPERATE level cannot cross zone boundaries.

The single exception to the rule is the SPEED INC and SPEED DEC function keys. These keys will still broadcast the command to increase or decrease speed to all the drives in the line. Again, this referencing command was configured on this level for convenience. All other commands are limited to a single zone of drives as selected by the "tabbed" drive. The function of the keys shown on the above level are the same as those described for the LINE OPERATE LEVEL.

c. Drive Operate Level

The drive operate level only allows function key commands on this level to be sent to a single drive as selected by the "tab". This level has several different commands available because of the constraint that commands on this level will be sent only to a single drive.

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1 NEW SETPT	2INDEPEN	3 RUN FWD	4 JOG FWD	5 SLOW FWD	6 DRIVE REFRNCE	7 SETPT INC	8CSTP
ACCEPT	COORDIN	REV	REV	REV	RESET	DEC	STOP

This level of function keys also contains referencing commands besides the normal movement commands, again for the convenience of the user. The referencing command keys will only effect the active regulator setpoint for the drive the operator has selected. Function key 1 allows the direct entry of the active setpoint. The entry sequence would be :

1. The operator uses the "+" or "-" key to select the SDCS display on which is the desired drive.

2. The operator then uses the TAB or Shift TAB key to select the drive.

3. The operator then presses the F1 function key (NEW SETPT).

4. The data entry area will show the current active setpoint value followed by a "?"

5. The operator enters the new setpoint but does not press the ENTER key, instead, he presses and holds the ALT key and then presses the F1 key (ACCEPT).

6. The new setpoint is first checked against the configured entry limits. If the new setpoint is over either the high or low limits, a system error message is written over the function key area which tells the operator that the limits were exceeded and will also show the operator what the limits are. The new setpoint will not be sent to the data base or to the selected drive, therefore; the old setpoint will still be in force. If the new setpoint is within the entry limits, the setpoint is sent to the drive and the setpoint column will be updated with the new setpoint value.

In the above example, the operator could have used the ENTER key after typing the new setpoint desired. The data entry area will show the entered setpoint in a different color, and the data entry limits will be checked at this time. If the new number is within limits, the data entry area will now ask the operator to confirm the new number by showing a "OK ?". To allow the number to be sent to the drive, the operator must type a "Y" (without the ENTER key). If an "N" is entered, the new setpoint is discarded and the original setpoint is restored. The ACCEPT key (ALT-F1) is a shortcut to the confirmation process in data entry.

The second function key allows the operator to change the mode of the selected drive from COORDINATED to INDEPENDENT mode or the

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other way. The definition of COORDINATED and INDEPENDENT modes was detailed when the mode column of SDCS was described earlier.

The next three function keys will allow a single drive to RUN, SLOW or JOG in both the forward and reverse directions. The eighth function key will send either a coast stop or normal stop. The coast stop command may be replaced by the quick stop command if desired. Remember that these commands will only be sent to the single drive that the operator has selected.

The sixth function key will allow the selection of the DRIVE REFERENCING LEVEL. This level will be discussed in the next section. The bottom key of the sixth function key (ALT-F6) will send a SOFT RESET to the selected drive. In most cases, this command will reset minor faults. Severe drive faults will require the attention of maintenance personnel and will require a HARD RESET which may only be accomplished at the drive panel.

The seventh key is a very special referencing function key. Because only one drive will be affected by this command, the active regulator is first determined. If the regulator is SPD and the drive is the "PEG" drive, the SETPT INC or DEC command will be broadcasted to all drives so that all master machine speed numbers may be updated. If any of the other four regulators are active, (DRW, CUR, XXX or BDP) only the setpoint for the selected drive will be increased or decreased. For these regulator setpoints, the drive itself will increase or decrease the setpoint at the INC/DEC slew rate configured in the drive.

d. Change Regulator Active Level

1TRNSFER	2 DRIVE	3CURRENT	4PROCESS	5	6BIAS DP	7MODIFY	8DRAW
SMOOTH	ON	ON	ON		ON	SETPTS	ON
BUMPLES	OFF	OFF	OFF		OFF	DRV REF	OFF

This is a special OPERATE LEVEL of function keys available for SDCS. This level will allow the operator to change the active regulator of a single drive. In addition, the operator may setup how the setpoint for the new regulator will be applied at the time the new regulator is made active.

Function key 1 allows the operator to choose a bumpless or smooth transfer of setpoints upon a regulator change. In actuality, this key will only be effective if the drive has been configured to obey these commands. The drive may be configured to always transfer regulator setpoints either bumpless or smooth depending on which regulator is being made active. If this is the case, the function key 1 commands cannot override the drive configuration. The definition of bumpless or smooth transfer can be best illustrated by an example.

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DRIVE X is running speed regulated with the DRW reference as the active running mode. DRIVE X has the capability of running as a current tension regulator. The current setpoint, although not active, is 30%. The active DRW setpoint is 10345 and the load of the drive is 25% of rated load. The operator presses ALT-F1 to make the next regulator transfer bumpless and then presses F3 to transfer the active regulator from DRW to CUR. At the time DRIVE X receives the transfer command, it reads the load and presets the IRATIO setpoint such that the new current setpoint will be exactly 25% upon the completion of the transfer. The drive is now running current regulated at the same load as it was running in DRW. This illustrates the BUMPLESS regulator transfer where the regulator setpoints are preset to match the current running conditions. If the operator then pushes F1, the next regulator transfer will be made under the SMOOTH condition. The SMOOTH regulator transfer incorporates features of the bumpless transfer by first preconditioning the new regulator to make the transfer match the existing running conditions. DRIVE X is now running as a current tension drive with its load at 25%. The speed of the drive has increased to allow this load to be regulated. The operator presses ALT-F3 to turn the CUR regulator off which then allows the DRW regulator to take control. The operator could have pressed F8 (DRW ON) for the same results. DRIVE X reads the current speed it is running and calculates a new DRW setpoint based on the difference of the master machine speed setpoint and the actual speed it is currently running when the transfer command is received. DRIVE X transfers regulators from CUR to DRW and makes this calculated DRW reference active. Immediately, the drive will read the DRW setpoint at which the drive had last been using, 10345 in this example, and applies this reference to the DRAW reference ramp. The speed of DRIVE X will then change from the setpoint calculated and applied when the transfer was first made to the stored setpoint on the DRAW reference ramp.

The system designer configures the drives with the regulators under which the drives could be run. Only to those regulators will those regulators be available for the operator to transfer. The regulators that are unavailable for transfer will have their ON descriptor boxes shown in black. Only those transfer descriptors shown in green will be allowed to be in effect.

A special regulator transfer is actually a mode change for the drive. The function key 2 allows an operator to turn a drive off from listening to any further commands from the LAN except the one command to turn it back on. This is a special mode where the drive appears to other devices on the network as being available, however; any command except for DRIVE ON will be ignored. This mode is for those times the operator requires that the line be reconfigured so the drive cannot be activated into any running state through any type of command.

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WARNING: This command, although it will prevent the drive from acting on any movement type of commands, is not meant to replace the safety "TAG OUT" procedures to allow maintenance personnel to work inside or around the mechanical equipment being driven by the drive.

The commands under function key 7 allow the operator to select the SETPOINT ACTIVE LEVEL and DRIVE REFERENCING LEVEL without the need to return to the HOME LEVEL first. These levels are put here with the anticipation of the operator requiring to do some setpoint work prior to or just after changing regulators and is put here to facilitate those functions.

3. REFERENCING FUNCTION KEY LEVELS

The referencing levels of function keys are primarily for altering the setpoints for the different regulators in one or more drives. The three referencing levels are much like the OPERATE levels, where the commands on each of the three levels are similar and the difference between the levels are the range of drives that the commands can reach.

a. Line Referencing Level

1PROGRES	2OPERATE	3 ZONE	4 SLACK	5 SPEED	6PROCESS	7IRATIO	8DRW
.	LINE	OPERATE	TAKE-UP	INC	INC	INC	INC
REGRESS	DRIVE	REFRNCE	TAKE-DN	DEC	DEC	DEC	DEC

Like the LINE OPERATE LEVEL the LINE REFERENCING LEVEL commands are sent to a minimum of one entire zone. The commands may also be sent to as many as all the drives in the coordinated line as well. If the key labeled PROGRES is selected, the command will be sent from the "tabbed" drive to the last drive in the coordinated line. If the key labeled REGRESS is selected, the command will be sent from the "tabbed" drive to the beginning of the line. Again, for the progressive and regressive modes of this level, the from/to register will define the beginning and ending drives in the coordinated line. Normally, this will be the first drive listed on the first SDCS display for the beginning drive and the last drive listed on the last SDCS display for the ending drive of the line. If neither of the progressive nor regressive keys are selected, the zone which is currently selected will receive the commands from this level.

The next two function keys will allow the user to select the OPERATE LINE, OPERATE DRIVE, ZONE OPERATE, or ZONE REFERENCING function key levels without first having to select the HOME LEVEL first.

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The fourth key will allow the commands SLACK TAKE UP or SLACK TAKE DOWN to be sent to the selected drives. SLACK TAKE UP enables an additional speed reference to be temporarily added to the current speed reference. The reference will be applied as long as the key is held down by the operator. This additional reference may or may not be applied through a timed ramp and the amount of reference is programmed at each drive. Sending this command will simply activate whatever reference that has been programmed at each drive by each drive that receives this command. SLACK TAKE DOWN is a similar function, except the reference will be in the opposite direction.

The next four keys allow references for different regulators to be increased or decreased. For the SPEED INC/DEC keys, the command will be broadcasted to all drives in the coordinated line. The INC and DEC keys for the PROCESS, CURRENT, and DRAW setpoints will send these commands to all drives selected. If those regulators are not active, the setpoints will still increase and decrease when the command is received. For example, the operator has selected the PROGRESSIVE key and has tabbed to DRIVE X. There are only two more drives until the end of the line, DRIVE Y and DRIVE Z. Any command sent from this level will go to all three of these drives. DRIVE X is DRAW, DRIVE Y is CURRENT and DRIVE Z is PROCESS regulated. If the operator sends the DRAW INC command, all three drives will receive the DRAW INC COMMAND and the DRAW setpoints will increase in all three drives even though DRIVE Y and DRIVE Z are not currently utilizing these setpoints actively. If the operator then sent a command to DRIVE Y to switch its regulation from CURRENT to DRAW, DRIVE Y will utilize the new DRAW reference.

There are different key types utilized on SDCS function keys. This level has all three different types of keys: LATCHED, JOG, and MOMENTARY. The PROGRESSIVE and REGRESSIVE keys are latched type keys. If the F1 key is pressed, the PROGRESS descriptor box will change from light blue to dark blue and then stay dark blue to indicate that the progressive mode is in effect. If F1 is pressed again, the progressive mode will be turned off. If ALT-F1 is pressed, the progressive mode will be turned off and the regressive mode would be in effect. Progressive and regressive modes are mutually exclusive. The level selection keys are momentary keys. The command to select the new level is invoked when the function key is pressed. This key only needs to be depressed momentarily to execute the command. The INC and DEC, JOG, and SLACK TAKEUP/TAKE DOWN keys are all examples of JOG type keys. JOG type keys will execute a command when the key is pressed and will stop the execution when the key is lifted. For example, when the DRAW INC key is pressed the DRAW references in the selected drives will begin to increase and will keep increasing until the DRAW INC key is released. The action of lifting the DRAW INC key actually sends a STOP DRAW INC command.

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When pressing keys described by the bottom descriptor boxes, the operator must depress the ALT key as well as the function key desired. For JOG type keys, it is important that both the ALT key and the function key be released to properly stop the jog type function being performed. The function will stop if only the function key is released, however; if the ALT key is not released and another function key is pressed, the function sent will be the command in the top descriptor box of that function key. It will appear as if the ALT key was not pressed even though the operator is still holding it down. The ALT key should be released after each command is sent to prevent erroneous operation by SDCS.

b. Zone Referencing Level

1PROGRES	2OPERATE	3 LINE	4 SLACK	5 SPEED	6PROCESS	7IRATIO	8DRW
.	ZONE	OPERATE	TAKE-UP	INC	INC	INC	INC
REGRESS	DRIVE	REFRNCE	TAKE-DN	DEC	DEC	DEC	DEC

The zone referencing level is, again, very similar to the LINE REFERENCING LEVEL. The ZONE REFERENCING LEVEL may only send commands, at the most, to all drives within a single zone. The same range of drives may be addressed as the ZONE OPERATE LEVEL. Again, all function key commands on this level have the same meaning as the keys on the LINE REFERENCING MENU. The Speed INC/DEC keys will increase and decrease the Master Machine Speed Reference for the entire coordinated line. Like the LINE and ZONE OPERATE LEVELS, the drives must be in the COORDINATED mode to be able to respond to the commands on this level as well as the commands on the LINE REFERENCING LEVEL.

c. Drive Referencing Level

1 DRIVE	2 SLACK	3TENSION	4TENSION	5TENSION	6PROCESS	7IRATIO	8DRW
OPERATE	TAKE-UP	ON	AVAIL	CUR	INC	INC	INC
REG CHG	TAKE-DN	OFF	NONE	PRO	DEC	DEC	DEC

The DRIVE REFERENCING LEVEL includes the normal referencing command keys and includes the TENSION mode setup and command keys as well. The TENSION mode of a drive is a special mode that requires special configuration of the drive. If a command was broadcasted to all drives to turn TENSION on, only those drives configured with TENSION AVAILABLE will respond to this command. This level allows the operator to make that configuration with key number 4. Key number 5 allows the operator to configure which regulator will control tension in the tension mode. These two keys are special in that they will show what has been selected by their color. If the top descriptor box of function

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key 4 is dark blue, the drive on which the operator is currently "tabbed" has tension available. If the bottom key is dark blue, the drive does not have tension mode available. Similarly, if the top descriptor box of function key 5 is dark blue, the regulator which will control tension for that drive is current. If the bottom descriptor box on function key 5 is dark blue, the process regulator will control tension in tension mode for that drive.

Function key 1 on this level will select the DRIVE OPERATE LEVEL and the CHANGE REGULATOR ACTIVE LEVEL. Function key 3 will send the command to turn TENSION mode on in the drive selected by the operator. Again, the drive will only respond to the TENSION ON command if the tension mode has been made available to that drive by function key 4. Function key number 5 chooses which regulator will control tension if tension mode is on. If tension is not available, the selection of key number 5 is irrelevant.

Function keys 2, 6, 7 and 8 are the normal referencing keys found on the previously discussed REFERENCING LEVELS. The commands from these keys on this level will only go to a single drive as selected by the operator with the "tab". Again, sending a regulator setpoint INC or DEC to a drive will INC or DEC the setpoint commanded regardless if that regulator setpoint is the active regulator or not.

d. Change Setpoint Active Level

1 ENTER	2 SPEED	3 DRAW	4CURRENT	5PROCESS	6	7BIAS DP	8SETP
. NEW	SETPT	SETPT	SETPT	SETPT		SETPT	INC
ACCEPT							DEC

This function key level is considered a REFERENCING LEVEL, however; it is a special level in that the function keys themselves will display data from the drive that has been selected by the operator. Like the previously discussed level, this level will only send data to a single drive, and that drive is the currently "tabbed" drive.

As the operator "tabs" from drive to drive, all current values of all available setpoints for each drive will be displayed in the bottom descriptor boxes under the appropriate regulator heading. As shown on this function key level, a drive may have up to five different setpoints, any one of which could be made the active setpoint. The system designer will configure each drive to allow the regulators that a drive may have. This level will show only those regulator setpoints with which a drive was configured. Those regulators not configured will have the top descriptor box for that regulator setpoint shown in black. In addition, although the SPEED setpoint will be configured for all drives in

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the coordinated line, only the SPEED SETPT descriptor box for the line "PEG" drive will be the normal color. All other SPEED SETPT descriptor boxes for all other drives in the coordinated line will be shown with a light blue background. If the operator "tabs" to the "peg" drive, this level will show the master machine speed setpoint in the bottom function key box under the SPEED SETPT descriptor box. If the operator selects this function key box by pressing the F2 key, the operator will be allowed to enter, increment, or decrement a new master speed setpoint. Any changes made to this number will be broadcasted to all drives in the coordinated line (again, only those drives that are in the COORDINATED MODE will receive this new master machine setpoint). Changes to the master speed setpoint can only be done at the "peg" drive.

If the operator has selected any drive other than the "peg" drive and that drive has a draw setpoint, the speed setpoint function key box may be utilized to modify the draw setpoint by entering a new speed setpoint. The number shown in the speed setpoint descriptor box will show a number corresponding to the master machine setpoint modified by the current draw setpoint. The equation is:

$$\text{SPEED SETPOINT} = \frac{\text{DRAW SETPOINT}}{10000} \times \text{MASTER MACHINE SPEED SETPOINT}$$

If the drive is running speed regulated with the draw modifier active (DRW in the mode column), the drive should be running at or very close to this speed setpoint unless the speed is further modified by another parameter (such as DROOP).

When the operator enters a new speed into the speed setpoint function key box (or increments or decrements the existing speed) a new draw setpoint is calculated from the new speed number and that new calculated draw setpoint will be sent to that drive. The equation used is:

$$\text{NEW DRAW SETPOINT} = \frac{\text{NEW SPEED SETPOINT}}{\text{MASTER MACHINE SPEED SETPOINT}} \times 10000$$

If the operator utilizes the SPEED SETPOINT descriptor box to modify draw, entry limit checking will be done on the resulting draw calculation and not on the speed entered. In other words, if the operator enters a speed setpoint that results in a draw which exceeds the draw setpoint entry limits, the error message showing the high and low limits will show the limits in terms of the draw setpoint value and not the speed value. Changes to the speed setpoint if the drive is not the "peg", actually only modifies that drive's draw setpoint.

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All other setpoint changes, including directly changing the draw setpoint, will be done directly to the value in the descriptor boxes without any interposing calculations. Changing the draw setpoint will update the value in the speed setpoint descriptor box to show the actual speed at which the drive will run. The setpoint values for the other regulators will be handled directly. For example, if the operator "tabs" to DRIVE X and that drive has been configured as a SPEED regulated drive with DRAW and has the capability of a CURRENT TENSION mode, the function key level would appear as follows (with example reference values):

1 ENTER NEW	2 SPEED SETPT	3 DRAW SETPT	4CURRENT SETPT	5PROCESS SETPT	6	7BIAS SETPT	8SETP INC
ACCEPT	2069.0	10345	45.0				DEC

Because DRIVE X is not the "peg" drive, the SPEED SETPT keys will be shown in light blue, however; the current calculated speed setpoint based on the master machine setpoint (in this example is 2000) and the current draw setpoint, will be shown in the bottom descriptor box. The current DRAW setpoint is shown and can be changed as well as the CURRENT setpoint. The function keys 5, 6, and 7 will be shown in black with no values in the bottom boxes because these regulators have not been configured as part of this drive. The active regulator setpoint will be automatically selected by the active regulator setpoint descriptor box shown in dark blue. If DRIVE X's active regulator was DRW, the top function key 3 descriptor box would be shown in dark blue. The operator may only alter the setpoint under the dark blue regulator setpoint descriptor box, in this case, the DRAW SETPT. The operator may also change the CURRENT setpoint by pressing the F4 function key which will switch the selection from the DRAW setpoint to the CURRENT setpoint. This switch in selection does not change the active regulator; it only changes which setpoint will be adjusted. The draw setpoint may also be altered if the SPEED SETPT is selected. The selected setpoint may be increased or decreased by the function key 8 or may be directly changed with a new value entered directly with the function key 1. The entry sequence is identical to the previous examples of new setpoint entry.

This function key level, then, is to adjust the setpoint of any regulator configured with a drive regardless if that regulator is active or not. It will be mostly used for setpoint setups after a shutdown or prior to making a new product without utilizing the recipe system. The CHANGE SETPOINT TEMP LEVEL allows the operator to change the setpoints stored in the TEMP recipe area and will be discussed in detail later in the next section.

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4. RECIPE FUNCTION KEY LEVELS

The RECIPE LEVEL for SDCS allows the operator to access all the recipe functions of SC300. All normal recipe functions are supported by this level of function keys and is configured here as part of the SDCS function key levels for the convenience of the operator. There is one difference in operation from the standard SC300 recipe system to that of SDCS related recipes and that is that all SDCS recipes transferred to the TEMP memory area which contain drive setpoints as ingredients, should only be edited from the SETPOINT TEMP LEVEL of function keys of SDCS rather than the COMPARE SCREEN as earlier described. Recipe editing drive setpoints is specially handled at this function key level and allows master machine setpoint changes in the TEMP recipe to be coordinated with all drive ingredients.

a. Configure Recipe Level

1OPERATE	2REFRNCE	3REPORT	4RECIPE	5RECIPE	6RECIPE	7COMPARE	8ABRT
.LINE	LINE	ACTIVE	DSK>ACT	ACT>TMP	DSK>TMP	ACT/TMP	RCP
.ZONE	ZONE	DISK	ACT>DSK	TMP>ACT	TMP>DSK	DSK/TMP	

The first two function keys allow the operator to select the OPERATE LINE, OPERATE ZONE, LINE REFERENCING and ZONE REFERENCING levels without first selecting the HOME LEVEL. Again these keys are configured here only for convenience to the operator.

The next four keys, from function key 3 to function key 6, allow the operator to invoke the recipe transfer action codes as described in the Recipe System Section of this document. The commands under function key 7 allow the operator to go to the COMPARE SCREEN as was again described in the Recipe System Section of this document.

The color coding of these function keys follow the SDCS convention of active changes to the control system in green keys, data manipulation by light gray keys and level or display changes by brown keys. The two green keys will be any recipe transfer to ACTIVE. All other recipe transfer keys will be light gray as the recipe data is simply moved to other memory locations within the system and not to the active process. The COMPARE SCREEN commands and the SDCS function key level commands will all be brown keys to denote that the display or function key level will be changed by these commands. The last key, (F8), will be a green key and is a special recipe command to the drives. This command to ABORT RECIPE is broadcasted to all the drives so that while the drives are actively changing setpoints, they will hold where they are.

The ABORT RECIPE command will actually do some coordination of

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setpoints within the drives. If the drives were changing speeds on an S-CURVE type ramp, the "S" part of the ramp would be impressed on the speed change when the ABORT RECIPE command was issued. In other words, the drives would not stop ramping in speed immediately, but round into a higher or lower speed (depending on the direction of the original recipe change in speed) and then round back to the speed where the ABORT RECIPE COMMAND was issued. All setpoints at the time the ABORT RECIPE command was issued will capture their current values and then transition to those values over a period of time dependent on the various ramp rates applied to those setpoints. To allow continuation of the recipe to the final values, the operator must re-transfer the original recipe, that was aborted, by the normal means. The ABORT RECIPE command only affects transfers of recipe to the ACTIVE recipe.

ABORT RECIPE can also be used on a manual change in setpoint. If the operator broadcasted a new master machine setpoint, and decided to hold the speed before the drives reached this new setpoint, ABORT RECIPE can be used for that purpose even though there was not an actual invocation of a recipe to the system. The ABORT RECIPE command only acts upon drive setpoints and not system recipe values, as the ABORT RECIPE command is only executed within the drives receiving the command. The command cannot be directed to any particular drive or zone as this command is broadcasted to all the drives in the coordinated line. Like all other commands and reference changes made as broadcasted or zone commands, only those drives in the COORDINATED mode will act upon an ABORT RECIPE.

b. Change Setpoint TEMP Level

1 ENTER NEW	2 SPEED SETPT	3 DRAW SETPT	4CURRENT SETPT	5PROCESS SETPT	6	7BIAS SETPT	DP 8SETP INC
ACCEPT							DEC

This level is similar to the CHANGE SETPOINT ACTIVE LEVEL of function keys and data entry procedure is identical. The operator must first load a recipe into the TEMP area from DISK or ACTIVE before invoking this level. If TEMP is empty, a system error message will be written over the function keys alerting the operator that he must first load TEMP. When this is done, the values in the bottom descriptor boxes under the appropriate regulator setpoint header in the top descriptor boxes will show the values in the temp recipe that was loaded as the operator "tabs" from drive to drive. Only the values for the drive that is "tabbed" will be shown. The regulator which was active when

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the recipe was first captured will be highlighted on the top descriptor boxes. The operator must press the function key corresponding to the setpoint desired to change prior to entering a new setpoint with F1 or increasing or decreasing the setpoint with the F8 keys. The altering of the draw setpoint may, like the CHANGE SETPOINT ACTIVE LEVEL, be accomplished by selecting the SPEED SETPT. Changing the speed setpoint for the "peg" drive will change the master machine speed setpoint for all the drives in the TEMP recipe. Changing the speed setpoint for any other drive will only change the draw setpoint for that drive.

Again, it is very important that all recipe editing be done with this level of function keys for drive setpoint ingredients . If other ingredients besides setpoints exist in the recipe class definition, those ingredients can be edited using the normal COMPARE SCREEN editing procedures already outlined.

Once the operator has completed making the necessary changes to the recipe, he may then use the CONFIGURE RECIPE LEVEL to transfer this edited TEMP recipe to DISK or ACTIVE. The TEMP recipe remains in memory unless the computer is restarted through a power up initialization or bootstrapped from a reset, or the current TEMP recipe is overwritten by another transfer. The operator should always move the TEMP recipe to DISK if he requires that the newly edited values be saved. This transfer can create a new DISK recipe or over write a previously created recipe as described in the Recipe System Section of this document.

5. CONFIGURE SPEED AND DRAW LEVEL

This function key level allows the operator to configure the SPEED and DRAW columns of the SDCS display. The SPEED column of the SDCS display shows the actual speed of a drive if the feedback is available. If the feedback is not available or is approximated from voltage feedback, the speed will be blanked from this column. This level of keys may be used to temporarily show the approximated speed for a drive, or the display of this speed may be made a permanent addition to the SDCS display. The configuration of the lead and follow drives for the DRAW column display may be changed by using this level as well as the display of draw itself from linear draw to percent draw or blanked.

1OPERATE	2SELECT	3 BLANK	4LINEAR	5 MAKE	6 SHOW	7OPERATE	8
.KEYS	LEAD	DRAW	DRAW	PERM	SPEED	LINE	
ACCEPT	FOLLOW	SPEED	PERCENT			DRIVE	

The SPEED and DRAW column displays were detailed previously in this document. The configuration procedure for these two columns are accomplished utilizing this level of SDCS function keys.

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As mentioned, the speed of a drive may be shown or blanked. To blank the speed display of a drive, the operator must "tab" to the desired drive on the SDCS display. Then from this level, he must press ALT-F3 which will immediately blank the speed display for that drive. To show the speed display he must press F6 which will immediately show the value of speed for that drive that SDCS has in its database.

The draw column configuration is slightly more complicated because of the many options available. When the operator "tabs" to a drive on the SDCS display, the draw column will show the draw between the two drives as configured by this level if the draw for that drive is not blanked. To change the draw display, the operator has three display options. The operator may blank the draw or show draw as a percent speed difference or a straight speed difference in the speed column engineering units. To blank the draw display for the "tabbed" drive, the operator simply presses the F3 function key. To show draw as a percent speed difference, the operator presses ALT-F4. To show draw as a linear speed difference the operator presses F4.

To change the follow drive configuration the operator must "tab" to the lead drive he wishes to configure and then press F2. This alerts SDCS that the operator wants to change the lead/follow drive configuration and selects the "tabbed" drive as the lead drive. When the operator presses F2, a message is written over the function keys to ask the operator to confirm that the lead drive is what was selected. The operator presses the "Y" keyboard key or the ALT-F1 key combination to accept the lead drive selection. The operator presses the "N" key if it is not what he wants. The operator must then "tab" to the follow drive. The follow drive can be on any SDCS display, so it is legal for the operator to call a different SDCS display and then tab to the follow drive. Once the name of the desired drive is highlighted with the TAB colors, he presses ALT-F2. Again, the operator must confirm his selection with the "Y" key or the ALT-F1 keys. If the operator hits the "N" key, he must start from the beginning to reconfigure the lead and follow drives.

Once the follow drive has been confirmed, the TAB marker will return to the drive name configured as the lead (the display will change to the SDCS display containing the lead drive if on another SDCS display) and the draw column will now show the speed difference between the new follow drive and the "tabbed" drive.

The new configuration is held temporarily and will return to the original configuration upon the next reset of MMI2. To retain the new configuration, the operator must press F5 (MAKE PERM). Once this is done the new configuration will be saved and will be used on subsequent resets of MMI2.

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6. DRIVE HISTORY LEVEL

This level of function keys controls the DRIVE HISTORY subsystem of SDCS. These function keys interface to the background program which retrieves locked drive history data buffers from the drives. The Drive History feature is described in the next section along with this function key level.

7. END LEVEL

As mentioned previously, this is the name of the function keys associated with the SUPER SCREEN display in the normal SC300 system. The END LEVEL is a single row of function keys and has none of the SDCS attributes previously described. This level of keys and its associated MORE level contains keys to allow the operator to call other displays of SC300 in addition to the HOME level of keys of SDCS. This level of keys is very dependent on how the particular MMI2 system and SC300 is configured. The END level shown below is a typical example of an END level of function keys.

```
1OPERATE2WET END3DRY END4RECORDR5 ALARM 6ANLYSIS7 MAINT 8MORE|
|.KEYS   | PERMS | PERMS |CONFIG |CONSOLE|CONSOLE|CONSOLE|     |
```

The MORE level of function keys is the level brought to the SUPER SCREEN display if the operator presses the MORE key (F8) on the END level of keys. This level contains, in general, the same type of display dispatch function keys and auxiliary functions for SC300 and SDCS and may appear as below.

```
1OPERATE2  LOG  3  LOG  4SC-300 5       6       7       8MORE|
|.KEYS   |  IN   |  OUT  |STATUS |       |       |       |     |
```

The END LEVEL of keys have convenience commands to allow the operator select other SC300 displays without typing the associated DIS command. In addition, the switching of virtual consoles may be programmed as part of these keys as well as the ability to LOG IN and LOG OUT of the SC300 system. There may be additional functions included here for particular applications, however; the keys here are to support all functions of MMI2 and SC300, and not only SDCS functions. The OPERATE KEY (F1) function key in the above example levels duplicate the HOME key on the keyboard. The EXTD FUNC key (F8) on the example HOME level of function keys, shown previously, duplicates the END key of the keyboard.

E. DRIVE HISTORY

Drive History is a feature of the DC300 drive itself. The drive will keep a buffer of internal memory available for the storing

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of variables at a very fast rate. The user may select which drive parameters may be stored in this buffer and the rate at which the values of these variables are stored. In addition, the drive may be programmed to lock the buffer upon a programmable trigger.
The drive history retrieval tool is not actually a resident of SDCS. This program resides in the primary SC300 console of MMI2 and runs detached (in background) from the view of the operator. Operator control of this program is provided by a menu level in the Super Drive Control Screen (SDCS) menu system. This program will scan all drives on the DRIVE LAN to see if the internal drive history (circular buffer) has been locked. If for any reason, a drive history has been locked, HISTORY will record the circular buffer memory into a raw history data file. The data file has an automatic naming convention :

DHMDD###.XXX.

The DH is constant to mean a (D)rive (H)istory file. The M is a number or letter of the month this file was saved to disk (1 - January, 2 - February, ..., 9 - September, O - October, N - November, D - December). The DD is a number corresponding to the day of the month the file was saved (1 - 31). The ### corresponds to the file number saved on the date specified by MDD, where 000 is the first file saved and 999 is the maximum number allowed. If more than 1000 files were saved under the same day, the first file will be overwritten with no indication in the file name that an overwrite has occurred. The XXX is the drop number from where the history came. Since each drive is assigned a drop number for communications on the LAN, this unique number is used to identify the drives in many of the tools and programs of MMI2.

Once the drive history is recorded to disk, HISTORY will attempt to unlock the history buffer in the drive. If the condition which locked the history is still present, the unlock will fail. HISTORY will remember the failed unlock attempt and on each subsequent scan, will attempt to unlock it again. Another History recording will not be done unless HISTORY sees that the drive history buffer has been unlocked for at least one scan.

1 BROAD CAST	2HISTORY LOCK	3 SCAN START	4HISTORY STATUS	5OPERATE LINE	6REFRNCE LINE	7 DRIVE OPERATE	8EXTD FUNC
ZONE	UNLOCK	STOP		ZONE	ZONE	REFRNCE	

The operator has the control to start and suspend scans for locks, and has the ability to manually lock or unlock the history in one drive, a group of drives (zones as defined in EEROM), or all drives currently listening on the LAN (broadcast). The function keys for HISTORY LOCK and HISTORY UNLOCK will be

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sent to the drive "tabbed" without either the BROADCAST or ZONE keys selected (F1 or ALT-F1). If Broadcast is selected, the command is sent to the entire coordinated line, and if ZONE is selected, the command is sent to the zone of drives that is currently "tabbed".

HISTORY initializes with its scanning disabled and the operator must manually start the scanning before HISTORY becomes active with the F3 key. HISTORY will NOT retrieve drive history data if the user fails to SCAN START. In addition, the communication from HISTORY to the drives is not based on the same data driven command structure as previously described. The drive history data and commands from the function key level is communicated by a serial message format. This format requires a private message buffer be constructed in the master control device. Other programs, primarily in the maintenance console (VC4), share this serial message buffer. The HISTORY communication has the lowest priority for using this buffer, and therefore; history retrieval and command sending will be disabled if the buffer is being used. An event message will be issued if HISTORY scanning has been suspended due to buffer use. The operator may send lock or unlock only while HISTORY is scanning. As mentioned, dedicated event and alarm messages are generated by the HISTORY program which will log when the HISTORY program is scanning for locked histories and when it is not. It will also log the time on which it has recorded a history to disk as well as the file name of the raw history file it recorded. It will log the time and date when a request to lock or unlock was made as well as the few fault conditions which could make the program not operate correctly (such as a full hard disk drive).

Notice on the DRIVE HISTORY function key level shown above that function keys four through eight simply select other SDCS function key levels. These keys can be and often are changed to allow different function selections for each particular application. The above examples of function keys are not meant to show any particular fixed configuration except the function keys dealing directly with the DRIVE HISTORY function. The function key four (HISTORY STATUS) will call a diagnostic display that will replace the SUPER SCREEN. This diagnostic display will show what the background HISTORY program is currently doing. This was meant to be a program debugging tool but was left as part of the SDCS auxiliary displays to allow users to see the HISTORY program status.

1. DRIVE HISTORY IN THE DRIVES

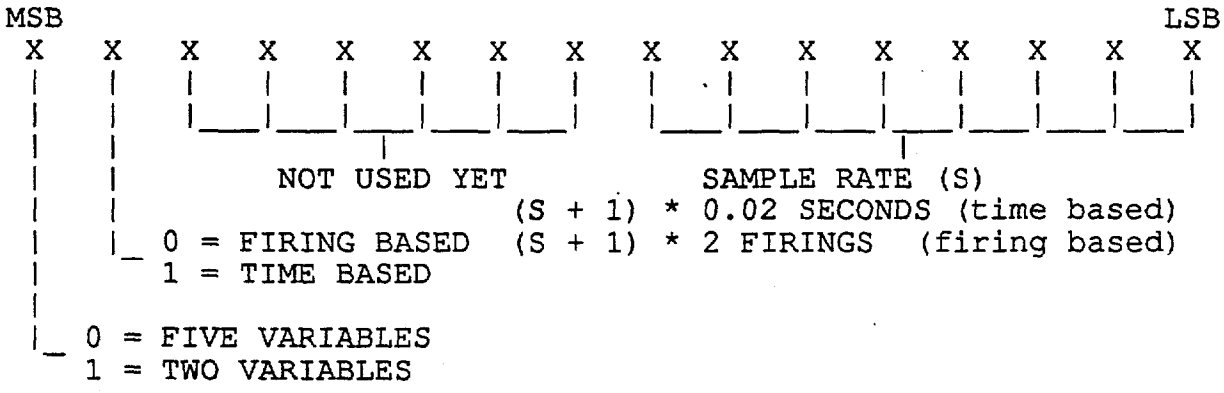
To fully define the drive history function, the drive history feature of the DC300 drives must be explained. The following is a description of history operation in the drive. The drive

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history feature is available to any drive containing the Micro Application Card or LAN Card. Drive history has several setup parameters that must be programmed into the drive's EEROM before it can be used.

HISTORY TYPE - EE378 HISTD

This EEROM location allows the user to select the type of history desired. The user can select a two variable history or a five variable history. A two variable history allows 102 samples to be taken on any two variables selected by the user. A five variable history allows five variables to be sampled 51 times with two variables of the five available selectable. Voltage feedback, current feedback, and RuP status (a word of bit status from the main regulating computer in the drive) are the fixed variables sampled and the two selectable variables are chosen the same way as the two variables in the two variable history. The user can select the histories to be firing or time based. Firing based samples are multiples of every other SCR firing (5.556 msec). Time based samples are multiples of 20 msec (0.002 seconds). The user then specifies the sample rate. If he selects 0, the samples will be at the minimum rate. If he selects 4, the samples are 5 times the sample base.



Example : EE378 = 5
Five variable History with Firing Based Samples and Sample Rate = 5 (samples will be on every twelfth firing of the SCRs)

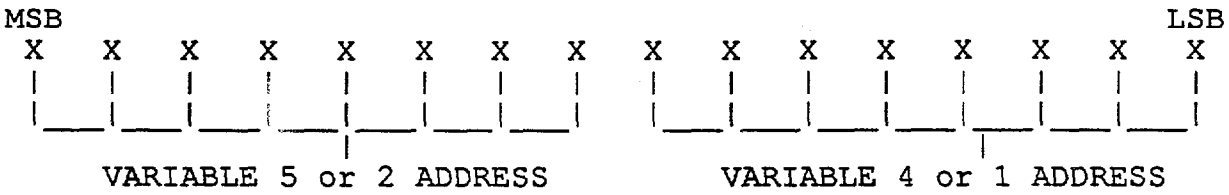
Example : EE378 = 49154
Two variable History with Time Based Samples and Sample Rate = 2 (samples will be taken every 60 milli seconds)

HISTORY POINTERS - EE379 HISPT

This EEROM location specifies the RAM location for the variables for history to use as variable 4 and 5 in a 5 variable history or variable 1 and 2 in a two variable history. Normally, the addresses specified in this EEROM location will point to a MuP

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pointer variable through which almost all variable data can be read through the history data buffer.



The addresses used here are for the location of the MVAR1 and MVAR2 variables respectively. These are selected because they are the fastest updating variable transfer area (MVAR1 updates every other firing and MVAR2 updates every fourth firing) and the data which can be programmed into these location are most of the drive parameters that could aid in the trouble shooting process. The value used in the above EEROM location (EE-379) will be F1EF hexadecimal or 61935 decimal where F1 (hex) is the MuP RAM address of the value pointed by MVAR2 and EF (hex) is the MuP RAM address of the value pointed by MVAR1. The actual variables selected for historical retrieval then are programmed in the EEROM for MVAR1 (EE-216) and MVAR2 (EE-217). These two EEROM locations are programmed with the RAM address for the particular signal desired. The RuP RAM addresses shown below is a partial list of signals and addresses available to be programmed into MVAR1 and MVAR2.

NUMERICAL DATA

SIGNAL	ADDRESS	SIGNAL	ADDRESS
ACOM	00494	LNSPD	01096
ANAJR	00465	LSCMP	01106
ASP0	00379	LTA	00413
ASP1	00380	MSR	00477
ASP2	00461	NET SPD REFERENCE	00063
ASR	00486	OR BIAS	01073
BLK COMP	01146	POSITION	00453
CEMF	00105	POS LOOP ERROR	00471
CFB	00055	POS LOOP INTEGRATOR	00480
CLA	00414	POS LOOP OUTPUT	00482
CT PHASE 1	00428	PSREF	00469
CT PHASE 3	00429	PULSE REF FREQ	00083
CURRENT LOOP ERROR	00020	REAL TIME CLOCK	00381
CURRENT LOOP OUTPUT	00061	REF TOTAL	01075
CURRENT OFFSET	00028	SFB	00071
CURRENT REF	00443	SFB WITH LEAD	00073
DIAMR	01092	SJ1	00459
DIG. TACH SPEED	00077	SJ2	00457
DRAW SPEED REF	00067	SPD DROOP	01038
DRAW OUT	00075	SPD LOOP ERROR	00101
DVM	00496	SPD LOOP INTEGRATOR	00051
ENCR PI POS	00091	SPD LOOP OUTPUT	00053

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NUMERICAL DATA (cont.)

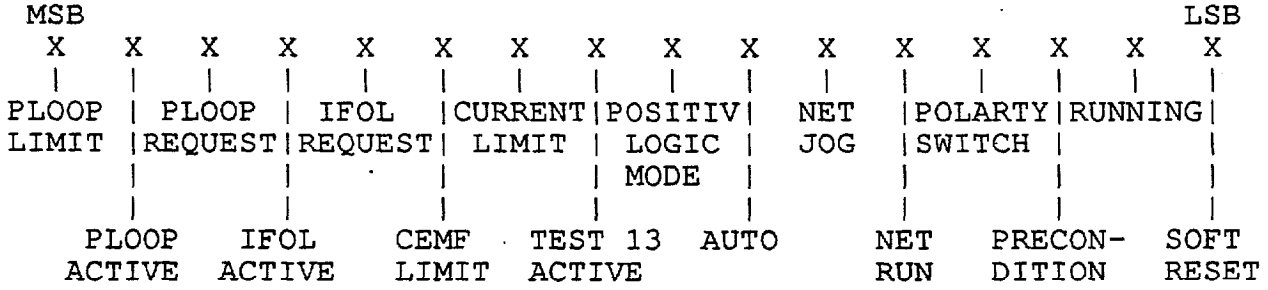
SIGNAL	ADDRESS	SIGNAL	ADDRESS
FAULT CODE	00023	SPD LOOP PROPORTIONAL	00048
FFLUX	01114	TACH FOLLOWER ERROR	00089
FIELD CURRENT	00463	TACH LINEARITY CRCTN	00450
FILTERED CFB	00448	TAPERED CURRENT LIMIT	00018
FILTERED SFB	00484	TENRF	01120
FILTERED VFB	00445	TIMED SPEED REF	00065
FIRING V-CMD	00030	TIR	01050
FRA	00467	TOC TIMER	00501
FuP DIAG VARIABLE	00093	TORQUE	00455
HORSEPOWER	00492	TuP DIAG VARIABLE	00087
IABS	00057	VFB	00103
IREF	01100	VLOPO	00475
JCOMP	01102	VOFST	00061
JOG REFERENCE	00069	VRATIO	01108

DIGITAL DATA

SIGNAL	ADDRESS	SIGNAL	ADDRESS
ADVANCE RATE LIMIT	00241	PRECONDITION	00152
AUTO	00146	REFERENCE ENABLE	00148
CEMF LIMIT FLAG	00141	RSTOP SATISFIED	00147
CURRENT LIMIT FLAG	00140	RUN 0	00190
CURRENT RATE LIMIT	00240	RUN 1	00191
DISCONT CURRENT FLAG	00157	RUN 2	00193
FIRING ENABLED	00139	SHUNT TRIP	00130
I FOLLOW REQUEST	00162	SLOW	00194
I FOLLOW ACTIVE	00172	TORQUE PROVED	00255
IN POSITION	00154	TRIP FAULT	00137
NO FAULT	00136	UP TO SPEED	00155
POS LOOP LIMIT	00251	ZERO CURRENT FLAG	00156
POS LOOP RUNNING	00250	XRAMP JR SELECT	00219
POS FDBK SELECT	00216		

In a five variable history, three variables of the five are preset and are not selectable. Those variables as mentioned above are voltage feedback (VFB = 00103), current feedback (CFB = 00055), and RuP status. RuP status is a packed word of bits that include many of the digital data in the table above.

RuP STATUS



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Each of the bits in the above word reflects the status of each item as indicated in the above breakdown of RuP status.

HISTORY TRIGGER SOURCE - EE251 UAPJP

Histories may be locked manually through SC300 by the DRIVE HISTORY menu level in SDCS or by issuing a STATUS - S message to lock by the SERIES SIX LADDERS (the latter method is fairly complex and not recommended unless special ladders have been allocated for this function). Histories may also be locked by selecting locking trigger sources through this EEROM location. The user may select the histories to lock any time the drive has a trip fault, if SLD 4 picks-up, or any time the drive has any type of fault (LATCHED or ANNUNCIATED)

```
MSB                                                                   LSB
 X   X   X   X   X   X   X   X   X   X   X   X   X   X   X   X   X   X
                         |   |
                         |   |
                         0   0   =  TRIP FAULT TRIGGER
                         0   1   =  PICKUP OF SLD 4
                         1   0   =  TRIP OR ANNUN. FAULT TRIGGER
                         1   1   =  PICKUP OF SLD 4
```

If the trigger source is SLD 4, the user must specify SLD 4 parameters to allow SLD 4 to trigger histories. This is the same as always but is listed below for convenience.

SLD PICKUP AND DROP OUT SELECTION - EE009 SLDJP

Remember that this EEROM location cannot be changed while the drive is in a running state. The options to tell how SLD 4 is to pickup or drop out is selected here.

```
MSB                                                                   LSB
 X   X   X   X   X   X   X   X   X   X   X   X   X   X   X   X   X   X
                                     |   |
 P.U. IF EE118 > EE119           =   0   0
 P.U. IF ABS(EE118) > EE119      =   0   1
 D.O. IF EE118 > EE119           =   1   0
 D.O. IF ABS(EE118) > EE119      =   1   1
```

SLD 4 INPUT SIGNAL - EE118 SLD4I

The user must enter in this location the signal for SLD 4 to watch. Again, this can be any RAM location listed above.

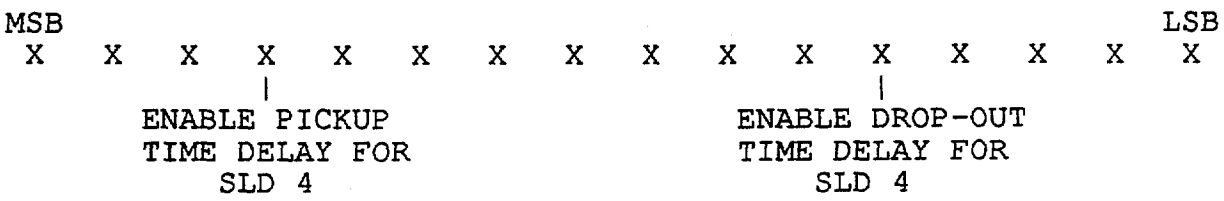
SLD 4 LEVEL - EE119 SLD4L

The user enters the level that the input signal must exceed to pick up the SLD or drop out the SLD. In addition to the above

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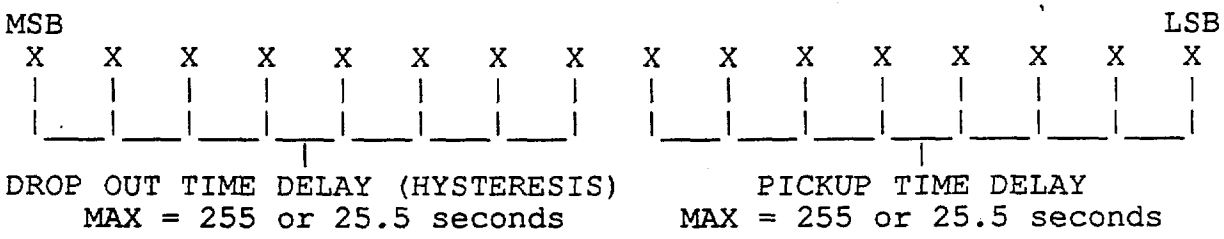
SLD parameters, a pickup or drop out time delay may be selected for SLD 4. This involves the setting of two more EEROM locations.

SLD TIME DELAY ENABLE - EE089 SLDTD



SLD 4 HYSTERESIS - EE120 SLD4H

This location sets the time delay for SLD 4 in tenths of seconds. The high byte value is for drop out time delay and low byte is for pickup time delay.



With the above EEROM parameters properly set in each drive, the drives themselves will record the variables selected and lock on the trigger source selected. The only function of HISTORY is to retrieve the raw drive history data from the drives when those history buffers are triggered to lock, and then to store them within MMI2. History will also annunciate the various status of itself as well as showing if and when a history was retrieved.

2. DRIVE HISTORY UNLOADER

The UNLOAD program is a companion of the HISTORY program except that it resides in the maintenance console (VC4) and must be invoked to run by the user. The sole purpose of this program is to convert the raw history files into a human readable form. There are three ways to unload a history file.

The user may select to print the raw history data to the system printer. The printout will show when the history was locked in terms of the real time and date of the occurrence. It will have a header which informs the user the type of history it is and the sample base as well as the variables recorded. All variable values will be in raw counts and trigger source will not be shown. The print format will be the same for both the two or five variable histories. DATA 4 and DATA 5 will be the second sample of DATA 1 and DATA 2 in a two

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variable history. DATA 3 is always RuP STATUS (a word of bit status from the main regulating computer in the drive) and is recorded on every other sample in a two variable history.

The user may choose to unload the history file into a text format that may be imported into Lotus 123 (TM). This allows the user to use all available Lotus (TM) tools to analyze the data. Again, variable values will be in raw counts. Scaling and inverting data will be left for the user to do.

NOTE : The LOTUS 123 package is not provided in the MMI-2 package, however; LOTUS (TM) may be executed in the MMI-2 environment concurrently with other MMI-2 programs.

The third way to unload histories is to unload them for viewing in the SCTREND graphics package that is provided with MMI-2. SCTREND allows the loading of up to seven graphs to be viewed on a common time base. UNLOAD will strip the history file of a selected variable and convert it into an SCTREND graph file. Once these graph files have been created, the user then creates an SET file to automatically load the SCTREND graph with the graph data. The user then loads this SET file into SCTREND and SCTREND automatically draws the data from the raw history file(s) onto the graph window. Unloading in this method normalizes variables to their Per Unit values rather than raw counts and inverts references and feedbacks so they are in the same polarity for better comparisons if the variables are known by the UNLOAD program.

Time based History variables may be unloaded from the different raw files with different lock times. The UNLOAD program will automatically space the graphs correctly in time and zero suppress (buck-out) the variables for maximum resolution. Firing based samples may be only unloaded from a single raw file for viewing since these variables can only be related to themselves in time and not to other histories locked at different times.

The unloading of history files for SCTREND will probably the most used option of the UNLOAD program. The UNLOAD for the printer or for LOTUS (TM) is fairly simple and only requires that the user know the name of the raw history file. For the LOTUS (TM) unloading, the file created by the UNLOAD program that may be then used for analysis with LOTUS 123 (TM) has the same name as the raw history file used except the first two characters (DH) is replaced with LT.

To unload a raw history file for SCTREND, the user must select the function key in that program that will unload a history file. When the user presses that key, he will be prompted to enter the raw history file name. The user can look at the event log from

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his event printout or ALARMCON, and enter the name in the exact form as read from the event line. UNLOAD will give a description of the file and a list of the variables in the raw history file the user may unload. The user then chooses the one variable to unload and presses the associated number of that variable. The variable is then shown under the function keys as being unloaded. The user may at this time create the SCTREND decides to unload more files, will be asked if he wants to unload from the same raw history file as the last variable. The operator enters "Y" or "N" in response to this question and enters a new file name or selects the new variable from the same file. Once he has unloaded from one to seven variables, the user then presses the MAKE SET FILE function key. UNLOAD will prompt the user for a one to eight character name and will create the for this file name as UNLOAD will force .SET as the extension.

The user now has a complete graphing set of files and may proceed to the analysis console (VC2) where he may start SCTREND and load the SET file just created. SCTREND will then graph the files as directed.

NOTE: For more information on the operation of SCTREND, refer to **GEK-????? SC-300 SCTREND Users Manual.**

F. DRIVE COMMAND PERMISSIVES

Another background program runs in association with SDCS to provide permissives for the function keys that send commands to the drives. This background program, NETPERM, relies heavily on the programming in the master control device and/or Drop In Operator Stations to provide permissive information to SDCS. Of the dozens of commands available to send to the drives from SDCS, only a few commands will be chosen for actual use in SDCS. In general, those commands are not checked by SDCS at any time as to the legality with respect to system permissives, of whether a command should be sent to the drives or not. The logic solution to allow a RUN command to be sent to a drive often depends on the mechanical and supporting systems to the drive system to be in certain states. These states are communicated to the master control device and/or Drop In Operator Stations expressly for this logical permissive solution for every available command that could be sent to the drives.

It is, therefore, reasonable to expect that these same permissive states be reported to SDCS so a command cannot be sent unless those permissive states allow it. The SDCS system addresses this need in a unique manner in that it requires the logic solving devices to send the results of their permissive algorithms so that those commands available at the function keys

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may also be allowed to or kept from sending commands when required.

The NETPERM program allows the system integrator to define the commands that are available to send from SDCS and will also update the permissive solutions from the master control device and/or Drop In Operator Stations. NETPERM keeps a copy of all the permissives for each command for each drive and resolves a NET PERMISSIVE for a command dependent on which drive(s) the operator has selected through the "tab" and the level on which the operator has selected.

The command keys that have permission to operate are shown in their normal colors. Function keys that do not have permission to operate will be shown with a black background and dark gray or light gray characters. As the operator "tabs" through the list of drives, the NET PERMISSIVES will change depending on the drives the operator is selecting. For example, the operator is currently on the DRIVE OPERATE level (remember, this level can only send commands to a single drive), and has the tab mark on DRIVE X. DRIVE X is a non reversing drive and therefore, NETPERM will keep the JOG REV function key black. DRIVE Y, the next drive on the SUPER SCREEN, is reversing and when the operator "tabs" to DRIVE Y, the JOG REV key will turn green showing that it is possible to jog DRIVE Y in the reverse direction.

If the operator was on a LINE or ZONE OPERATE level and had more than one drive selected (multiple drive zones), if just one of the drives in the group selected did not have permission for the function in question, the whole group will not have permission. Thus, the key representing that function or command will be black.

NETPERM serves as an operator guide through the function key color representation of permissives on any function key level of SDCS. In some cases, the NETPERM program may not show dynamic permissives if the system designer has excluded a command from being used at SDCS at all. In those cases, the function key representing that command will always stay black and therefore, cannot be used.

In addition to NETPERM, there are built-in permissives based upon the configuration of SDCS by the system designer. For example, each drive has been configured with only those regulators into which that drive may be switched. Only those setpoints for those regulators will have the permissive to be changed by SDCS. These built-in permissives are not addressed by the NETPERM program and are configured separately, however; the indication of the ability to invoke the function is still carried by the color of the function key representing that command or function.

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G. SYSTEM PERMISSIVE DISPLAYS

In association with the NETPERM program are custom SC300 displays to report the states of system permissives to the operator. System permissives are those inputs to the master control device and Drop In Operator Stations that determine whether a function is available to operate. In most cases, these permissives are those signals which are part of the permissive logic solution for allowing a drive to be initiated into a RUNNING state. These permissives may be thought of as the raw inputs to the READY TO RUN logic for the drives. In addition, these displays may have other inputs that, although are not part of READY TO RUN logic, may have a bearing on the operating state of the process and not necessarily pertinent just to the operation of the drives.

These permissive displays are custom created for each application. That is the permissive display(s) for one system will not be the same as the permissive display(s) of any other system although the information displayed on these displays have the same purpose or intent. The function key on the HOME level labeled DRIVE PERMS is used to call these displays to the screen. In addition, the keys on the END level may be configured to access these screens as well.

In general, the permissive screens will contain a background screen with a list of statements. To the left of each statement is an oblong colored symbol meant to show whether the statement adjacent to it is true or not. The color of the symbol will be configured such that when all systems are normal, all symbols on each permissive display will be green. This allows the operator to simply glance at the screen to be able to determine whether or not a problem exists. A non green symbol means that the adjacent statement is true and there is an abnormal condition existing in the process. Each symbol and statement will almost always be accompanied by an alarm configuration such that any time the statement becomes true an associated alarm or event will be recorded to ALARMCON and the system printer which always logs system events. The operator upon seeing a problem on the permissive display may switch to ALARMCON to ascertain when the problem actually happened and if it was part of another or group of events which could pinpoint the cause of the problem.

The statements on the permissive displays will be organized logically to show their pertinence to a particular drive or group of drives. The MODE column on SDCS displays will show whether a drive is READY TO RUN by displaying RDY as the first three character mnemonic in the column. If the column is showing STP instead of RDY, the operator should then proceed to display the permissives screen(s) and look for a non green symbol in the group of statements associated with the non ready drive. He can quickly determine why the drive is not READY TO RUN and direct

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maintenance personnel to the source of the problem and avoid time consuming permissive tracking.

Permissive displays will, in most cases, contain read only screen parameters. In other words, there will not be any parameters on the permissive displays that an operator can adjust or change. These displays can also contain states of system interlocks that are also pertinent to the process but irrelevant to the drives. Because the permissive displays are normal SC300 displays, all attributes and configuration options are allowed. These displays have a great potential if configured logically in aiding in the operation of a process.

H. SDCS SUMMARY

The Super Drive Control Screen (SDCS) is a complex subsystem of SC300 devoted to the control and monitoring of a coordinated drive system using General Electric DC300 drives. The main objective for the SDCS system is to reduce down time in a coordinated drive system and to allow more efficient operation of the process. Of secondary importance is the ability to operate the drive system similar to a master operator system.

The features of SDCS and SC300 to reduce downtime are:

1. System logger which logs alarms and events which may isolate a problem to smaller proportions for analysis.

2. Trend system which allows the collection of system data and the ability to view this data graphically.

3. Drive data trending which is triggerable upon internal and external events. This allows the "locking" of data upon a programmable event which captures data leading up to that event. Like the trending feature, this data may also be viewed graphically using the same graphing tool.

4. Specialized permissive displays which alert operation personnel and pinpoint system interlocks which prevent the drive system from operating.

5. Specialized screen displays which show the status of all drives and their associated operating setpoints and feedbacks. This screen display will highlight problems in the drive by highlighting those status and feedbacks which are abnormal.

The features for more efficient operation are:

1. The ability of capturing and recalling drive settings for more consistent operation and efficient execution of a change in operating modes of the process.

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2. "Single" screen data presentation which minimizes "display flipping" to monitor and operate drives.
3. Flexible command function key levels which guide operators as to what functions are available.

The unique architecture of SC300 and the drive system allow the efficient communication of data and commands throughout the system. This allows SDCS to process an extremely large amount of data for the purposes of displaying, trending, alarming, limit checking, decoding, and archiving this data and commands. The subsystems to SDCS which are both specialized and part of SC300 are:

1. Enhanced event and alarming specifically designed for drive system diagnostics

2. Data trending capabilities

3. Historical drive data trending

4. Special Recipe capabilities for drive parameters

5. System Permissive displays for both process and drives

6. Detailed Drive data display

7. Drive operation capabilities

Related to SDCS is the MMI2 maintenance console which has specific tools for the startup and maintenance of the drive system. These programs in the maintenance console is described in the companion document in this document's appendices.

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APPENDIX A - INSTALLATION AND STARTUP

The MMI2 programs have been specially configured for this particular application. MMI2 can be run in several computer hardware environments and the care and installation of the particular computer hardware is dependent on this environment. This section will describe the common components in every system and the installation and startup procedure required.

Minimum system components :

IBM AT COMPATIBLE (80286) COMPUTER
- 3 Mb RAM
- 80287 MATH COPROCESSOR
- 40 Mb FIXED DISK DRIVE
- 1.2 Mb FLOPPY DISK DRIVE
- ENHANCED GRAPHICS ADAPTER BOARD WITH 256 Kb RAM
- PARALLEL PORT (1 PORT MINIMUM)
- RS-232 SERIAL PORT (1 PORT MINIMUM)
- IC640BSS303 GE-FANUC WORKSTATION INTERFACE BOARD (WSIA)
- IC640BLD304 GE-FANUC WORKSTATION TERMINATION BOARD

ENHANCED GRAPHICS (EGA) MONITOR

DOT MATRIX PRINTER (EPSON FX, LX, LQ COMPATIBLE)

LICENSE PROTECTION KEY

SERIES SIX LOCAL TRANSMITTER BOARD IC600BF900L

MMI2 SOFTWARE DISTRIBUTION KIT
- CORE SOFTWARE KIT
- VERSION PATCH DISK (If necessary)
- CONFIGURATION DISK(S)

INTERFACE CABLES
- PARALLEL PRINTER CABLE
- TTL RGB MONITOR CABLE
- SERIES SIX PARALLEL I/O CABLE IC600WD### (### = LENGTH)

In most cases, the hardware in the system will have been pre assembled and tested as part of the system test for the drive system. In this case, all of the system software would have been loaded and all hardware would have been configured and tested, so the only tasks left is to interconnect the computer hardware pieces with the provided interface cables.

For the minimum system as described above, the computer hardware has been already configured and the hard disk has been prepared with all software. The user is required to connect the License Protection Key into the primary parallel port of the computer.

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The printer cable then connects to the end of this key device and then connected to the printer. The monitor cable connects to the EGA nine pin connector and then to the monitor.

The Series Six Local Transmitter Board (IC600BF900L) should be installed into either a CPU rack or a local I/O rack of the Series Six. The Transmitter Board has two BERG jumpers located on the board. The jumpers should be set for NORMAL I/O CHAIN and DISABLE CPU FAULTS. The DIP switch block for the slot in the Series Six Rack need not be set to any particular setting.

The Series Six parallel cable (this cable may be from 5 to 500 feet long) should be connected to the Interface Card connector on the MMI2 computer and then to the Local Transmitter Card in the Series Six Rack.

Once this interconnection has been completed, the power cables for the computer, monitor, and computer should be connected. The power requirements for each device and limits are listed in the manufacturer's documentation for each device. Common power usage for the computer, monitor, and printer would be 108 to 125 VAC, 60 Hz, at 15 amperes maximum. The power source is expected to be free of high voltage spikes and high frequency electrical noise. Power conditioning equipment may not be needed, however; is available as an option to the system.

The computer will have an automatic bootstrap program already installed so the MMI2 programs will commence to operate once power has been applied to the devices. Some computers may be configured to ask the users configuration questions as the bootstrap program progresses. These configuration options during startup will be explained by the on-site General Electric representative.

NOTE: As part of the automatic bootstrap program, the license protection key is checked before any of the MMI2 programs can execute. If a printer is connected to the key, the printer **MUST** be at least powered on. If no printer is installed, MMI2 will eventually give a system error message on the monitor :

PRINTER NOT READY ... Retry or Ignore

The user must press the "I" key on the keyboard to continue.

The communication program in MMI2 depends on the correct connection of the parallel interface to the Series Six Master Controller and the Series Six must be in the RUN mode. If these conditions do not exist, the automatic bootstrap program will halt execution and display that no communication was established. The communication will continuously try to become established unless aborted by the user or the communication eventually

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becomes established. If it is aborted, the MMI2 programs will appear to be started, however; no active data will be communicated from the MMI2 to the rest of the system. The user must reset the computer (most computers have a RESET button on the front panel of the computer itself) or cycle its power to begin the automatic bootstrap again.

If the MMI2 hardware was pre shipped from the factory without being configured, the on site General Electric representative will assemble the hardware, prepare the hard disk drive and load the software distribution kits.

The instructions for the hardware and software preparation for the MMI2 system utilizing the AST Premium 286 Model 140 computer system is included in the next appendix.

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APPENDIX B - AST PREMIUM 286 INSTALLATION

SC300 Installation Guide
By: T.A. Digate
June 3, 1988

The purpose of this document is to provide the information required to install the General Electric Man-Machine Interface II (SC-300) software package on the AST Premium 286/Model 140 computer. This document should be used with the installation kit of (4) high density floppy diskettes. If the demo software is to be loaded, an additional diskette will also be provided.

Prior to installing the SC-300 software the existing software already installed on the hard disk in both partitions MUST be backed up. **This installation procedure will erase ALL existing software on the disk.** The SC-300 software system does not utilize the MS-DOS operating system software in its standard configuration and uses the Concurrent DOS XM operating system instead.

I. Hardware Setup

In order for the software installation to proceed without errors, the hardware MUST be configured correctly. Failure to do so may result in this installation procedure failing to work correctly.

The AST should have the following hardware:

1. 80287 Math co-processor installed in empty chip socket on AST motherboard.

2. At least (1) high density floppy diskette (1.2MB)

3. AST 3G+ Enhanced Graphics Adapter w/ SC300 EPROM used for character generation. Replace the existing EPROM supplied by AST (can be identified by label stating (C) Interlink) with the GE supplied EPROM. Make sure that pin 1 of the EPROM is matched with pin 1 of the EPROM socket. Make sure that the toggle switch located on the back of the board is set to the EMU position.

4. One 40MB Hard disk

5. Two AST FASTRAM memory boards configured as follows:

   a. Board #1: <u>2MB</u> (x's mean ON)

```
SW 1 2 3 4 5 6 7 8 9 10
   x x x   x x   x x x
```

   Jumper(s) E1 and E2 installed.

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5. b. Board #2: 1MB or 2MB (x's mean ON)

SW	1	2	3	4	5	6	7	8	9	10
	x	x	x	x				x		x

Jumper(s) E1 and E2 installed.

Failure to setup the memory boards EXACTLY as stated above will cause SC-300 to operate incorrectly.

(Technical note: Board 1 is being configured to have 640K of conventional memory, 512K of extended memory, and the remaining memory defaults to expanded memory. Board 2 is being configured as entirely expanded memory.)

6. One GE supplied software license key. This key should be plugged into the parallel port connector attached to the main computer motherboard. It will not be effective if used on auxiliary parallel ports. Parallel printers should be plugged into the key.

7. When an optional Everex tape drive is installed, the tape drive hardware should be configured to use IRQ5 and the interrupt line for the secondary parallel port should be disabled. See tape drive manuals and auxiliary I/O port board manuals for further details on how to setup IRQ lines.

8. Install the GE-Fanuc WSIA board and the WSIA terminator board. The dip switches on the terminator board should be set to terminate the I/O bus. This is not the normal factory configuration as the switches should all be in the closed position. Refer to the GEK data sheet supplied with the WSIA hardware to verify the dip switch settings. Note that when an I/O cable longer than 50' is to be used, the use of a Local I/O transmitter board in the Series Six is required.

The AST SETUP program must be run after changing the memory configuration. Insert the AST supplied diagnostic disk into the floppy drive. Press the front panel reset button to 'bootstrap' the computer. Select the 'Run SETUP' option from the menu. Set the time and date and configure the memory as follows:

640 KB Main Memory
512 KB Extended Memory

(Technical note: The remainder will be used as expanded Memory by the SC-300 software.)

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II. Hard Disk Preparation

After configuring the memory hardware and running the AST SETUP program, the hard disk must be properly prepared to accept the Concurrent DOS operating system and the SC-300 operating programs.

The hard disk must be organized into two logical partitions using the procedures outlined below. **Failure to follow the procedures EXACTLY as described will make the installation procedure fail.**

PREPARATION OVERVIEW

The hard disk will receive a LOW LEVEL FORMAT to erase old partition table information and ALL data on the disk. The Concurrent DOS FDISK utility will be used to create a PCDOS partition on the first 512 cylinders. The Concurrent DOS HDMAINT utility will be used to create a CPM/86 partition on the remaining 512 cylinders.

A. Low Level Formatting

Place the AST supplied diagnostic diskette into the floppy drive. Reset the computer by pressing the front panel reset button on the computer. Select 'Hard Disk Format' menu option. Answer the menu prompts by typing the first character of the commands. Use the following answers when asked:

default interleave factor of (2)
quick verification (3 passes)
format without defects list

This procedure takes approximately 30-45 minutes to complete. When it has completed, the hard disk will be completely erased. The disk is now ready for partition creation.

B. PCDOS Partition Creation

Insert disk 1 of the SC-300 installation kit into the floppy drive. Press the front panel reset button to 'bootstrap' the Concurrent DOS operating system. If the operating system fails to bootstrap correctly the computer hardware has not been configured correctly. Check that the EGA graphics adapter has the toggle switch in the EMU position. Verify that all the available memory is error free.

At this point we need to run the FDISK program. Type FDISK at the A> prompt. Select the option to 'Create a DOS partition'. When asked if all 40MB are to be used for this partition, answer NO.

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Enter starting cylinder of 1. Enter size of 512 cylinders. When asked for a volume label use 'PCDOS'. This partition will become drive C: under Concurrent DOS XM.

C. Concurrent DOS Partition Creation

Insert disk 1 of the SC-300 installation kit into the floppy drive. Press the front panel reset button to 'bootstrap' the Concurrent DOS operating system. If the operating system fails to bootstrap correctly the computer hardware has not been configured correctly. Check that the EGA graphics adapter has the toggle switch in the EMU position. Verify that all the available memory is error free.

At this point we need to run the HDMAINT program. Type HDMAINT at the A> prompt. Select the option to display or change disk partitions. Select the option which creates a CPM86 partition.

Select maximum partition size (around 511 cylinders).

Select options which enables 2048 directory entries, and no verification of write operations.

This procedure takes approximately 30 minutes to complete.

IMPORTANT: After creating the partition, select enable CPM bootstrap from main menu.

This partition will become drive D: under Concurrent DOS XM. This partition will not be accessible when running PCDOS as it is a 'foreign' partition type.

III. Software Installation

Insert disk 1 of the SC-300 installation kit into the floppy drive. Press the front panel reset button to 'bootstrap' the Concurrent DOS operating system. If the operating system fails to bootstrap correctly the computer hardware has not been configured correctly. Check that the EGA graphics adapter has the toggle switch in the EMU position. Verify that all the available memory is error free.

To verify that the hard disk has been prepared properly, attempt to switch to the C: and D: drives by typing the commands:

C: followed by a <cr>
D: followed by a <cr>

If everything is correct, you should receive a new prompt after each of the above commands. If not, start this procedure again from step 1.

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Make sure that you are logged onto the A: drive by typing A:. You should have an A> prompt on your screen.

Now type INSTALL to restore the SC-300 software to the hard disk. This procedure will prompt you to change diskettes when needed and will echo a message on the screen when it has completed. Several questions may be asked concerning the application of this software. Answer with the appropriate data (normally a Y or N response).

After the installation procedure has completed, the files located in user 0 of drive D: named STARTUP1.BAT, STARTUP2.BAT, STARTUP3.BAT, and STARTUP4.BAT can be edited to change the startup behavior (i.e. printer numbers, etc.). Change user areas by using the command 'USER 0' where '0' can be any user area from 0 to 15.

Use the DREDIX text editor to comment out the lines that would automatically startup SC-300 software. Commenting out a command in this file can be accomplished by placing a semi-colon in column 1 of the line. The STARTUP files have comments in them to assist you in determining how to enable/disable SC-300 automatic startup. The default configuration of these files is set to NOT automatically start SC-300.

After editing the STARTUP files, remove the disk from the floppy drive and press the hard reset button on the AST computer. The Concurrent DOS operating system should boot and the SC-300 software should attempt to run as you directed in the previously edited STARTUP files.

IV. Preliminary System Configuration

A. Concurrent DOS Operating System

In order to run the SC-300 software, the operating system should be properly configured. The SC-300 installation procedure previously described details how to install the software. The Concurrent DOS manual describes in more detail the various operating system commands.

SC-300 is normally configured to use a ram disk labeled drive E:. The CCONFIG.SYS file is used to create the ram disk using the AST provided FASTDISK.SYS device driver. The supplied CCONFIG.SYS file is normally correct to provide this feature. When the General Electric Intelligent Communications processor is used instead of the WSIA parallel interface, the expanded memory manager software must be configured differently. Change the line that contains 'a000' to 'b800' using the DREDIX text editor.

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Printers connected to the system are given numbers depending on the serial or parallel port used to connect them. The following numbers can be used:

LPT1:	0
LPT2:	1
LPT3:	2
COM1:	3
COM2:	4

Edit the STARTUP?.BAT files to select the default printers used for each virtual console.

Run the SETPORT command to configure the serial port baud rates, handshaking protocols, etc. If no serial ports are being used in this application it is not necessary to configure the serial ports. The default configuration of these ports when first installing the operating system is 9600 baud, 8 bits, no parity, XON/XOFF handshaking on input and output.

After setting the port parameters, run the SETUP command to save system parameters (i.e. serial port configuration). Make sure that you run SETUP from user 0 of drive D:. If SETUP is not run, the SETPORT command must be performed after each reboot if changes to the defaults are required. The SETPORT utility commands can also be placed in your STARTUP batch files if desired. See the Concurrent DOS manual for more information.

B. Virtual Console Layout

The physical CRT screen of the computer is actually capable of running four programs. Each program is assigned a 'virtual console'. These four virtual consoles share the same physical console---the CRT and keyboard. Concurrent DOS automatically manages which programs can read and write from the physical console at any point in time. Pressing the control key while simultaneously pressing the numeric keypad buttons 1 through 4 will switch the control of the screen to the respective virtual console.

The SC-300 software dedicates the four virtual consoles to a specific application program. The commands in the STARTUP?.BAT file(s) are used to start the application programs.

a. Console 1 runs the SC-300 main program using the START menu page of the SC-300.MNU file. See the Concurrent DOS manual for more information about the RUNMENU facility. This menu page invokes the batch file procedure named START.BAT. After START.BAT completes, it invokes START1.BAT. START1.BAT actually runs all of the SC-300 background and foreground tasks.

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b. Console 2 runs the SC-300.MNU menu system using the ANALYSIS menu page.

c. Console 3 runs the ALARMCON alarm console program.

d. Console 4 runs the SC-300.MNU menu system using the MAINTENANCE menu page

While it normally is not necessary to change the startup procedures, the Concurrent DOS EDITMENU facility can be used to change the actions and screens performed/displayed in the SC-300.MNU file.

C. SC-300 Configuration

In order to configure SC-300 for a real application, the configuration system should be used. All configuration data is normally stored in user area 1. Before attempting to configure the SC-300 system for application specific information, be sure to change to user area 1. Virtual console 4 is usually used to configure the system.

To get the configuration menu, type CONFIG at the D> prompt. Refer to the appropriate documentation for information on each of these configuration choices.

D. I/O Templates

The normal I/O template used by the SC-300 system is a program named DAC.CMD. This program performs the necessary communications between the Series Six parallel I/O bus and the WSIA board set located in the IBM PC.

When the DAC program cannot communicate with the Series Six, it disables the ability to change values in the memory resident database of SC-300. In order to allow development of software and testing without a Series Six, a dummy I/O template is provided as part of the SC-300 software. This program is named NOIO.CMD.

The START1.BAT file is used to run the I/O template. Edit this file and comment out the DAC command with a semi-colon and insert the NOIO command. Restarting SC-300 will then use the dummy I/O template.

Warning:

Do not run BOTH the DAC and the NOIO templates simultaneously. If so, changes intended to go to the Series Six may get lost.

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APPENDIX C1 - MAINTENANCE CONSOLE - EECONSLE

MMI2 MAINTENANCE CONSOLE
STARTUP/MAINTENANCE COMPUTER TOOLS - PAPER MACHINE DRIVE SYSTEMS

Introduction

The MMI2 Maintenance Console is a dedicated virtual console for the MMI2 computer system. This console contains a collection of computer tools to aid in the startup and maintenance of coordinated drive systems.

NOTE: This is a preliminary description of the tools available today.

EECONSLE

This program allows direct communication of MMI2 and any drive on the LAN. This program allows the manipulation and storage of EEROM in the drives. This program incorporates several different ways of manipulating a drive's configuration and all these will communicate to the drive itself through a special communication buffer in the Master Control Device. This is the same buffer used for the Drive History function. When EECONSLE is run, it will stop the scanning for locked Drive Histories by the HISTORY program, essentially terminating HISTORY until the user quits EECONSLE.

The central communication buffer in the Master Control Device allows EECONSLE to read and write several EEROM values from a single drive or read and write one value from multiple drives. The power of EECONSLE is the data presentation of EEROM values and the many different automatic calculation facilities in each of the functions.

When the user runs EECONSLE from the screen menu residing on VC4, the program will show a function key menu at the top of the screen giving program options. It will also display messages as the program initializes and will be ready for user input when the bottom line of the screen displays the prompt "Please Enter the Desired Function : ". EECONSLE can only be run while the MMI2 system is on-line and communicating to the Master Control Device. If the system is not communicating, EECONSLE will display several error messages and will appear to "hang" in execution. It is important that the user insures that the MMI2 system is fully operational before starting EECONSLE.

Currently there are five separate programs that may be called from the EECONSLE menu. The first program is EEROM CONSOLE which is invoked by pressing F1. This program has several sub-functions available.

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EEROM CONSOLE

READ EE
This function allows the user to read 36 consecutive EEROM locations from a single drop starting at a user specified address. The data presented will include for each value, the EEROM address in decimal, the mnemonic for the value, the value in unsigned decimal (just like the programmer card on the drive), and the value in hexadecimal. These values are stored in memory of the MMI2 and the user may now alter the value in MMI2 memory by entering the new value in unsigned decimal OR hexadecimal in the new value column of the display.

When the READ EE function is invoked, the program will ask for the drop number from which to read the EEROM values. The user must enter the desired drive's drop number. If this drive is not communicating on the LAN, an error message will be shown on the bottom line of the display. If the drive is on the LAN, the program will ask for the starting address of the EEROM values to read. The user enters a number from 0 to 988, and the 36 values starting from that address will be shown on the screen. The user may now edit any one of these values by using the TAB key or the cursor arrow keys to select the number to change. Once the desired value has been selected, the user simply enters the new value. The user may enter this value in hexadecimal by preceding the value with a "$". This alerts the program that it should interpret the next four characters as hexadecimal. Decimal entry is done by entering up to five numbers. Decimal entry can only be done as an unsigned number from 0 to 65535.

Once a number has been entered, the function key line will now show that F2 will write the new value to the drive's EEROM.

WRITE EE
This function allows the user to write back to EEROM, the up to 36 values entered (as described above). This function, prior to writing, checks to see if the software protection is disabled and the hardware jumper on the drive is in the write enable location. It also sets the most significant bit of EE251 to force the drop to echo back the EEROM value it has changed for verification at MMI2. This function allows the writing of EEROM as long as power is on the drive and the hardware protection jumper is enabled for EEROM writing. It is up to the user to insure that a running drive will not be affected. Only the values that are changed will be written. The value shown for the address that has been written will be updated to show the new value. The current value in hexadecimal and decimal should now match the value in the new data column of the screen.

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EE<DISK
This function will allow all of a drop's EEROM values be stored to disk. This function also allows a batch archive mode that will save EEROM values from all the drives listening on the LAN unattended. The user will be asked if he wants a batch archive or a single drive to be recorded before proceeding. The user then has an opportunity to name the EEROM disk file. For batch archive files, each drop will have an individual file. The user is allowed to enter an eight character name and the file name will be appended with an extension relating to the drop number from which EEROM was recorded. Example - User enters EEDRV and the full name will be EEDRV.### where ### is the drop number of each drop recorded.

If storing EEROM values from a single drive, the screen display will be blanked and a single statement will be written showing the file name and the drop number involved in this function. As each address is read from the drive and written to the file, the current address will be updated on the screen. It takes approximately 10 to 15 minutes to record a single drive's EEROM to disk fully.

If using the batch archive technique, the screen will show all the expected drop numbers that the program will attempt to archive. The drop number that is being currently read will show the address that is being currently read and written to the file. Drops that have not been done will be shown as blank. Drops that the program has attempted to archive will show "DROP OFF" if the drop was not communicating on the LAN. "LAN FAULT" will be shown if the drop had a problem sending the EEROM values, or "DONE" if the archive was complete and successful for that drop.

The user may terminate the archive process by pressing F8 at anytime during the process. During a batch archive, the question "Skip drive or Abort function ? :" will be displayed. If the user enters "S" that drive being archived will be skipped and any disk file started for that drive will be erased. If the user enters "A", the entire archiving process will be aborted. If the function was archiving a drop when the abort function was issued, the disk file for that drop will be erased. Any other file created prior to the abort will be retained.

DISK>EE
This function allows a disk file of EEROM values to be written back to a drop. There is no batch mode for this function. Each file must be written to a drop individually. Again, checks are made prior to writing for the ability to write the EEROM, and all values written will be verified. The actual procedure this function uses is to compare the file to the drops EEROM and store away all locations that are different. It then writes to only those locations that are different to extend the longevity

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of the EEROM part. Like archiving a single drive, the screen display will be blanked and a single line will be written on the screen that will show the file name and the drop number involved in the writing process. The line will also show the address currently being compared during the compare phase and the address being currently written during the write phase.

WARNING

The drive which is being written should **NEVER** be in a running state during this function. The drive may be running during the compare phase but never during the write phase.

COMPARE EE

This function allows the user to compare a file of EEROM values to any drop on the LAN. The differences can be printed to the system printer or sent to a disk file (MISCOMP.DAT) or both. The printout and file will list the time of the compare, the drop and file being compared, the address, mnemonic, and values of the miscompared location in the file and the drop. This function can be used to insure that no EEROM values have been changed since a known point in time, and that is when the file was originally saved. The screen will show the drop number and the file name involved in the compare process and the address currently being compared.

BROADCAST READ

This function allows the reading of a single address from all drops listening on the LAN. The user will be asked for the address to read. Once entered, the screen display will show all the expected drops on the LAN to respond to this command. The drop numbers will be listed followed by either a "DROP OFF" or by the address and value. If "DROP OFF" appears next to a drop number on this display, it means that at the time the message was sent to read the EEROM to the drives, that particular drop was not on the LAN. The drives that are able to respond to this command will first show the address and then the value in that address. Both the address and the value will be shown in unsigned decimal.

BROADCAST WRITE

This function allows the user to broadcast a value to a single address in all drops listening on the LAN. The user will be asked the address he desires to write in all available drops. The screen shows all the expected drop numbers followed by the status of the write (NOT ENABLED - drop echoed wrong value, DROP OFF - drop not listening, NOT VERIFIED - drop did not echo value). If the broadcast write was successful and verified, the display will show the address in decimal and the value written in unsigned decimal. This is the same display format showing the value when broadcast read from the drops.

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EDIT EE FILE

This function allows the editing of a EEROM disk file in a similar fashion as the READ EE and WRITE EE functions provide. Values will be changed in the disk file and the time and date of the last time the file was edited will be stored as part of the EEROM file along with the time and date of when the file was originally made.

The procedure to edit an EE FILE is the same as editing a EEROM location in the drive. Only one file may be edited at a time. Again, the user must read the file at a specified starting address. He may then use the TAB key or the arrow cursor keys to point to the value he wishes to change. He then may enter the new value in decimal or hexadecimal as outlined above.

When the user quits the edit function, the user is asked whether or not the file should be saved. If the values have been edited but not saved in the file, the user may choose to save the file or not. If the user wishes not to save the new values he must press the F8 (ABORT FUNC) key. This will abort the saving process prior to quitting back to the main menu level.

PRINT EE
Under the EDIT EE FILE menu, this function allows the printout of the EEROM values in an EEROM disk file. Header information includes the time and date of the original saving of the file along with the time and date the file was last edited. EEROM values are shown in unsigned decimal with their address in decimal and their mnemonic.

EE JUMPERS

This program is similar to the EEROM CONSOLE except that it will present the EEROM values in BINARY as well as decimal. The only way to edit these values will be to toggle the state of a bit within the binary representation of the value.

When the user selects this function, a new function key menu will be displayed. The user may then read up to 36 values from the EEROM of a single drive. He may do this by selecting the standard jumper display which shows the commonly used addresses that will benefit by the binary representation of the numbers. The user may also select a custom jumper display which he can configure to include up to 36 of any addresses available in the drive. When the custom jumper list is selected and a custom list already exists, the program will ask the user if he wants to create a new list or use the previously configured list. If the user selects to configure a new list, he is instructed on how to do so.

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Once the jumper list is displayed, the user may select which address to edit by using the cursor arrow keys. The first bit of each word as they are selected will be highlighted. Using the TAB and/or shift TAB keys, the user may select the particular bit in the word. As the user "tabs" from bit to bit, the decimal weight of the bit and the bit number currently "tabbed" is displayed on the bottom right of the screen display. The user may use the TOGGLE BIT function key to set or reset the selected bit. The decimal representation of the word will be updated as he toggles the bits. The user, by toggling bits in the jumpers displayed, changes the value in MMI2 memory. He may then write the jumpers he has changed by pressing the WRITE EE function key. The drive to which the new values are being written must NOT be in a running state or an error message will be displayed. Again, as in the other tools, the drive will be checked to insure that the EEROM is enabled for writing before the values are actually written to the drive.

GEARIN CONSOLE

This program will automatically calculate the speed scale factors and write the results to the EEROM for drives with digital tachometers. The two scale factors are TCPPR and RPMSF. The user enters global data to be used (i.e. top speed counts and top speed FPM). He then has a choice of two methods to calculate the scale factors. Method 1 allows the entry of tachometer PPR data, gear ratio, and roll diameter. Method 1 then calculates the scale factors based on those entries. Method 2 allows the entry of a hand tachometer reading and interpolates the current scale factors to what must be used so that the drive is normalized to the hand tach. Method 2 requires that the drive is running a constant speed while taking the hand tach reading and entering this reading. It takes speed data from the SC300 console to calculate the new scale factors.

When this function is selected, the display screen will show the blank GEAR IN form. The user must "tab" to the drop number cell and enter the drop number of the drive he wishes to "GEAR IN". The current TCPPR and RPMSF will be read from that drive. The gear in data such as tachometer PPR, roll diameter, gear ratio, etc. will be displayed from what was last entered. The operator must tab to the desired cell to modify one of these parameters and then choose the method to calculate the new scale factors. The user presses the appropriate calculate function keys (method 1 uses the gear in data and method 2 uses a hand tachometer entry). When the new scale factors are calculated, the user may then write the new scale factors to the EEROM in the current drop. The drive must be stopped to be able to write these new values or an error message will be displayed. When the new values are written, the current values of TCPPR and RPMSF will match the new calculated values.

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ACC-DEC CONSOLE

This function allows the user to change the acceleration and deceleration rates of the drives. The DC300 system utilizes multi-motor parameters to specify acceleration and deceleration rates during different modes of drive operation in a coordinated system. In most coordinated systems a drive has at least two ramp rates. Those are the independent rates utilized when the drive is started and stopped, and the coordinated rates utilized for coordinated speed changes. The coordinated rate is the known as the MASTER RAMP and the parameters pertaining to this ramp should be identical in all the drives in the coordinated system. The independent ramp is the ramp used for independent acceleration and deceleration when a RUN or STOP is invoked to that drive and is known as the DRIVE RAMP. DRIVE RAMPS need not be the same in each drive if the line is not required to stay coordinated with RUN and STOP commands. There are two additional ramps called AUX1 and AUX2 ramps that may or may not be used depending on the application of the drives. The DRIVE RAMP is used when a RUN command is issued to the drive and if the ramps are configured in this manner, the ramp will automatically switch to the MASTER RAMP when the UP TO SPEED function is satisfied. The UP TO SPEED function reads the EE104 parameter and uses this value to determine how close to the speed setpoint the speed feedback must be before UP TO SPEED is satisfied. In addition the EE105 parameter specifies a hysteresis value which prevents the UP TO SPEED switch to "chatter" around the pick up point of the function.

When the user invokes the ACC-DEC CONSOLE, a partial form is displayed showing a cell for the drop number, drive name, and global data (Top Speed Counts and corresponding Top Speed FPM). The user must tab to the drop number cell and enter the drop number from which will be read the ramp parameters. The program will read the ramp parameters for all four ramps regardless if all four are used. The user is prompted to choose the ramp desired to display (MASTER, DRIVE, AUX1, or AUX2). When the operator selects the desired ramp to view, the rest of the ACC-DEC form is displayed for that drive.

The form will indicate whether the ramp is an S-Curve type ramp or a linear type ramp, and whether the UP TO SPEED function or the automatic switching of the ramps based on START/STOP control is active. The bottom part of the form shows the ramp times and rates as programmed in EEROM. The left side shows the raw values which show the time in seconds to go from 0 counts of speed to 6553 counts in speed and back again. If the ramp is an S-Curve type, the rounding times will also be shown; that is how much time will be spent rounding into the linear part of the ramp and from the linear part of the ramp to the final speed for the acceleration side of the ramp and the deceleration side. Also

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the total time of the ramp will be displayed based on the global top speed counts entered at the top of the form. The right side of the form shows the linear portion acceleration and deceleration rates in feet/minute/second (FPM/sec), and if an S-Curve, the speed at which the rounding will end in FPM.

The user has the ability to change the type of ramp from S-Curve to Linear and Linear to S-Curve. He may also activate or deactivate the UP TO SPEED function for the drive. The ramp parameters may be changed on either side of the form, where the user may specify the raw ramp times and the rounding times or the linear acceleration and deceleration rates and the rounding speeds. The total ramp time for an S-Curve type ramp is a read only parameter and may not be changed. After each new value is entered, the rest of the values on the form is recalculated and updated. Because the acceleration and deceleration rates and the rounding speeds create a "circular" calculation, several re entries may be necessary to obtain the exact values desired.

Once the parameters are entered, the user may write the results to EEROM of the single drive being currently viewed or may broadcast the values to all the drives by pressing the appropriate function keys. All parameters associated with the ramps will be written including any setup jumper parameters that select the type of ramp and the UP TO SPEED function. If the broadcast write option is selected, the jumpers are first written individually to all the drives. The data form on the screen will switch to the broadcast form showing all expected drop numbers and showing the results of the writing of the jumpers. As the jumpers are written into each drop, the status of "JUMP DONE" is written next to the drop number written. A status of "DROP OFF" will be shown next to those drops that are not on the LAN and the status "LAN FAULT" if the jumpers are not, for any other reason, written and/or verified for that drop. After the jumpers have been written, the numerical parameters will be broadcast to all the drives one address at a time. The status shown next to each drop number on the screen is the same as what is shown when executing a broadcast write.

If the UP TO SPEED function is selected, the UP TO SPEED SENSITIVITY (EE104) is defaulted to a value of 2, and the UP TO SPEED HYSTERESIS (EE105) is defaulted to a value of 10. In most cases, these values will be adequate for this function. If one or more drops have a DROOP parameter (EE209) of more than 0.5 %, the UP TO SPEED SENSITIVITY will be required to increase or the UP TO SPEED switch will never be satisfied. It is recommended that, even though all drives do not have DROOP, the worst case UP TO SPEED SENSITIVITY value be used for all drives. This different value can be programmed by the BROADCAST WRITE function in the EEROM CONSOLE program after the ACC-DEC CONSOLE is completed.

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APPENDIX C2 - MAINTENANCE CONSOLE - LOG CARDS
MMI2 MAINTENANCE LOG

The maintenance log is called up from the sub-function key menu available on the Maintenance Console (VC4) system screen. This log can be considered like a "Cardfile" where each log entry is a "card" in the file. The log cards can be added, edited, searched for, and printed.

VIEWING THE LOG CARDS

To move through the maintenance log cards, use the following keys:

HOME - Moves to first card in log
END - Moves to last card in log
PAGE UP - Moves one card closer to the beginning of log
PAGE DOWN - Moves one card closer to the end of the log (or to first blank card if already at the end of log).

ADD LOG CARD

To get to a blank card, press the "End" key on the keyboard. This will display the last entered card in the log. Press "Page Down" key on keyboard. This will cause a new blank card to be displayed. Enter the information on the card. NOTE: The "name" field cannot be left blank because without a name, the card will not be stored. It is suggested that a "Name convention" is followed to make it easier to search for data later on. An example would be to have each person use their initials. Note that access to the green fields is not possible. These fields are filled in automatically by the log. Each time a new card is created, the date and time are recorded on the card.

EDIT A LOG CARD

To edit a card (one that already exists with data on it), first find the card required. Using the cursor arrow keys and/or the "Enter" key, move the cursor to the data to be changed. Modify the data. Next, press the "Edit Log Card" function key. The card will be stored and the "Change Data" and "Time" will be updated to the current date and time. Note: Cards cannot be deleted. If a card no longer applies, edit out the information and replace with the words deleted. This method is used to keep the log modifications available.

SEARCH FOR CARD

Cards can be searched for by persons name, fault number, system, and date. The search process only works from the current card

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onwards to the end of the log. Therefore, if the user wants to search the whole log, press the " Home" key first. Pressing the Search function key will prompt the user to enter the "search criterion". Note that the program will only find cards that match exactly. All spaces and words must be identical thus the importance of having "standardized" naming conventions and entry methods is important. Once a card is located, the search can be repeated by again pressing the search function key, and following the prompt.

PRINTING THE LOG

Pressing the Print function key will prompt the user with a selection similar to the search function. The "ALL" selection will print the entire log and should only really be done if the system is not busy, as it may tie up the printer for some time.

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APPENDIX D - EVENTCON - EVENT RECORDER/ANALYZER

I INTRODUCTION TO EVENTCON

Eventcon is the main foreground program of Virtual Console 3 (VC3). This program serves many purposes in the MMI-2 system, its main purpose being the primary analysis tool for the entire control system. Eventcon receives alarm and event messages from the background program ALARM running in VC1, and stores these messages in two disk files dependent upon the type of message received. A small file of 100 active alarms is maintained and is displayed separately from the other file of messages. Active alarms are considered to be those critical and urgent alarms still existing on the system (alarms that have not been cleared or reset). The second and main file of messages contain the latest two thousand (2000) event and alarm messages with the oldest one thousand (1000) messages being limited to critical and urgent alarms only. All types of event messages will be contained in the first one thousand (1000) messages in the file. This main file forms the basis of the event message database on which this program operates.

The user may view all 2000 messages from the CRT, sort the messages into specific subsets of messages, totalize, print and obtain further detailed information from any of the messages in this data base. Naturally, these data base management functions can also be used on the 100 active alarm message base.

The EVENTCON screen consists of three main parts: the function key area, the message area, and the program status line. The screen will have the operating system status line just as the SC300 program has as well.

A. FUNCTION KEYS

The function key line offers the description of operating options available to the user. The function keys will change automatically to present operating options that are pertinent to the functions that are being used at any particular time. The function keys can be thought of as being context sensitive in that, based upon how the user is operating EVENTCON, the function keys will display only those operating options which are available. In some cases more than the seven options are available and therefore, any additional options are shown on function keys which are accessed by the eighth "MORE" function key. If no more than eight functions are available, the "MORE" key will become the eighth operating option.

B. MESSAGE AREA

The message area has a list of alarm/event messages recorded from

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the MMI-2 system. This area can contain up to 21 messages at any time. The entire file of messages may be viewed by using the appropriate NEXT PAGE or PREV PAGE or LATEST EVENTS function keys. The messages themselves are of the format described in the ALARM program section of this manual. Any one of the 21 messages displayed on the screen can be selected using the TAB or SHIFT TAB keys for further operations. The DOWN ARROW and UP ARROW keys duplicate the functions of the TAB and SHIFT TAB keys respectively.

C. STATUS LINE

The status line on the twenty fourth line of the screen display contains information on the mode of EVENTCON. The status line displays which file the user is currently displaying and which message the user has selected in that file. The selected message number corresponds to the chronological attribute of the message. If the line shows that message 1 is currently selected, the user has chosen the latest message available. If the line shows that message 100 is selected, there are 99 messages that have occurred after the selected message. If the user is viewing the screen and the this selected number increments to a higher number, EVENTCON is alerting the user that new messages have been received and by pressing the "LATEST EVENTS" function keys, those new messages may be viewed.

II EVENTCON DATABASE

There are actually three message files which form the EVENTCON database. The ACTIVE ALARM file can be thought of as a separate database from the EVENT file and the SUBSET FILE. The SUBSET FILE is actually a 126 message buffer that is created by the user from either the EVENT or ACTIVE ALARM files. The SUBSET file is the key memory area which allows the user to sort messages from the larger EVENT file for detailed analysis. All files are circular which means that they are fixed length files containing the newest messages that have been received by EVENTCON. Once any of these files become full, the oldest message is deleted from the file and the newest is added. The main EVENT file can hold up to 2000 messages. The first 1000 messages can be of any type and the last 1000 messages in this file will only be critical and urgent alarms. The circular limit for non-alarm event type messages is 1000. The ACTIVE ALARM file can contain up to 100 critical or urgent alarms. The SUBSET file can contain up to 126 messages of any type, dependent on the SUBSET file loading criteria specified by the user.

A. MESSAGE TYPES

There are several different types of messages. Messages are categorized by the following criteria:
HIGH ALARMS (Analog points exceeding a fixed setpoint)

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LOW ALARMS (Analog points exceeding a fixed setpoint)
DIGITAL ALARMS (Digital points in an alarm state)
DRIVE FAULT ALARMS (Faults declared from the DC300 drives)
MMI-2 PROGRAM ALARMS/EVENTS (MMI-2 program messages)
RETURN TO NORMAL (Any of the above alarms returning to normal)
DRIVE MODE/SETPOINT CHANGE EVENTS (Special logging messages)
SETPOINT CHANGE EVENTS (MMI-2 initiated setpoint changes)
GENIUS ALARMS (Genius I/O fault messages)

In addition, for each of the above categories, each type of alarm/event message can be one of five priorities or grades of alarms/events : URGENT, CRITICAL, ATTENTION, EVENT and NORMAL. These priorities are listed in order of the message severity.

Messages can be distinguished into these categories by the text in the message themselves. Here are some examples :

1. DIGITAL ALARMS

```
15:20:38 03Feb88 DA U 272 COMFAIL   COMMUNICATIONS                STALLD
```

Digital alarms are denoted by the DA mnemonic after the time and date fields. The next character defines the message's priority (URGENT in this case), and the next number is the configuration alarm number for this message. The next field is the signal name of the digital point on which this alarm is based. This name specifies the data base point in MMI-2 as well as the Series Six I/O nickname if the bit is in the Series Six program. The field after the name contains the description for the digital point. The last field is the bit's current state (the user may configure a name for the state of a bit, in this case STALLED is the name for one state and OK may be the name for the other state). Digital return to normal messages will show the non alarmed state, otherwise, the alarmed state will be shown in an alarm/event message.

2. ANALOG HIGH/LOW LEVEL ALARMS

```
15:27:58 03Feb88 HI C  12 FCUFB13  1ST DRYER LOAD                123.0   %
15:45:12 03Feb88 LO U 129 COU_VAC  COUCH VACUUM (INCHES of H2O)    1.9   IN
```

Analog alarm messages are generated whenever an analog point exceeds a configured high or low limit value. The HI or LO mnemonic after the time and date fields distinguish the HIGH and LOW level analog alarms from the other available messages. Like the digital alarm, the priority is shown next (C = CAUTION, U = URGENT) followed by the configuration alarm number. The signal name again is the data base location in the MMI-2 as well as the Series Six register nickname. The signal description follows the signal name which is followed by the value of the signal when the alarm message was generated. The last field is the engineering units of the signal. The return to normal message for a high and

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low level alarm will be generated when the signal has returned within the alarm limit thresholds plus the configured deadbands.

3. DRIVE FAULT ALARMS

```
15:27:00 03Feb88  4 U FIRST PRESS   LOSS OF AN AC PHASE                F22
```

The drive fault alarm message is a special class of messages only generated when MMI-2 has been configured to operate with a coordinated line of DC300 drives. The DC300 drive will report faults through the drive local area network (LAN) and MMI will build a text message describing the fault. The number after the time and date field is the drive number for MMI-2. This number is not necessarily the drive's drop number on the LAN. The next field is the priority of the message (URGENT) and the following field is the drive's name. This name matches the drive's name shown on the Super Drive Control Screen in SC300 (SDCS). The description of the fault is next followed by the fault number. The help facility that is part of EVENTCON (fully explained later) has complete help for drive faults, however; additional information of the fault may be found in the custom instruction book for that drive under the specified fault number.

4. MMI-2 PROGRAM ALARMS/EVENTS

```
15:23:49 03Feb88 72 E SC-SETUP     Execution from DISK complete     F93
15:27:00 03Feb88 75 E RECORDER     Pen channel started and sampling F203
```

MMI-2 consists of many programs which run concurrently. Most of the programs are actually executing in a background mode so that they cannot communicate to the user by showing something on the console CRT, nor can they get commands directly from the user from the console keyboard. When these background programs have something to tell the user, they do this by forming a data packet and sending it to a foreground task which can then display the message in the form of an alarm/event message. This class of messages from the MMI-2 background programs, have similar formats as shown above. The number after the time and date field is a unique program number for each of the background programs. The next field is the message priority. The next field is the name of the program that sent this message. The actual message is next and that is followed by the message number.

5. DRIVE MODE/SETPOINT CHANGE MESSAGES

```
15:27:12 03Feb88 MDCH PRESS O.S. to PRESS SECT                NORMAL STOP
15:29:21 03Feb88 SPCH MASTER CTL - BROADCASTED      MMS       3460.0  FPM
```

Drive mode and setpoint change messages are a special class of messages. They are always of the ATTENTION priority and the format of the messages change slightly depending on whether it is a mode or setpoint change, and what the destination(s) are. Mode

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changes are commands sent from one drop on the LAN to another drop. These are boolean type or single bit type commands such as RUN, STOP, JOG, etc. Setpoint changes are value changes to a drop. In other words, setpoint change messages will specify the new value of a setpoint sent by a drop to another drop(s). Both types of messages begin with the source of the change just after the time and date fields. The next several fields are devoted to the destination of the change and can denote a single drop, a group of drops or a specific from/to destination. For mode change messages, the last field is the command that was sent. In the mode change example above, the entire message can be interpreted as : The press operator station sent the command, NORMAL STOP, to the entire press group of drives. For setpoint change messages, the second to the last field is the setpoint being changed, and the last field is the new value that was sent. In the above setpoint change message example, the master control device (Series Six) sent to the entire line of drives (BROADCASTED) a new master machine speed reference (MMS), and that new reference is 3460 feet per minute (3460 FPM). The EVENTCON help facility will show the format of the messages for other destination designations.

6. INTERNAL SETPOINT CHANGE MESSAGES

```
15:26:01 03Feb88 SPCH      E1MXTSP   EXT 1 M-PIPE MELT  0.0000 to  230.0 DEG
```

Internal setpoint change messages are generated when any value is changed at the MMI-2 SC300 console regardless of whether it is a drive parameter or a system parameter. The designation SPCH after the time and date fields is the same as a drive setpoint change message, however; these messages will always have an EVENT priority while drive setpoint change messages will always have an ATTENTION priority. The field after SPCH is the nickname of the data base point that is being changed. Again, if this point exists in the Series Six, this name will be common to both the Series Six and the MMI-2 system. The data point's description follows its nickname. For these messages, the old value is recorded as well as the new value followed by the engineering units.

B. MESSAGE PRESENTATION

Messages have a color on the EVENTCON screen which are dependent on the message priority. This allows the user to easily distinguish the severity of an event by just recognizing the color of the message. The colors are configured to a default scheme but may be modified by the user using the MMI-2 system configuration tools provided with the system. The default message colors (text color/background color) are:
URGENT - RED/BLACK
CAUTION - YELLOW/BLACK
ATTENTION - WHITE/BLACK

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EVENT - LIGHT BLUE (CYAN)/BLACK
RETURN TO NORMAL - GREEN/BLACK

Messages are displayed and stored in chronological order. On EVENTCON, the screen will show the latest events lower on the screen and the older events as the user scrolls up the screen.

III USING EVENTCON

A. VIEWING THE MESSAGES

1	2	3	4	5	6	7	8	
ACTIVE ALARMS	**EVENT SUBSET**	**NEXT PAGE**	**PREV PAGE**	**LATEST EVENTS**	GRAPH EVENT	RETURN SC-300	MORE	

When EVENTCON is invoked, the main EVENT file is initially displayed along with the initial function key set of descriptors. If the file has any messages, the latest message is automatically selected. The status line will show that EVENTCON is viewing the EVENT file and message 1 is selected. As additional messages are received, the message selected number on the status line will increment showing that there are newer messages now in the file.

The user has a choice to view one of the three files kept by EVENTCON. The main EVENT file is shown first and the user may switch to the ACTIVE ALARM file or the SUBSET file by pressing the appropriate function keys. Once the desired file is selected and displayed, the user may now scroll through the contents of the file by several different methods. To simply view the messages in the file, the user may press the NEXT or PREV PAGE function keys. The page up and page down keyboard keys mimic these function keys. The user may scroll backwards in time, to the oldest message in any of the respective file. Once the oldest message is displayed, the scrolling will stop even if the maximum message number is not reached. The user may then scroll forward in time until the latest message is reached.

The LATEST EVENT function key is a quick way of moving the display to the latest 21 events or alarms in any of the three files. A description of the other function keys on the above function key line will be given later in this document. When the user presses the MORE function key (F8), another function key line is displayed. The functions on this line is described next.

B. PRINTING MESSAGES

1	2	3	4	5	6	7	8	
LOAD SUBSET	ALARM TOTALS	**PRINT EVENTS**	HELP	CONFIG HELP	ACK ALARMS	QUIT		MORE

Once the user is displaying the desired file, the user may print any of the messages. By pressing the PRINT function key, a new function key line is displayed. When the user is viewing the main EVENTCON file the following three options are displayed

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1 PRINT TAB-NOW	2 PRINT SCREEN	3	4	5	6	7	8 EXIT

1. PRINT TAB-NOW

If the user presses this function key, all messages in the displayed file from the currently selected message (shown on the display with a blue background and the normal message text color) to the latest message is printed.

2. PRINT SCREEN

This function key will print the currently displayed (21) messages.

3. EXIT

This function key will return the user to the main function key descriptors and leave the PRINT function keys.

4. PRINT ACTIVE ALARMS/SUBSET BUFFER

When the user is viewing the ACTIVE ALARM file or the SUBSET buffer, the PRINT TAB-NOW option is replaced by the PRINT ALL key. This means that the only option the user has to print messages from the ACTIVE ALARM file or the SUBSET buffer is to print the entire ACTIVE ALARM file or the entire SUBSET BUFFER. The PRINT SCREEN key, however; does exist for the ACTIVE ALARM file and the SUBSET buffer so the user may print portions (21 messages at a time) of these two areas.

When EVENTCON is printing, it entirely uses the printer resource. EVENTCON will not relinquish the printer resource to any other task until it has completed the printing request. The user may be required to schedule the best time to use the print function for EVENTCON especially if a large portion of the EVENT file is printed.

C. LOADING THE SUBSET BUFFER

1 **LOAD SUBSET**	2 ALARM TOTALS	3 PRINT EVENTS	4 HELP	5CONFIG HELP	6 ACK ALARMS	7 QUIT	8 MORE

Function key 1 of the "MORE" function keys will allow the user to load the subset buffer based upon the message selected. The LOAD SUBSET function will take the type of message that is selected and then with the users input from context sensitive SCOPE function keys, the subset buffer will be loaded. The SUBSET buffer may be loaded by selecting a message in any of the three file areas, the EVENT file, the ACTIVE ALARM file or even a previously loaded SUBSET buffer. The user would then press the

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LOAD SUBSET key which would then present the appropriate SCOPE keys for the message selected. The scope function keys allow the user to broaden the scope of the selected message type to allow the subset buffer to be loaded with more varying messages available in the database. Each different type of message (see the above description of message types), has its own set of scope keys that is only pertinent to that type of message. When the user presses the LOAD SUBSET key (F1), the particular scope keys for the message selected are automatically presented. Here are some examples:

1. DIGITAL ALARMS

1 NAME MASK	2	3 ALL LEVELS	4RETURN NORMAL	5 PREV 5 EVENT	6	7 DONE	8 EXIT

The scope keys allow the broadening of criteria for messages to be loaded into the subset buffer. If the user presses the DONE function key without making any other selections, only messages that had exactly the same attributes as the one selected except for the time and date will be loaded into the subset buffer.

```
15:20:38 03Feb88 DA U 272 COMFAIL   COMMUNICATIONS                STALLD
```

As an example, look at the above digital alarm message. If no scope options are chosen, only those digital alarm messages with an URGENT priority, and with the signal name COMFAIL will be loaded into the subset buffer. If the NAME MASK scope is selected, the user is prompted to enter a name mask criteria for EVENTCON to use. If a name mask of ???FAIL was entered, and no other scope keys were selected before the user pressed DONE, all digital alarm messages with an urgent priority and having a signal name ending with FAIL will be loaded into the subset buffer. The ALL LEVELS key allows messages of any priority to be loaded (but they must match all other scopes of the selected message) into the subset buffer. RETURN NORMAL scope allows the return to normal messages to be loaded into the subset buffer along with the alarm messages. The PREV 5 EVENT key will force EVENTCON to load into the subset buffer for each message that matches the desired scope, the preceding five event messages that occurred prior to that message. The user can then use this to analyze what was happening just prior to the alarm messages being generated.

If the user has selected a return to normal message, and presses the LOAD SUBSET key, the scope keys will be presented with the RETURN NORMAL key automatically selected. This key cannot be deselected and the alarm messages which the return to normal messages showed as being cleared will be loaded into the subset buffer along with the return to normal messages.
Select all scopes desired and then press the DONE key. When the DONE key is pressed, EVENTCON will search the EVENT file for all

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matching cases and will then load and display the SUBSET buffer. Note that the status line will show that the SUBSET BUFFER is being displayed with the latest message in that buffer selected. The EXIT key allows the user to return to the main function keys after the subset buffer is displayed. The user may use the EXIT key to abort the selecting of scopes and showing the subset buffer as well. If the subset buffer has been displayed, the EXIT key will leave the current message displays on the screen. To return to the main EVENT file or the ACTIVE ALARM file, the user must select the desired display from the newly displayed function keys.

2. ANALOG HIGH/LOW ALARMS

```
1 NAME  2 HI+LO 3  ALL  4RETURN 5 PREV  6        7 DONE  8     |
|.MASK  |ALARMS |LEVELS |NORMAL |5 EVENT|        |       |EXIT|
```

The scope keys for analog high/low alarms are very similar to digital alarm scope keys except an additional scope key was added. The user may choose to load the subset buffer with occurrences of messages that exceeded the high limit and low limit as well. If the key is not selected, only the messages that have the same type (HI or LO) will be loaded into the subset buffer. If the user had selected a return to normal message when loading the subset, both HI and LO alarms will be loaded into the subset buffer.

3. DRIVE FAULT ALARMS

```
1  ALL  2 TRIP  3ANNUNC 4RETURN 5 PREV  6        7 DONE  8     |
|DRIVES |FAULTS |FAULTS |NORMAL |5 EVENT|        |       |EXIT|

15:27:00 03Feb88  4 U FIRST PRESS   LOSS OF AN AC PHASE          F22
```

Drive fault messages have different criteria for subset buffer loading. Using the above message as the selected message in this example, if the user selects ALL DRIVES, the subset buffer will be loaded with LOSS OF AN AC PHASE fault messages for all drives in addition to the FIRST PRESS. If the user selects TRIP FAULTS, the subset buffer will be loaded with all URGENT drive fault messages for the FIRST PRESS. If the ANNUNC FAULTS key is selected, the subset buffer will be loaded with all URGENT faults and CAUTION fault messages, even if the TRIP FAULT key is NOT selected. This is because the message that is selected is an URGENT alarm message, and selecting ANNUNC FAULTS will also select all TRIP FAULTS. Similarly, if the user had selected a CAUTION drive fault message, pressing the TRIP FAULTS key will again load all CAUTION faults as well as URGENT faults into the subset buffer. The other scope keys operate similarly to what is described above.

4. MMI-2 PROGRAM ALARMS/EVENTS

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```
 1ALL SYS2ALL SYS3ALL SYS4RETURN 5 PREV  6        7 DONE  8     |
 |PROGRMS|ALARMS |EVENTS |NORMAL |5 EVENT|        |        |EXIT|
```

The scope keys for MMI-2 program alarms/events will allow the subset buffer to be loaded with messages from all system programs with either the EVENT and ATTENTION priorities or URGENT and CAUTION priorities or both. Selecting ALL SYS PROGRAMS by itself will not give any additional scope to the message selected. This scope key requires that either ALL SYS ALARMS or ALL SYS EVENTS or both keys be selected as well. If just ALL SYS ALARMS is selected, all URGENT and CAUTION messages for the program in the selected message will be loaded into the subset buffer. If ALL SYS EVENTS is just selected, all EVENT and ATTENTION priority messages for the selected program will be loaded. If either of these keys is selected and ALL SYS PROGRMS is also selected, all of the selected priority messages from all MMI-2 programs will be loaded into the subset buffer. The RETURN NORMAL key allows return to normal messages to be included in the SUBSET buffer. Remember that EVENT priority MMI-2 program messages have no corresponding return to normal messages.

5. DRIVE MODE/SETPOINT CHANGE MESSAGES

```
MODE CHANGE
 1  ALL  2  ALL  3  ALL  4        5 PREV  6        7 DONE  8     |
 |SOURCES| DEST  | CMDS  |        |5 EVENT|        |        |EXIT|

SETPOINT CHANGE
 1  ALL  2  ALL  3  ALL  4        5 PREV  6        7 DONE  8     |
 |SOURCES| DEST  |SETPNTS|        |5 EVENT|        |        |EXIT|
```

```
15:27:12 03Feb88 MDCH PRESS O.S. to PRESS SECT                NORMAL STOP
15:29:21 03Feb88 SPCH MASTER CTL - BROADCASTED        MMS     3460.0  FPM
```

The drive mode\setpoint change messages has similar options for loading the subset buffer. These scope keys allow the user to load only the selected source or messages from all sources and/or only the selected destination(s) or messages directed to any destination. For mode change messages, the user may choose to load the subset with messages for only the selected command or messages with any command. Setpoint change messages may be chosen for any setpoint or all setpoints. The setpoint field pertains to the setpoint name and not the value. In the above example message, if the user does not choose ALL SETPNTS, the subset buffer will only be loaded with messages that send MMS. If the user does choose ALL SETPNTS, the messages loaded into the SUBSET buffer can be any available STATUS-S setpoint (DRAW, IRATIO, etc.).

6. INTERNAL SETPOINT CHANGE MESSAGES

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```
1 NAME  2        3        4        5        6        7 DONE  8      |
|.MASK  |________|________|________|________|________|________|EXIT|
```

```
15:26:01 03Feb88 SPCH      E1MXTSP    EXT 1 M-PIPE MELT  1.0000 to  230.0 DEG
```

Internal MMI-2 setpoint change messages can be loaded into the subset buffer with one option. The NAME MASK key will allow the user to load the subset buffer with setpoint change messages pertaining to other signals than the signal in the selected message. All setpoint change messages may be selected by entering a name mask of ???????.

7. TOTALIZING ALARMS

```
1 LOAD   2 ALARM  3 PRINT 4 HELP   5CONFIG 6  ACK   7 QUIT   8      |
|SUBSET  |TOTALS |EVENTS |        | HELP   |ALARMS |        |MORE|
```

A special method of loading the SUBSET buffer is to select the ALARM TOTALS key from the "MORE" set of function keys. This option is independent of the currently selected message because it is a direct function to search through the entire EVENT file data base (all 2000 messages) and load the SUBSET buffer with the 126 most frequently occurring URGENT and CAUTION priority messages. The messages are shown in the SUBSET buffer in their entirety with number of occurrences shown to the left of each message. The most frequently occurring message is shown at the bottom of the SUBSET buffer screen and the least frequently occurring message is stored at the top of the buffer. In other words, the user may scroll up the buffer with the PREV PAGE key to display the messages that have been occurring with less frequency.

This function allows the user to analyze the alarms that have been most prevalent over a period of time and then concentrate the effort in correcting those alarm conditions. The user may reload the subset buffer from this display by selecting the message desired and then pressing the LOAD SUBSET key. The normal scope keys are presented and the SUBSET buffer will be reloaded with the SUBSET buffer load request. As an example, the user selects a message that has occurred ten times. He then presses the LOAD SUBSET key and then the DONE key on the scope function key line. The SUBSET buffer is then loaded with the ten alarm messages as they were actually recorded.

D. USING THE HELP SYSTEM

```
1 LOAD   2 ALARM  3 PRINT 4 HELP   5CONFIG 6  ACK   7 QUIT   8      |
|SUBSET  |TOTALS |EVENTS |        | HELP   |ALARMS |        |MORE|
```

The Help system resident in EVENTCON is a context sensitive help

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system to provide additional information on alarm/event messages. The help system is accessed by pressing the HELP function key or the F10 function key on the console keyboard after the user has selected the message on the eventcon screen. Any message of the three message file areas may be accessed.

Once the user has pressed the HELP key or F10, the message display is replaced by another screen with the selected message shown at the bottom of the screen in place of the status line. The rest of the screen is devoted to a text display with two parts. The first part is four lines of text devoted to a general description of the message selected. The second part is fifteen lines devoted to explaining the possible causes and cures for the selected message or a detailed description of the message itself.

The help system contains several help files. Each type of message has a help file devoted to it. For the help system, the message types are categorized as follows:

```
SYSTEM ALARMS   : DIGITAL ALARMS
                  ANALOG HI/LO LIMITS
DRIVE FAULTS    : DRIVE FAULTS
DRIVE MODELOG   : DRIVE MODE CHANGES
                  DRIVE SETPOINT CHANGES
MMI SYS ALARMS  : MMI-2 PROGRAM ALARMS/EVENTS
                  INTERNAL MMI-2 SETPOINT CHANGES
```

The help files for DRIVE FAULTS, DRIVE MODELOG, and MMI SYS ALARMS are completed and included with every MMI-2 system. Each type of possible message in these categories has a preconfigured help screen already configured, however; each help screen has been configured with some extra space (in most cases) and the user may modify the help screens with his own explanations. The user may even delete lines from the preconfigured help screen to remove help descriptions that may not pertain to his particular application.

In some cases, a preconfigured help file may be delivered with EVENTCON for SYSTEM ALARMS, but in most cases the help file for this category will be empty. It is up to the user to create the help file for this category since the majority of messages in this category is very dependent on the application involved.

When the user presses the HELP key, the help screen is displayed along with a new function key line.

```
1 EDIT  2       3       4       5       6       7       8     |
|.HELP  |_______|_______|_______|_______|_______|_______|EXIT|
```

If the user is satisfied with the help screen now displayed, the EXIT key will allow the messages to be displayed from the file area that was being viewed prior to obtaining help. If the user

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wishes to modify the help screen displayed he must press the EDIT HELP key before any modifications can be done. There is security involved with this key. The user must have a security level of 6 or higher in the SC300 system or the EDIT HELP key will not even be displayed. If the user has an appropriate security level, the EDIT HELP key will be displayed and the user may edit the help screen.

1. EDITING THE HELP SCREEN

1INSERT LINE	2DELETE LINE	3	4	5	6	7 SAVE	8 EXIT

When EDIT HELP is selected, EVENTCON changes its mode and becomes a simple text editor. In this mode, the arrow keys become full function cursor keys which will move the text entry cursor to the editable portions of the screen. The editor is always in an overwrite mode, that is when characters are typed from the keyboard, the new characters overwrite the existing text on the screen.

If the user wants to delete text he may use the delete key or the backspace key on the keyboard to delete individual characters or he may use the DELETE LINE function key to delete an entire line of text. When any of these keys are used the space taken up by the deleted character or line is removed and taken up by the next character or line. To insert text, the user must press the insert key on the keyboard which will insert a space at the cursor position and push the character over the cursor and the characters to the right of the cursor one space to the right. If any character moves off the right edge of the screen, it is lost forever. If the user presses the INSERT LINE function key, a line is inserted in the text on the line of the cursor. Any lines of text at or below the cursor is moved down one line. Any lines moved beyond the bottom the screen is lost forever as well.

When entering text, the cursor will automatically wrap to a new line when the right edge of the screen is reached, however; if the cursor wraps to a new line in the middle of a word, the word will be broken in two pieces, half on the previous line and the rest on the present line.

Once the user has entered the desired text, he may press the SAVE function key which will save the new help screen permanently for the selected message. The user may EXIT the edit mode at any time, even after modifications has been made, without saving the screen and the original help screen will return if help is invoked for the selected message again.

The description above is valid for editing a help screen after it has been displayed by pressing the HELP key while "tabbed" on a message in one of the three message files. If the user desires

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to enter a new or modify an existing help screen without selecting a message in the EVENTCON display and viewing help, (as when the message desired does not exist) he may select the CONFIG HELP key from the "MORE" level of function keys. Here, again, the CONFIG HELP key requires that the user have an appropriate security level for this function to be available (level 6 or higher). If the security level of the user is not high enough, the CONFIG HELP key will not be displayed on the function key line.

2. CONFIGURE HELP

1 EDIT HELP	2IMPORT HELP	3	4	5	6	7	8 EXIT

When the CONFIG HELP key is pressed a new set of function keys are displayed (shown above). The EDIT HELP key allows the user to select the help file to be edited.

1 DRIVE FAULTS	2 DRIVE MODELOG	3 SYS ALARMS	4MMI SYS ALARMS	5 USER ALARMS	6	7	8 EXIT

Remember that help screens for DRIVE FAULTS, DRIVE MODELOG, and MMI SYS ALARMS have been preconfigured prior to installation. The SYS ALARMS help file and the USER ALARMS help files are the files that are most likely to need modifications.

For all help files except for the SYS ALARMS help file, the user is required to enter the fault number. When the fault number is entered the help screen corresponding to that number is displayed. The user then uses the EDIT HELP function keys and keyboard to modify the displayed screen. This procedure is exactly like the editing procedure described above.

For the SYS ALARM file, the user will be prompted to enter the signal name. The signal name is the seven character mnemonic in DIGITAL ALARM messages and ANALOG HI/LO ALARM messages that specify the signal on which the message is based. Once the signal name has been entered, the help screen for that signal name (and therefore the appropriate alarm) is displayed. The user may then edit this help screen as described above.

The IMPORT HELP key allows an entire help text file to be imported into EVENTCON. This function requires a specific text file format and is reserved for the use of system designers. In some cases it would be more convenient for a user to enter all his custom help information in a text format and import this help into EVENTCON. The text file format will not be described in this document but its requirements may be obtained on request. Consult your GE representative.

E. GRAPH EVENT

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1ACTIVE ALARMS	2 EVENT SUBSET	3 NEXT PAGE	4 PREV PAGE	5LATEST EVENTS	6 **GRAPH EVENT**	7RETURN SC-300	8 MORE

This function integrates with the program SCTREND normally running in Virtual Console 2 (VC2). The user may select a message in any of the three EVENTCON files and press the GRAPH EVENT key. When this is done EVENTCON checks the message for a legal signal name mnemonic. If the selected message contains a valid signal name, EVENTCON will search the trending data base to determine if the signal is being actively recorded by the background program RECORDER. If the RECORDER program is not running, an error message will be displayed or if the signal is not being actively trended an error message will be displayed.

If EVENTCON finds that the signal is being trended, it will automatically switch the CRT to VC2 and send a message to SCTREND to graph the appropriate PEN file that contains the selected signal. If SCTREND is not running at the time the switch to VC2 is made, SCTREND will display the signal when it is started.

SCTREND graphs the entire PEN file and attempts to place the graph cursor at the time specified in the alarm/event message selected. If the PEN file does not contain any samples at the specified time, an error message is shown on the SCTREND screen.

Those messages not containing a valid trendable signal name can not be graphed with the GRAPH EVENT function. There is one exception and that is the MMI-2 PROGRAM EVENT that alerts the user that a drive history has been recorded to disk.

If the user selects a message that states that a drive history has been recorded to disk and presses the GRAPH EVENT key, a message is sent to the background HISTORY program to unload the specified raw history file as specified in the event message. If HISTORY cannot unload the history file into graph files, it will send an alarm message back to EVENTCON that it could not unload the file (the history file may not exist or may be damaged). If the HISTORY program successfully unloads the raw history file, a message is passed to SCTREND to graph the unloaded history file. Here, again, the console CRT is automatically switched to VC2, and if SCTREND is running, the history file(s) will be shown. If SCTREND is not running, the history files will be shown when SCTREND is started.

F. ACKNOWLEDGE ALARMS

1 LOAD SUBSET	2 ALARM TOTALS	3 PRINT EVENTS	4 HELP	5CONFIG HELP	6 **ACK ALARMS**	7 QUIT	8 MORE

In some MMI-2 systems, an alarm attribute may be configured to set a bit in the Series Six to enable an audible alarm on the

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receipt of an alarm message in MMI-2. The ACK ALARM key will set another bit in the Series Six so that a Series Six rung may be programmed to turn the audible alarm off. In actuality the bits being set by MMI-2 can be used in whatever manner the user desires. This function simply manipulates a bit in the I/O table of the Series Six.

G. RETURN SC-300

1 ACTIVE ALARMS	2 EVENT SUBSET	3 NEXT PAGE	4 PREV PAGE	5 LATEST EVENTS	6 GRAPH EVENT	7 **RETURN SC-300**	8 MORE

This function will switch the user to VC1 where the SC300 programs should be executing.

H. QUIT

1 LOAD SUBSET	2 ALARM TOTALS	3 PRINT EVENTS	4 HELP	5 CONFIG HELP	6 ACK ALARMS	7 **QUIT**	8 MORE

This function will terminate the operation of EVENTCON. The user must have a security level of 6 or greater for this function to be valid. If the user does not have a high enough security level, this function key will not be presented as an option on the function key line.

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APPENDIX E - SCTREND OPERATIONS MANUAL

1.0 Introduction

The SCTREND program is used to graphically display data that is stored in SC300 historical files. These files are generated by both the SC300 pen recorder and the SC300 drive history unload function. SCTREND will allow up to seven pens (files) of data to be displayed simultaneously. All pens share a common time base (X axis), however each pens vertical scaling (Y axis) is independent. Time on the X axis is displayed in one of 3 different ways, these are MM/DD/YY, HH:MM:SS or MM:SS.SSS. All user action is initiated through the use of function keys, similar to those on the SC300 operator console. Two sets of seven function keys exist these sets are selected by depressing the more key (F8).

2.0 SCTREND Features

2.1 Seven pens/graphs displayed simultaneously
2.2 Time resolution - years to milliseconds
2.3 Data cursor - allows precise value determination from the graph
2.4 Pan function - allows data to be scrolled through from newest to oldest
2.5 Zoom function - allows data to be focused into a smaller period of time
2.6 Save and load setup function
2.7 Draw latest function - displays the newest data available in the pens
2.8 Pen On/Off - allows a pen to be temporarily disabled
2.9 Automatic vertical best fit scaling
2.10 Display printing in either black and white or color depending on hardware

3.0 SCTREND Operation

SCTREND is started from the ANALYSIS menu in console 2 of the SC300 system. This is the only recommended method of starting SCTREND. The program will initially display function keys and a blank graph. At this time a previously saved setup file may be loaded or a pen may be enabled manually.

3.1 "TABBING" to data entry fields

The current tabbed location is where the blue box is. This box is moved from field to field by using the TAB key and shift TAB key. This box is used to select the field upon which the CHANGE POINT function key (F1) will act. If using a touch screen the desired location may be "touched" instead of tabbing.

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3.2 "CURSORING" on the graph

The graphic cursor is a white vertical line on the graph display. This cursor is moved by jumps of 10 with the left and right arrows. Fine positioning can be accomplished with the up and down arrows. If using a touch screen the desired location may be "touched" instead of cursoring.

3.3 Enabling a pen

A pen is enabled by tabbing to one of the seven possible pen positions and depressing the change point key (F1). The pen positions are the seven tab fields directly to the right of the text "Name:". At this time the program will prompt for a filename. This filename varies depending on the source of the historical data file. If the data was taken by the pen recorder program the filename will be of the form "PEN.01" through "PEN.64". Data entry is always terminated by either the enter key to accept changes or the ESC key to abort and disregard the entry. During data entry the following keys are functional: left arrow, right arrow, insert, delete, and backspace. If a valid filename is entered, and data exists in the file for the current period and start time, SCTREND will then prompt for the high limit. The high limit shown is the largest data value found. This value can be either accepted or changed. Next the low limit will be prompted. After completing the low limit data entry, the program will proceed to graph the data extracted from the pen file. If no start time or start date has been specified, SCTREND will default to the time and date of the newest data in the file. The default period is the time span of the first file displayed.

3.4 Changing the Y axis high and low limits for a pen

Tab to the desired pen limits and depress CHANGE POINT. The new high and low limits will then be requested.

3.5 Changing the graph start date

The starting date is changed by tabbing to the date field and depressing the change point key. At this time the desired date will be requested. The date may either be entered in MM/DD/YY format or as an incremental date such as "-1D" meaning 1 day older. The valid time and date suffixes are:

D: day
H: hour
M: minute
S: second

If a blank date is entered, the date and time will be set to the most current time in any of the currently displayed pen files.

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3.6 Changing the graph start time

The starting time is changed by tabbing to the time field and depressing the change point key. At this time the desired time will be requested. The time may either be entered in HH:MM:SS format or as an incremental time such as "1H" meaning 1 hour later. See 3.2 for valid time and date suffixes. If a blank time is entered, the date and time will be set to the most current time in any of the currently displayed pen files.

3.7 Changing the start time and date by using PAN LEFT/PAN RIGHT

PAN is used by cursoring to a desired left or right axis time on the graph then depressing the PAN LEFT or PAN RIGHT function key. PAN LEFT causes the graph to be shifted so that the cursor position becomes the left axis of the screen. PAN RIGHT causes the graph to be shifted so that the cursor position becomes the right axis of the screen.

3.8 Changing the graph period

The period is changed by tabbing to the period field and depressing the change point key. At this time the desired period will be requested. The period is entered as "1D", "1H" or "1.02H" etc.

3.9 Changing the graph period by using ZOOM

ZOOM is used by cursoring to a desired end time on the graph then depressing the ZOOM function key. This causes the graph to be drawn again with the graph period being the time between the right axis and the cursor and the start time is unchanged.

3.10 DRAW LATEST function

This function will check each pen which is currently defined and determine the time of the freshest data available and set the graph start time equal to this. This function is typically used to update the display with any new data which has been logged to the file since the last Draw Latest.

3.11 UPDATE VALUE function

Update Value is used to obtain the value of the pen at the current cursor location. Cursor to the desired time on the graph and depress the Update Value function key. The values at the cursor of each pen will be displayed at the bottom of the screen below the pens.

3.12 LEFT and RIGHT AXIS function

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The LEFT AXIS function key causes the labels on the left axis of the graph to toggle through the left axis labels for each pen. The RIGHT AXIS function key does the same for the right axis.

3.13 SAVE SETUP function

This function will prompt for a filename (use no extension). The user will then be asked if the current start time and date is to be saved with the setup. If the start time and date is not saved, the start time and date will be set to the time and date of the most recent data in any of the files that were specified for loading. Otherwise, the time and date will be restored to the start time and date being viewed when the setup was saved. The current setup will then be saved to the file "FILENAME.PST".

3.14 LOAD SETUP function

This function will prompt for a filename (use no extension). The previously saved setup will then be loaded from the file "FILENAME.PST".

3.15 PEN ON/OFF function

This function is used by tabbing to the desired pen and depressing the PEN ON/OFF function key. The tabbed pen will either be turned off if it was on or on if it was off. This function allows pens to be easily temporarily removed from the graph and then added back again. A pen which is off has its text displayed in gray.

3.16 PRINT function

The print function will print the current display to the default printer for the analysis console (console 2). The type of printer and various print options are specified in the file ALARMCON.IBM. Currently SCTREND supports black and white printouts to an EPSON compatible graphic printer. Color printouts are possible if using an HP PAINTJET printer. The following options are specified in ALARMCON.IBM:

```
printer type: epson, HP paintjet
paper type: paper, transparency  (for paintjet only)
form feeds: none, before, after, before+after
```

How to specify these options is documented within the file ALARMCON.IBM.

3.17 RETURN SC300 function

This function will switch consoles to the SC300 main console.

3.18 SQC

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This function will display the frequency distribution histogram of the currently tabbed pen. The frequency distribution is evaluated over the span currently displayed on the graph. The normal distribution bell curve is superimposed upon the actual histogram as an aid in determining the "normality" of the data. The mean and standard deviation are displayed in the upper right corner of the graph. The x axis of the histogram is graduated in standard deviations from the mean as follows:
(-3dev,-2dev,-1dev,Xbar,1dev,2dev,3dev)
The y axis is in percent of the sample population evaluated.

3.19 REDRAW ON/OFF

This function will turn display updates off. This function is useful to speed up certain operations such as changing the limits of all the pens on the graph. The following is an example of how to use redraw on/off.

1. depress REDRAW ON/OFF (turns redraw off)
2. change limits pen1 (display does not update)
3. change limits pen2 (display does not update)
4. change limits pen3 (display does not update)
5. change limits pen4 (display does not update)
   ...
6. depress REDRAW ON/OFF (turns redraw on and updates display)

4.0 Operation Tips

4.1 To "blow up" an area on the graph.

1. cursor to right edge of area
2. PAN RIGHT
3. cursor to left edge of area
4. ZOOM

4.2 To autoscale pen limits

1. tab to desired pen name
2. CHANGE POINT
3. ENTER
4. ENTER
5. ENTER

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